PORTFOLIO

Ziqi Cui

-

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ARCHITECTURE - URBAN- COMPUTATIONAL DESIGN

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Weaving Octopus
A smart weaving prototype exploration

ML in Urban design/ Quantitative urban analysis

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A quantitative urban study by machine learning tools

What makes a park popular?
A quantitative urban study by machine learning tools

Go to Green
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Offsetting carbon emission of the city blocks by implementing a comprehensive greening plan

ML in architetcure design

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Generative AI in 3D
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Architecture Design

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How will urban facilities meet the diverse and dynamic needs of the public?

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Architecture Preservation

The MONZA Park
A preservation project of the royal park Monza

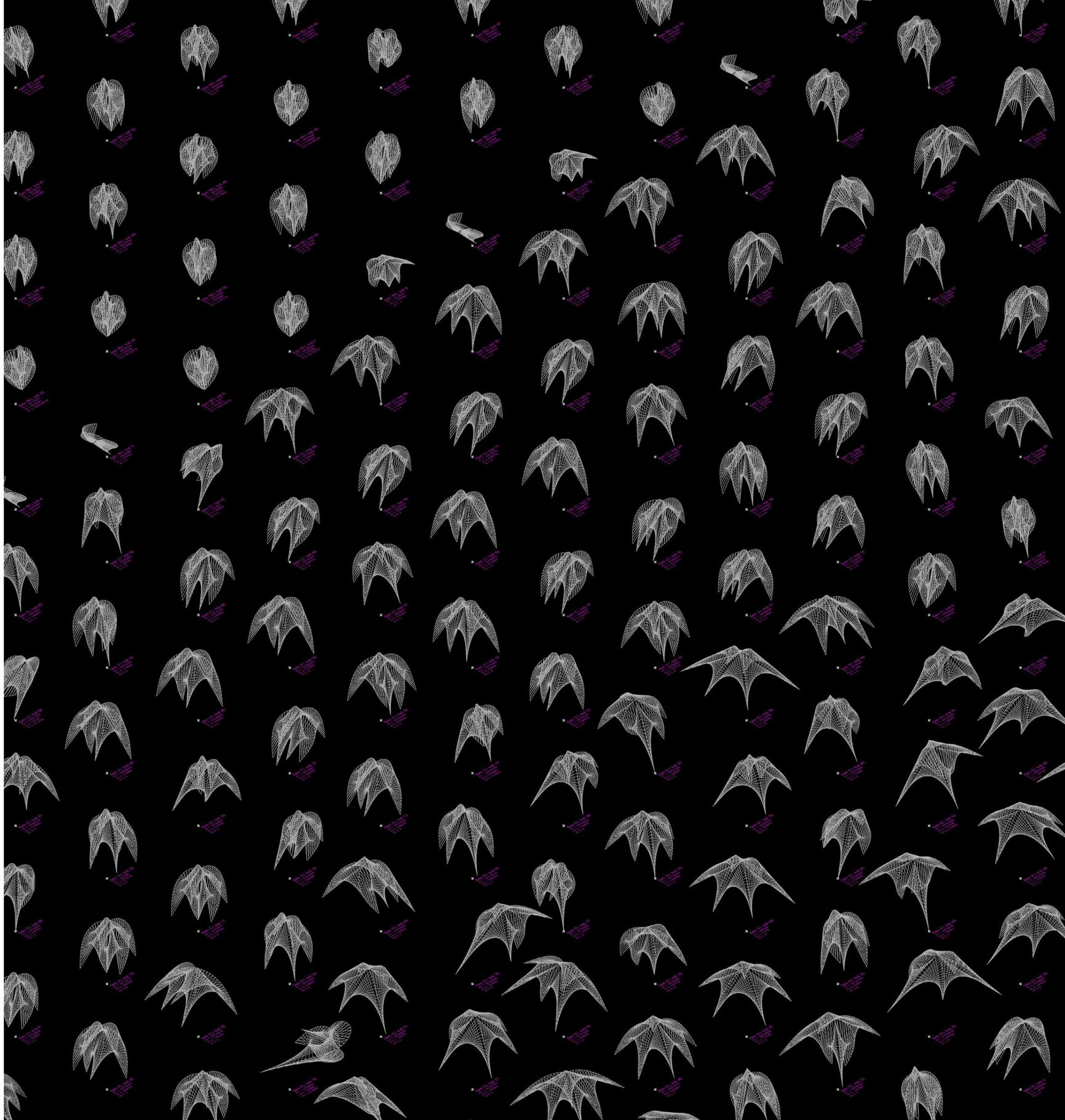
Others

Other works/ digital modeling/ drawing

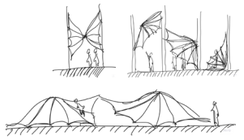
Weaving Octopus

A smart weaving prototype exploration

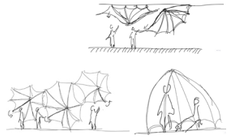
Ziqi Cui, Siman Zhang
2023 Spring
Final Thesis in Politecnico di Milano
Faculty: Prof. Alessandra Zanelli, Prof. Salvatore Viscuso



Aim and Goal



Adaptive lightweight structure
that can be applied according to different seniors
(plaza, alley etc.)



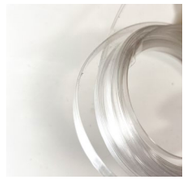
Interactive Device
that can be react to the human's behavior

Units Copiable
that can be flexibly assembled, eventually
forming a large surface

Hybrid Textile System

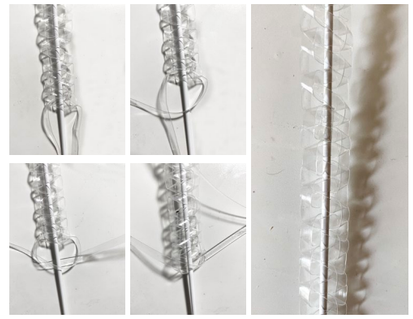
(a) Kit of components

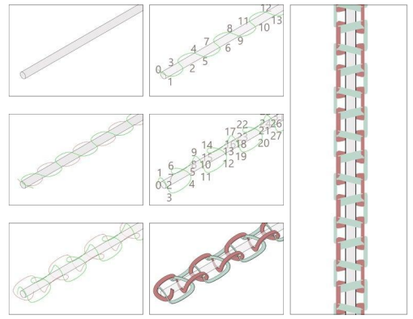
Bending element



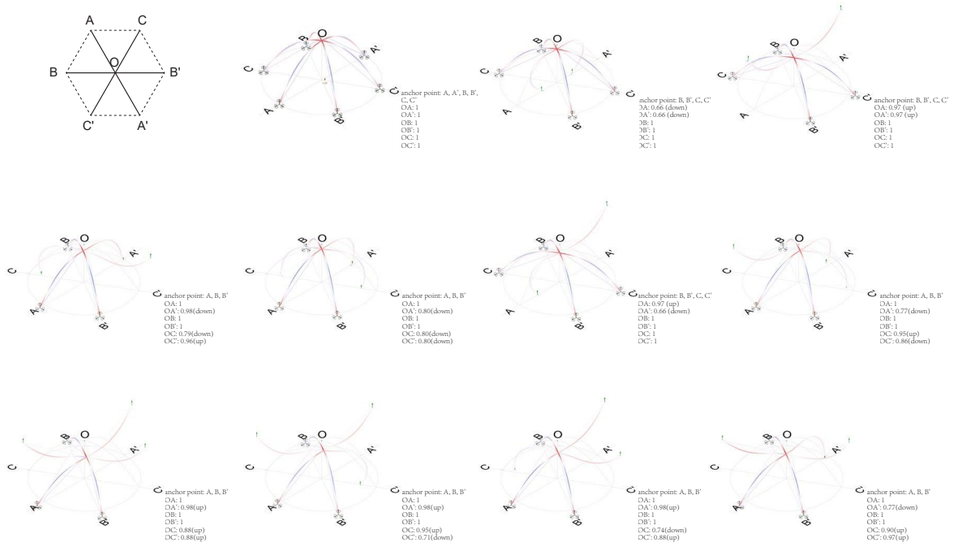
Textile element

Weaving System On Skeleton

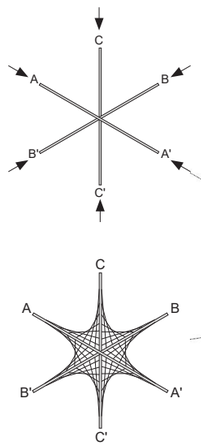


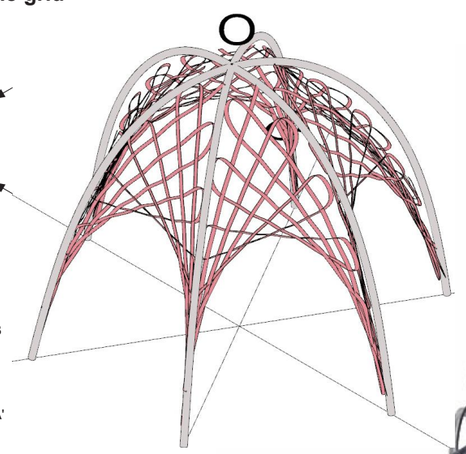


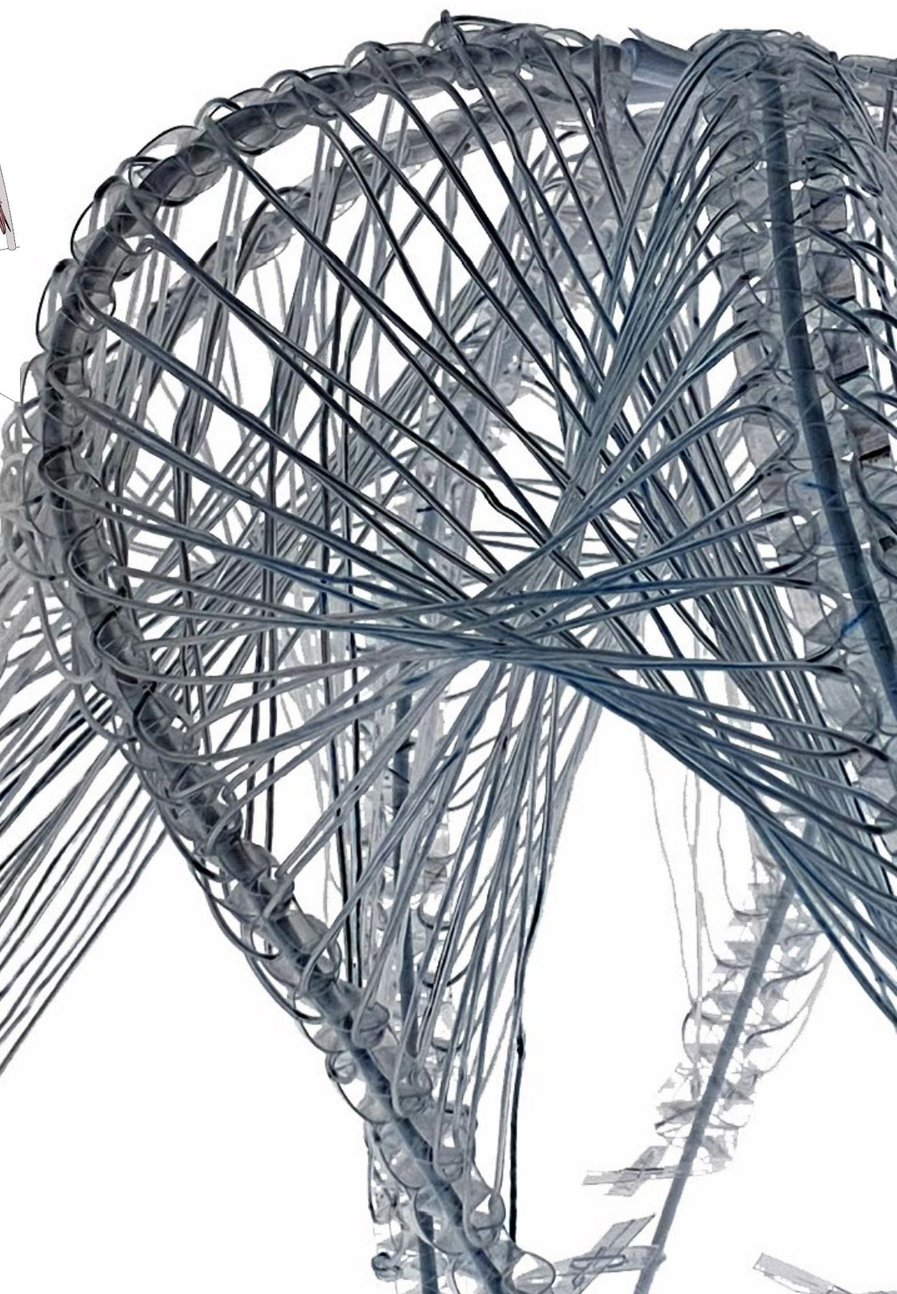
Skeleton(Bending-active)



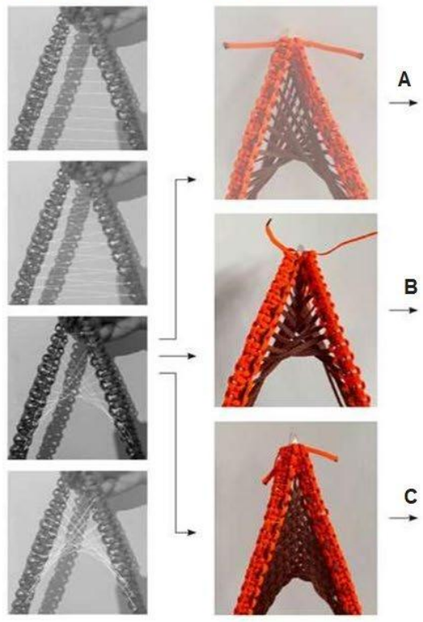
(b) Flat deployable grid

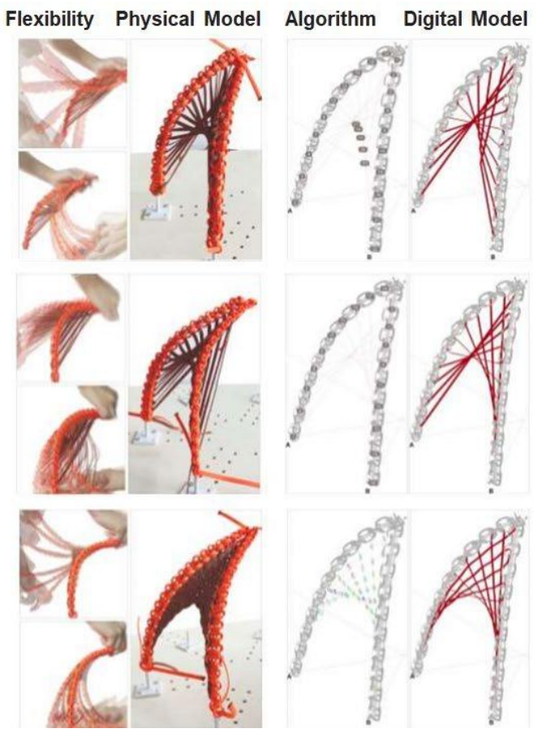






Weaving Method of Skin





Waving feature

☒ Material Usage : ● ● ○
☒ Stability of skin: ● ○ ○
☒ Flexibility of bar: ● ● ○
☒ Spatial form: ● ● ○

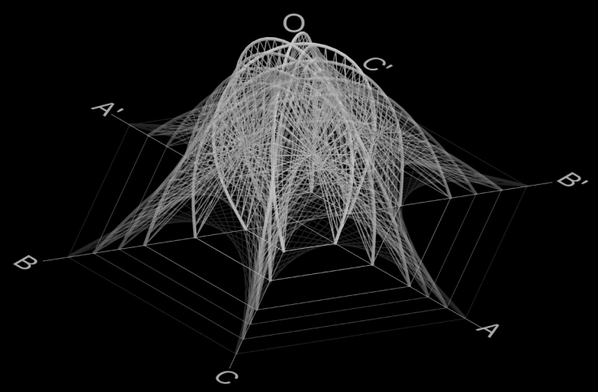
✕

☒ Material Usage : ● ○ ○
☒ Stability of skin: ● ● ○
☒ Flexibility of bar: ● ○ ○
☒ Spatial form: ● ● ●

✓

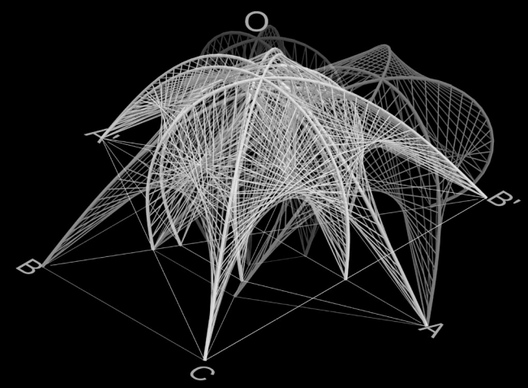
☒ Material Usage : ● ● ●
☒ Stability of skin: ● ● ●
☒ Flexibility of bar: ● ● ●
☒ Spatial form: ● ○ ○

✓



Experiment_01 Evenly Study

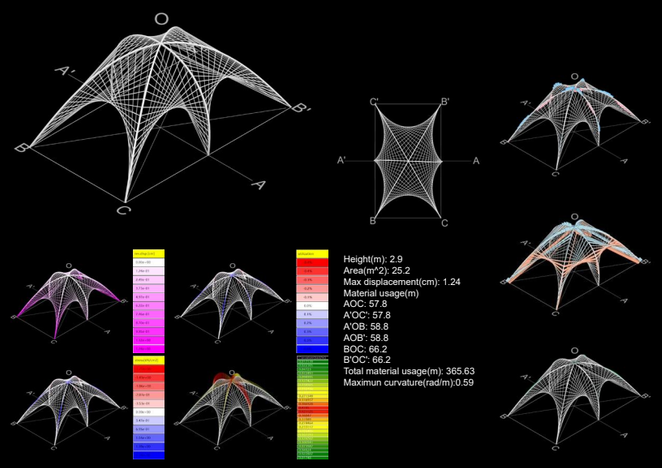
Using typeB, observing what effect it has on the final shape while reducing the weaving material in each two skeleton evenly at the same time.



Experiment_02 Unevenly study

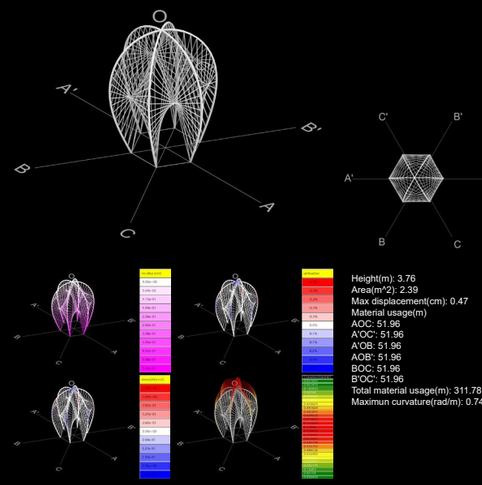
Using typeB, observing what effect it has on the final shape while reducing the weaving material in each two skeleton unevenly.

Iteration 24



Height(m): 2.9
Area(m^2): 25.2
Max displacement(cm): 1.24
Material usage(m)
AOC: 57.8
A'OC': 57.8
A'OB: 58.8
AOB': 58.8
BOC: 66.2
B'OC': 66.2
Total material usage(m): 365.63
Maximun curvature(rad/m):0.59

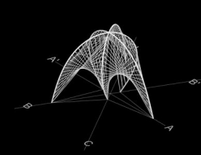
Iteration 5



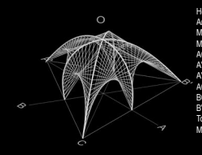
Height(m): 3.76
Area(m^2): 2.39
Max displacement(cm): 0.47
Material usage(m)
AOC: 51.96
A'OC': 51.96
A'OB: 51.96
AOB': 51.96
BOC: 51.96
B'OC': 51.96
Total material usage(m): 311.78
Maximun curvature(rad/m): 0.74

Form Variation

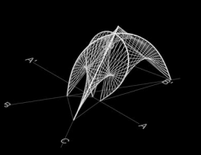
Input: material usage for each two skeleton
Out put: form/height/area/curvature/displacement of prototype



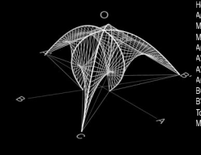
Height(m): 3.52
Area(m²): 3.59
Max displacement(cm): 0.79
Material usage(m)
AOC: 56.03
A'OC': 55.99
A'OB: 55.79
AOB': 56.03
BOC: 55.79
B'OC': 55.99
Total material usage(m): 335.58
Maximun curvature(rad/m): 0.70



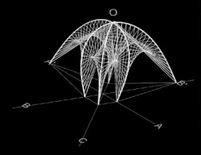
Height(m):3.1
Area(m²):18.9
Max displacement(cm):1.35
Material usage(m)
AOC:59.8
A'OC':59.54
A'OB:59.54
AOB':59.8
BOC:59.8
B'OC':59.8
Total material usage(m): 358.46
Maximun curvature(rad/m):0.58



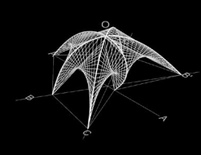
Height(m): 3.25
Area(m²): 9.10
Max displacement(cm): 1.99
Material usage(m)
AOC: 54.68
A'OC': 57.00
A'OB: 47.21
AOB': 59.75
BOC: 56.16
B'OC': 63.84
Total material usage(m): 338.63
Maximun curvature(rad/m): 0.78



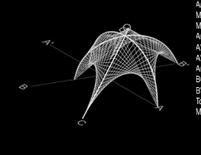
Height(m): 3.27
Area(m²): 7.92
Max displacement(cm): 1.27
Material usage(m)
AOC: 58.72
A'OC': 59.48
A'OB: 59.40
AOB': 58.28
BOC: 59.18
B'OC': 56.84
Total material usage(m): 353.91
Maximun curvature(rad/m): 0.75



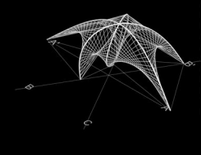
Height(m):3.1
Area(m²):7.13
Max displacement(cm):1.13
Material usage(m)
AOC:52.3
A'OC':58.4
A'OB:59.8
AOB':59.8
BOC:52.25
B'OC':58.35
Total material usage(m): 340.84
Maximun curvature(rad/m):0.81



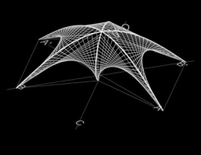
Height(m)2.99
Area(m²):20.5
Max displacement(cm):1.26
Material usage(m)
AOC:56.5
A'OC':61.9
A'OB:61.77
AOB':56.23
BOC:62.5
B'OC':62.35
Total material usage(m): 361.24
Maximun curvature(rad/m):0.79



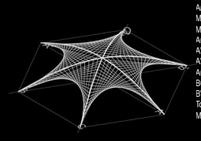
Height(m):3.138785
Area(m²):18.53391
Max displacement(cm): 0.93
Material usage(m)
AOC: 62.77
A'OC': 51.62
A'OB: 51.62
AOB': 62.77
BOC: 60.19
B'OC': 60.19
Total material usage(m): 349.181005
Maximun curvature(rad/m):0.547



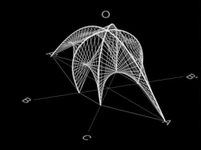
Height(m): 3.06
Area(m²): 12.45
Max displacement(cm): 2.63
Material usage(m)
AOC: 63.07
A'OC': 64.38
A'OB: 63.24
AOB': 67.19
BOC: 45.12
B'OC': 55.88
Total material usage(m): 358.87
Maximun curvature(rad/m): 0.807



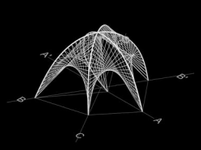
Height(m): 2.57
Area(m²): 30.12
Max displacement(cm): 1.71
Material usage(m)
AOC: 55.29
A'OC': 63.26
A'OB: 69.56
AOB': 69.35
BOC: 57.55
B'OC': 64.57
Total material usage(m): 379.59
Maximun curvature(rad/m): 0.95



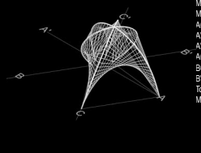
Height(m): 1.24
Area(m²): 47.78
Max displacement(cm): 2.20
Material usage(m)
AOC: 63.10
A'OC': 63.10
A'OB: 63.10
AOB': 63.10
BOC: 63.10
B'OC': 63.10
Total material usage(m): 378.73
Maximun curvature(rad/m): 0.16



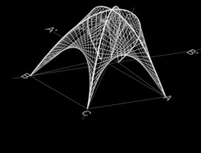
Height(m):3.4
Area(m²):11.26
Max displacement(cm):1.44
Material usage(m)
AOC:60.8
A'OC':61.0
A'OB:60.8
AOB':61.0
BOC:50.9
B'OC':51.2
Total material usage(m): 345.94
Maximun curvature(rad/m):0.62



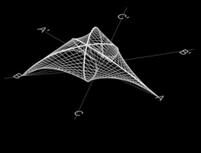
Height(m): 3.19
Area(m²): 11.46
Max displacement(cm): 0.99
Material usage(m)
AOC: 59.10
A'OC': 55.79
A'OB: 55.55
AOB': 58.98
BOC: 59.42
B'OC': 59.56
Total material usage(m): 348.41
Maximun curvature(rad/m): 0.80



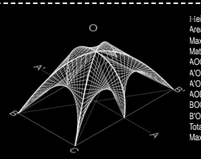
Height(m): 3.35
Area(m²): 8.21
Max displacement(cm): 0.84
Material usage(m)
AOC: 62.95
A'OC': 49.52
A'OB: 49.49
AOB': 59.51
BOC: 59.70
B'OC': 49.41
Total material usage(m): 330.59
Maximun curvature(rad/m): 0.83



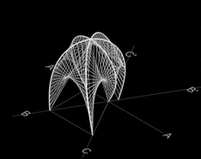
Height(m): 2.92
Area(m²): 14.40
Max displacement(cm): 2.14
Material usage(m)
AOC: 70.08
A'OC': 43.86
A'OB: 56.38
AOB': 56.11
BOC: 69.72
B'OC': 42.59
Total material usage(m): 338.74
Maximun curvature(rad/m): 0.84



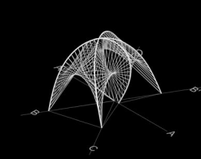
Height(m): 2.78
Area(m²): 3.39
Max displacement(cm): 1.35
Material usage(m)
AOC: 59.93
A'OC': 50.78
A'OB: 59.04
AOB': 59.04
BOC: 59.93
B'OC': 50.78
Total material usage(m): 339.51
Maximun curvature(rad/m): 0.85



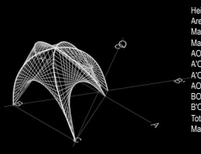
Height(m): 2.9
Area(m²): 25.2
Max displacement(cm): 1.24
Material usage(m)
AOC: 57.8
A'OC': 57.8
A'OB: 58.8
AOB': 58.8
BOC: 66.2
B'OC': 66.2
Total material usage(m): 365.63
Maximun curvature(rad/m):0.59



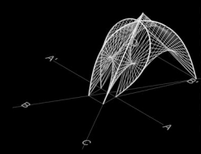
Height(m): 3.50
Area(m²): 7.72
Max displacement(cm): 0.96
Material usage(m)
AOC: 55.79
A'OC': 57.22
A'OB: 56.06
AOB': 47.08
BOC: 60.54
B'OC': 56.46
Total material usage(m): 333.17
Maximun curvature(rad/m): 0.72



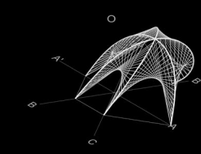
Height(m): 3.21
Area(m²): 7.53
Max displacement(cm): 1.25
Material usage(m)
AOC: 58.42
A'OC': 62.17
A'OB: 51.13
AOB': 48.54
BOC: 60.49
B'OC': 60.43
Total material usage(m): 341.19
Maximun curvature(rad/m): 0.81



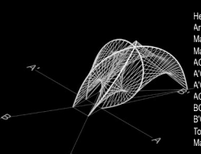
Height(m): 3.04
Area(m²): 17.94
Max displacement(cm): 1.16
Material usage(m)
AOC: 62.00
A'OC': 62.06
A'OB: 65.25
AOB': 48.29
BOC: 65.17
B'OC': 48.29
Total material usage(m): 351.09
Maximun curvature(rad/m): 0.61



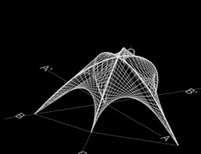
Height(m): 3.16
Area(m²): 10.58
Max displacement(cm): 1.03
Material usage(m)
AOC: 49.78
A'OC': 60.64
A'OB: 49.96
AOB': 60.61
BOC: 52.05
B'OC': 64.61
Total material usage(m): 337.67
Maximun curvature(rad/m): 0.79



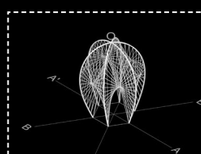
Height(m): 2.54
Area(m²): 20.89
Max displacement(cm): 2.20
Material usage(m)
AOC: 60.33
A'OC': 63.17
A'OB: 55.37
AOB': 56.09
BOC: 57.56
B'OC': 63.09
Total material usage(m): 355.62
Maximun curvature(rad/m): 0.89



Height(m): 3.27
Area(m²): 3.69
Max displacement(cm): 1.14
Material usage(m)
AOC: 49.65
A'OC': 50.59
A'OB: 52.00
AOB': 62.00
BOC: 51.63
B'OC': 60.93
Total material usage(m): 326.79
Maximun curvature(rad/m): 0.82



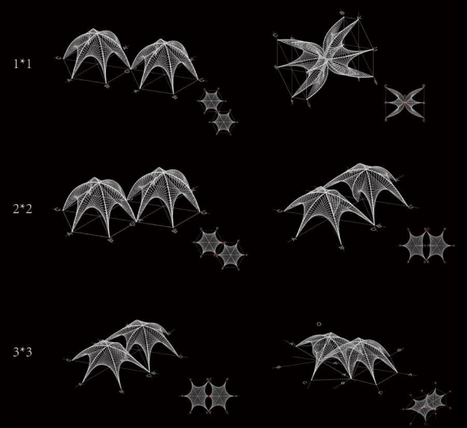
Height(m): 3.09
Area(m²): 16.76
Max displacement(cm): 1.36
Material usage(m)
AOC: 63.51
A'OC': 55.77
A'OB: 55.53
AOB': 63.49
BOC: 59.92
B'OC': 60.17
Total material usage(m): 358.40
Maximun curvature(rad/m): 0.73



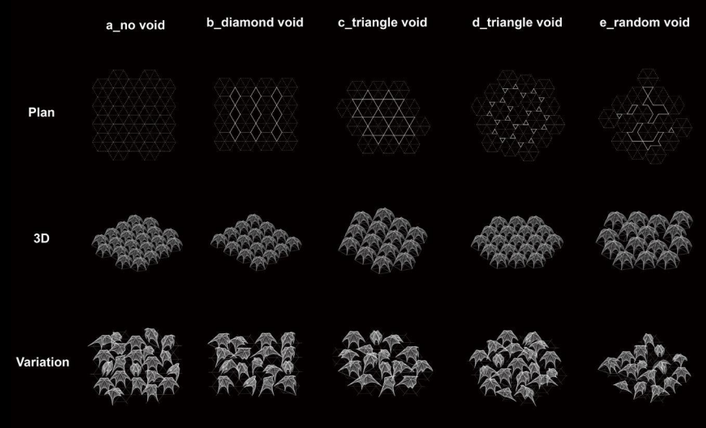
Height(m): 3.76
Area(m²): 2.39
Max displacement(cm): 0.47
Material usage(m)
AOC: 51.96
A'OC': 51.96
A'OB: 51.96
AOB': 51.96
BOC: 51.96
B'OC': 51.96
Total material usage(m): 311.78
Maximun curvature(rad/m): 0.74

Module Combination

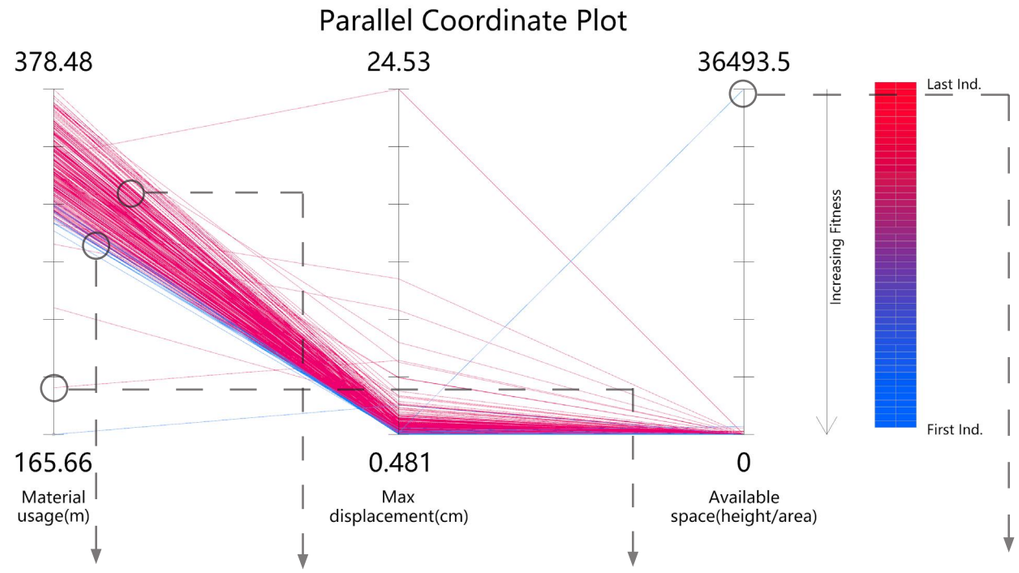
Flexibility of self-combination



Flexibility of combination layout(X-Y)



Results from Optimization

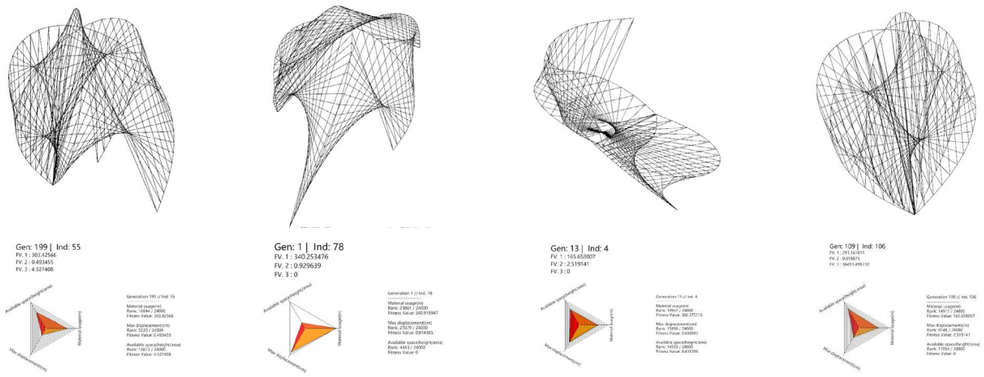


The last few generations: optimization results generated as the number of iterations increases.

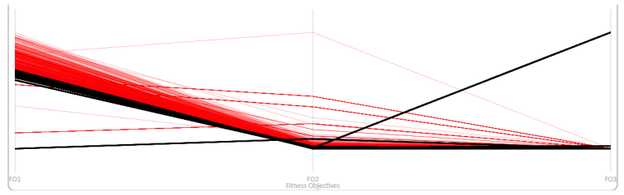
The first few generations: optimization results generated at the beginning.

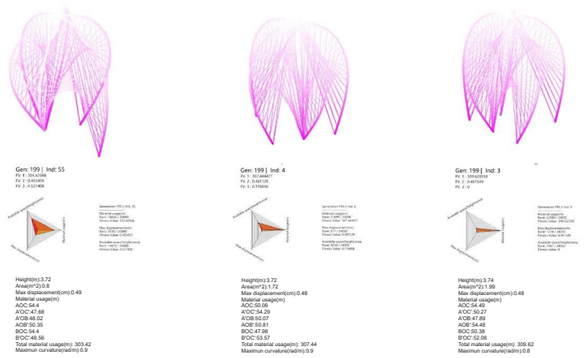
Outerlier: form crashes, unable to generate a valid shape.

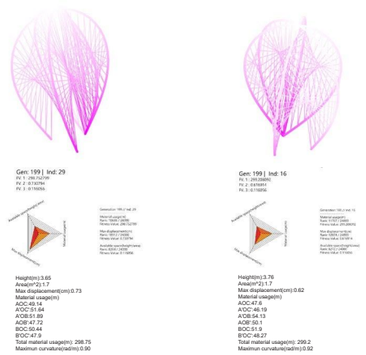
Outerlier: height/area not in a reasonable range.



Results from generation199(last generation)

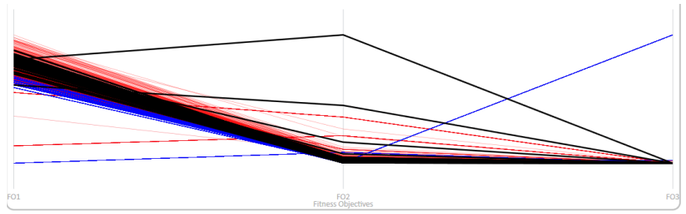


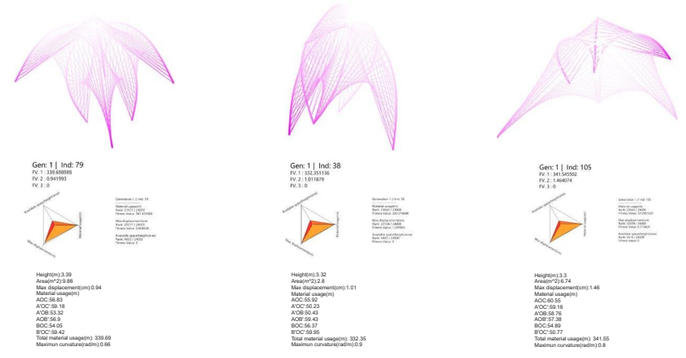


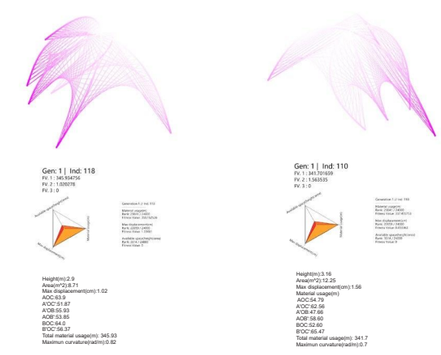


As the evolutionary direction moves closer to the fitness goal, in the last few generation results(for example generation199). The shape of each prototype is closer together, forming a similar cocoon shape. At same time, the material consumables and structural stability are optimized compared to the original generation results.

Results from generation01

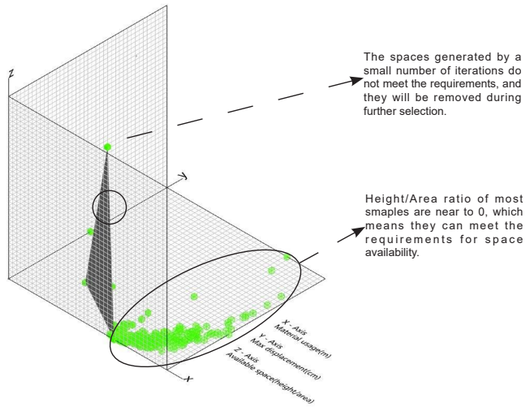




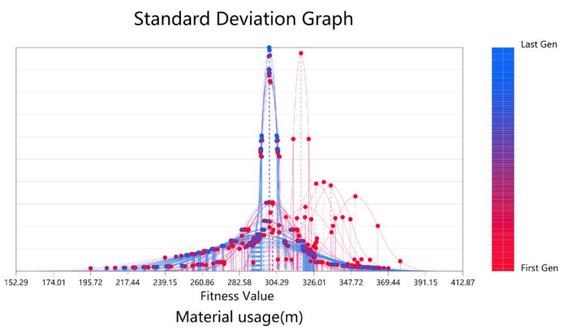


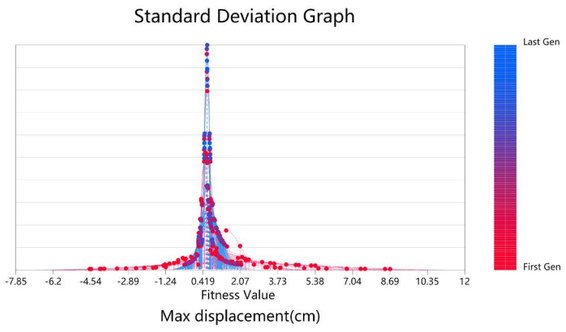
By observing the first few generation results, for example, in generation01 results, the shape of the prototype is closer to an exaggerated form. In these cases the value of the material consumables and the structural displacement are relatively large.

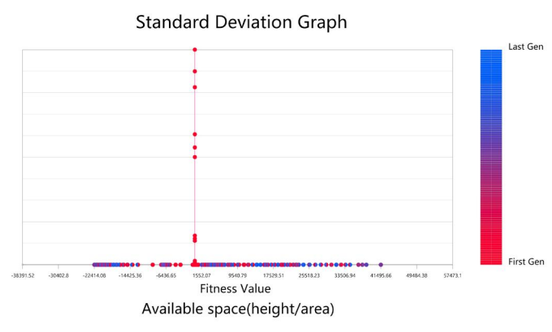
After 24thousands of iterations. the final generated results will gradually meet our ideal results.



Fittness03=0 means the H/A ratio in the ideal range, it can form availabe space for peopel use, which is the reslut what we want.





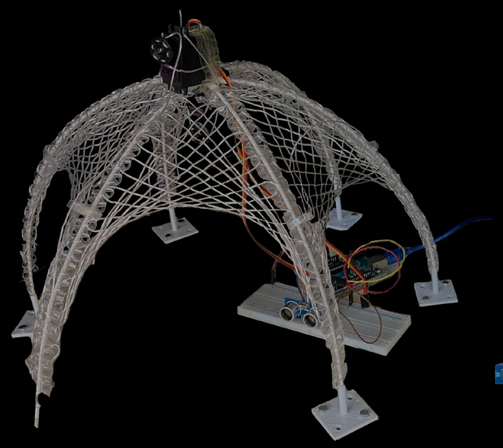


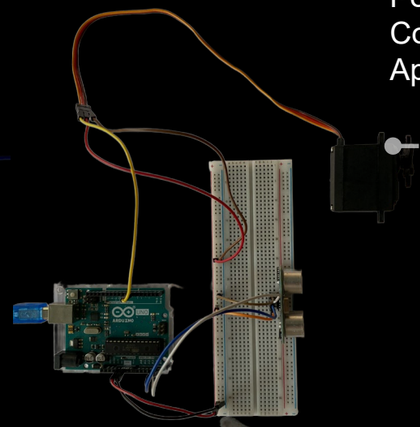
Material consumables(fittness01) for optimized results are approximately around 295-304m. Structure displacement(fittness02) for optimized results are approximately around 0.4-0.6cm.

Prototype C

Pons: high flexibility and interactive potential
Cons: not stable enough
Application: dynamic interactive system

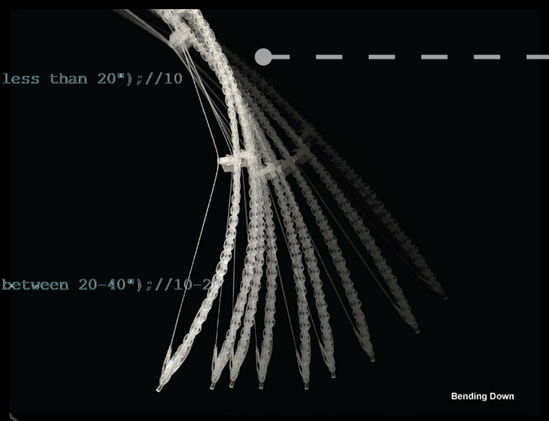
```
#include <Servo.h>

Servo servo;
const int trig = A4;
const int echo = A5;

void setup() {
  servo.attach(7);
  Serial.begin(9600);
  pinMode(echo, INPUT);
  pinMode(trig, OUTPUT);

}

void loop() {
  long duration,distance;
  digitalWrite(trig,LOW);
  delayMicroseconds(2);
  digitalWrite(trig,HIGH);
  delayMicroseconds(10);
  digitalWrite(trig,LOW);
  duration = pulseIn(echo,HIGH);
  distance = (duration/2)/29.1;

  if (distance<20){
    Serial.println("the distance is less than 20");//10
    servo.write(60);//86,
    delay(2000);
  }
  else{
    servo.write(90);
  }

  if (distance<40 && distance>20){
    Serial.println("the distance is between 20-40");//10-2
    servo.write(120);//98
    delay(2000);
  }
  else{
    servo.write(90);
    //Serial.println(distance);
    //Serial.println("cm");
  }
  delay(500);

    if (distance>40){
    Serial.println("the distance is more than 40");//20
    servo.write(90);
  }
  else{
    Serial.println(distance);
    Serial.println("cm");
  }
  delay(500);
}
```



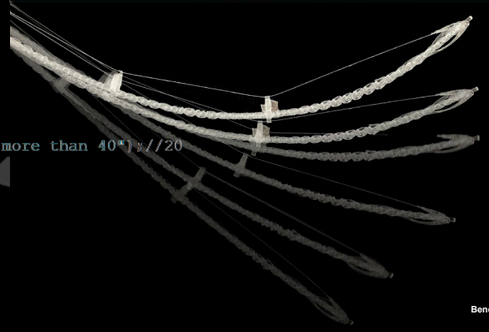


Servo

Dynamic Mechanisms

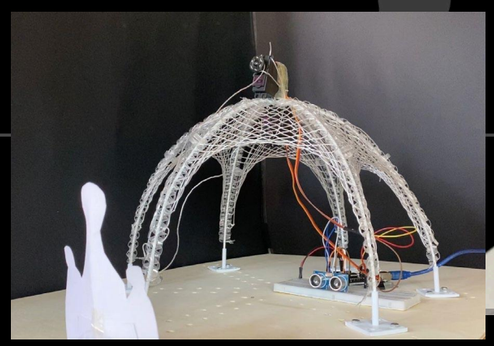


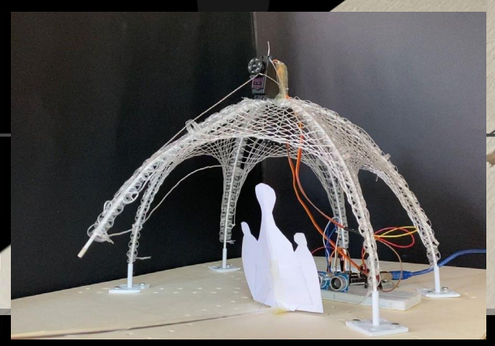
Bending Down

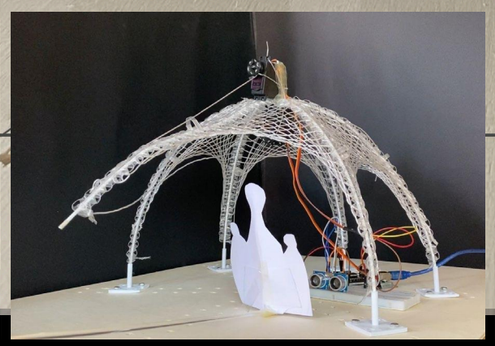


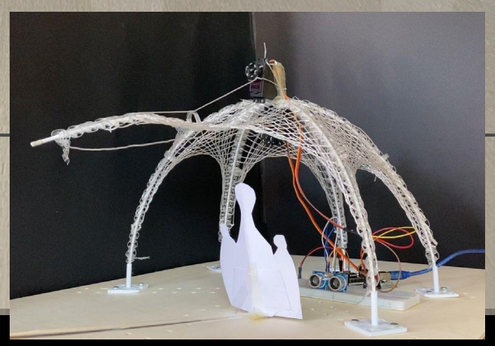
Bending Up

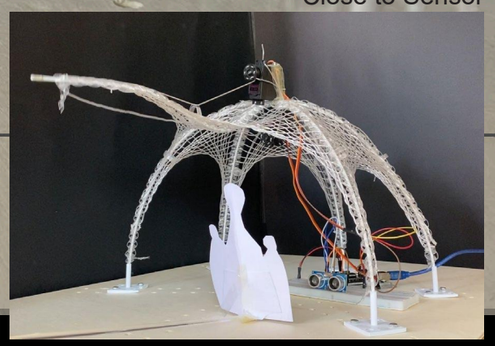
Close to Sensor

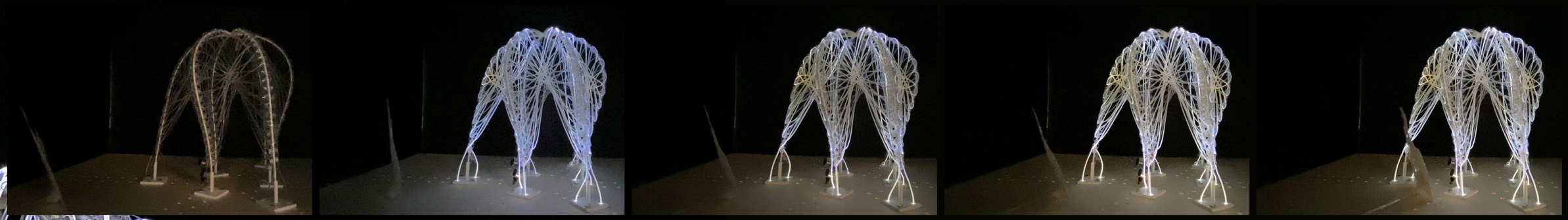








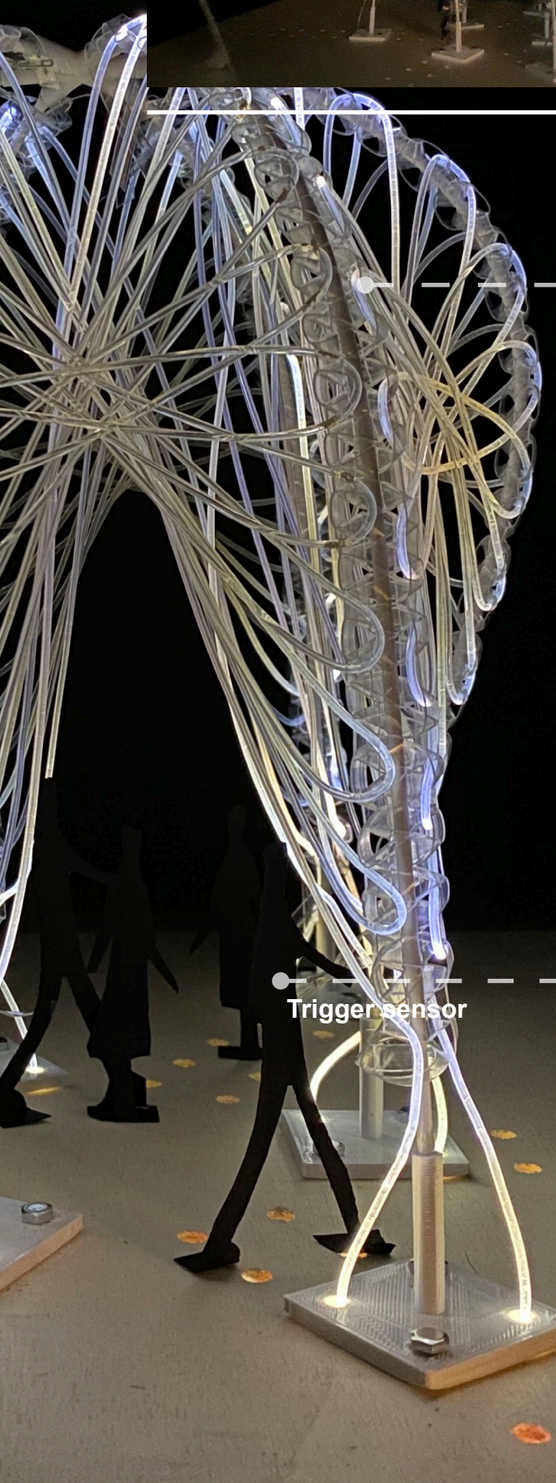


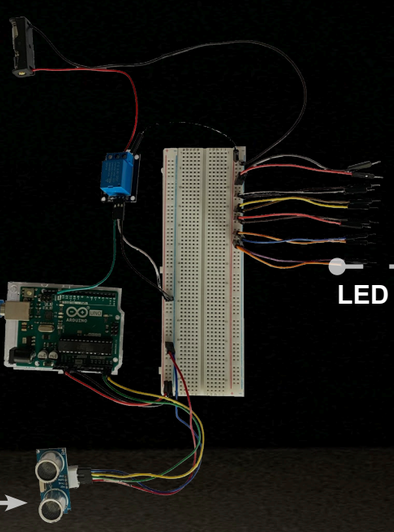


Close to Sensor

Prototype B

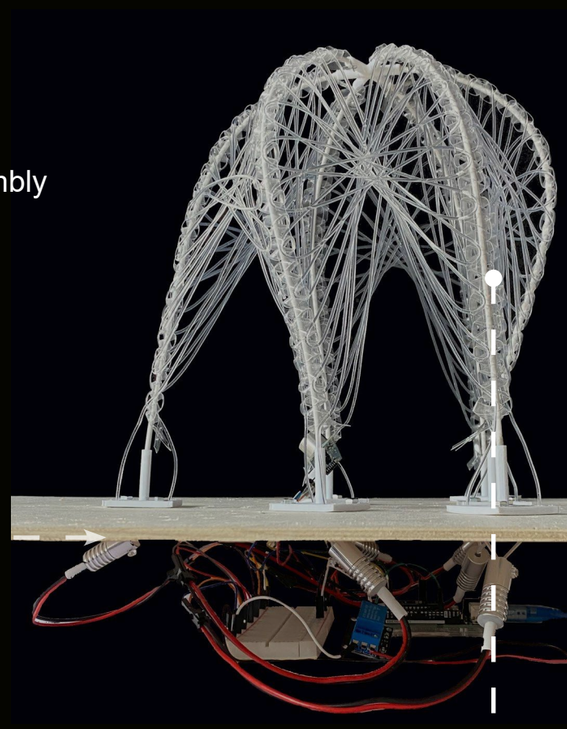
Pons: more stable, can be customized
Cons: the shape cannot be changed after assembly
Application: lighting interactive system

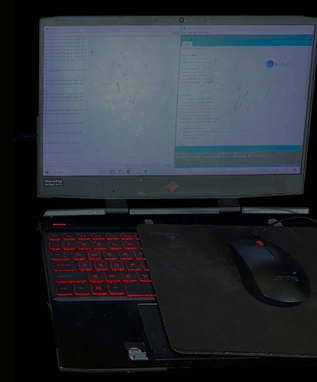




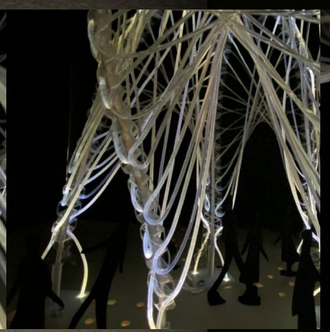
LED

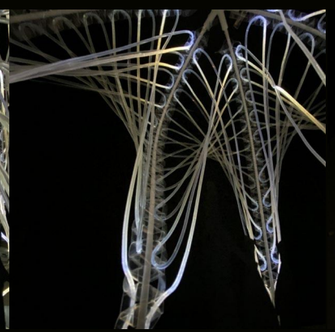
Trigger sensor

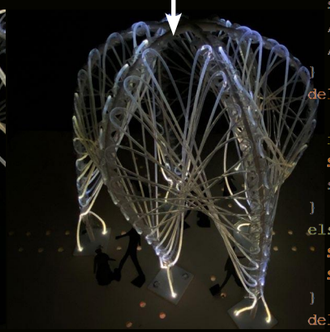




```
#include <Servo.h>

Servo servo;
const int trig = A4;
const int echo = A5;

void setup() {
  servo.attach(7);
  Serial.begin(9600);
  pinMode(echo, INPUT);
  pinMode(trig, OUTPUT);

}

void loop() {
  long duration,distance;
  digitalWrite(trig,LOW);
  delayMicroseconds(2);
  digitalWrite(trig,HIGH);
  delayMicroseconds(10);
  digitalWrite(trig,LOW);
  duration = pulseIn(echo,HIGH);
  distance = (duration/2)/29.1;

  if (distance<20){
    Serial.println("the distance is less than 20");//10
    servo.write(60);//86,
    delay(2000);
  }
  else{
    servo.write(90);
  }

  if (distance<40 && distance>20){
    Serial.println("the distance is between 20-40");//10-20
    servo.write(120);//98
    delay(2000);
  }
  else{
    servo.write(90);
    //Serial.println(distance);
    //Serial.println("cm");
  }
  delay(500);

    if (distance>40){
    Serial.println("the distance is more than 40");//20
    servo.write(90);
  }
  else{
    Serial.println(distance);
    Serial.println("cm");
  }
  delay(500);
}
```



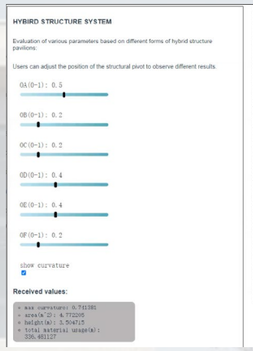


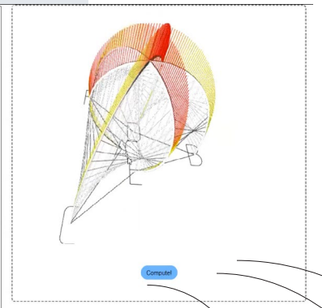


Application in Architecture Facade

Kinetic Textile Facade exampled with the case Pedagogy in Secondary Education
School Universidad Católica / Alberto Moletto + Sebastián Paredes

The "Weaving Octopus" Productive Cycles





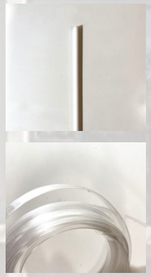
Customization design
Website（OP1,OP2,⋯）

Material use cost and
"*Weaving Octopus*"
Guidebook 1

Order materials
form factory

Materials recyceled
by customer

Customer
Requirement



Material production

Assembling according to
Guidebook 1

Assembling

Disassembling

Manufacture
OP1

Assembling according to
Guidebook 3

OPT2 Material use cost
and "*Weaving Octopus*"
Guidebook

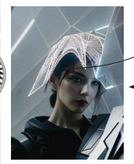
Assembling
according to
Guidebook 2

OPT3 Material use cost
and "*Weaving Octopus*"
Guidebook

Assembling

Disassembling

Manufacture
OP2



Assembling

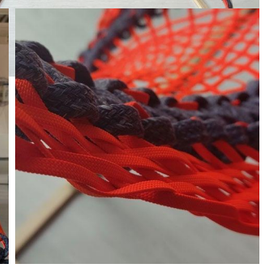
Disassembling

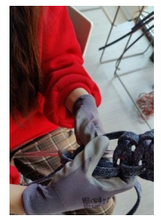
Manufacture
OP3⋯

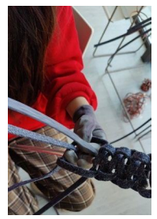
End of Life

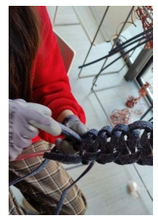
- Recycle
- Resuse
- Disposal

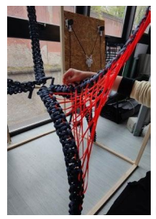
1:2 Scale-up Assembly Model

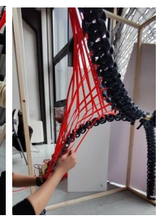








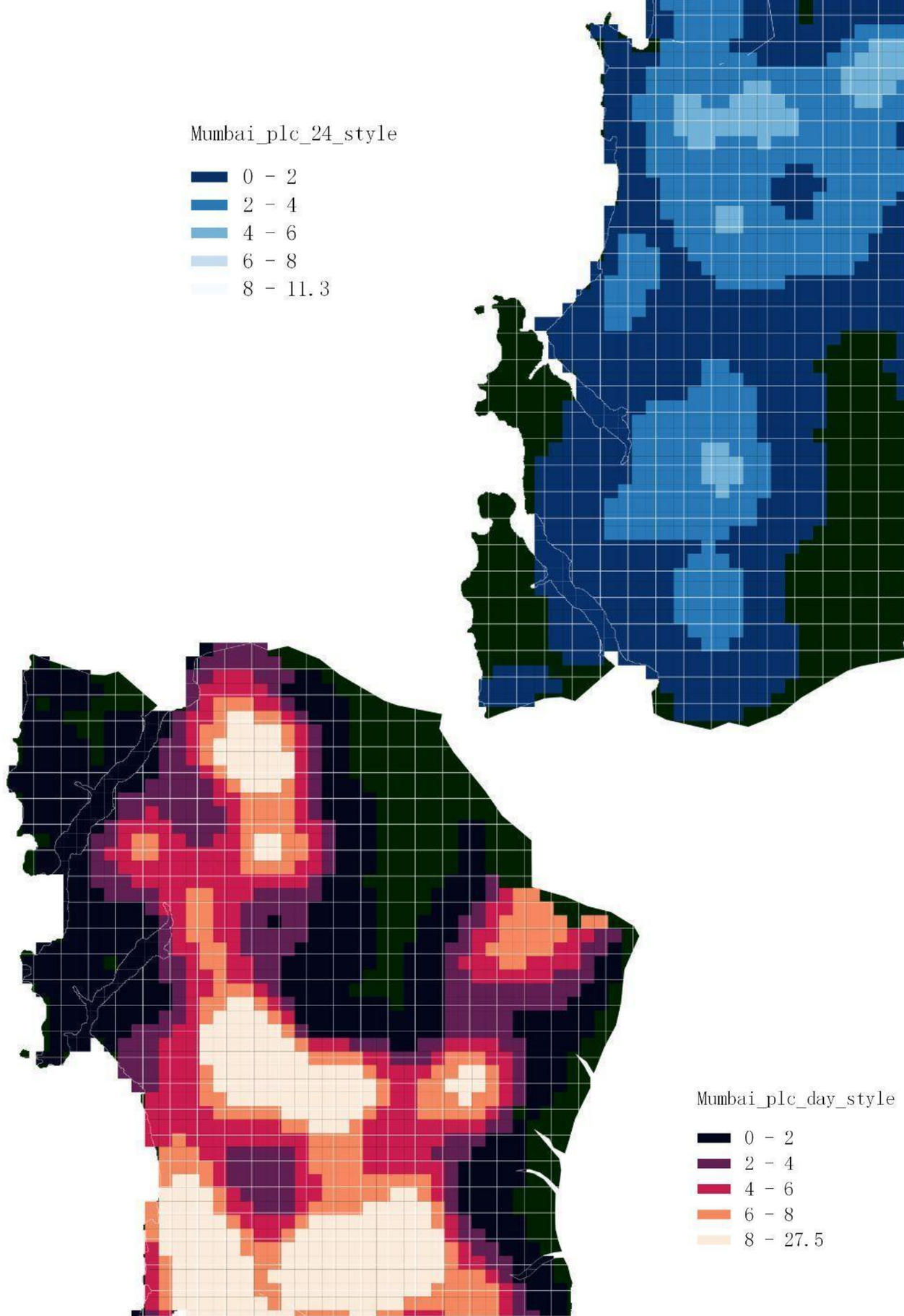




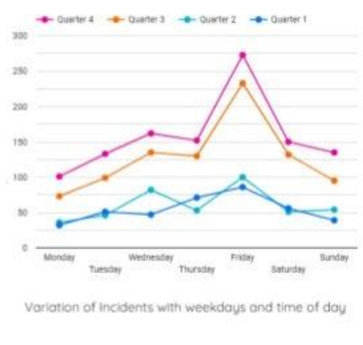
Spatiotemporal Prediction Model on VAW Incidents in Mumbai in relation to Urban Environmental Factors

A quantitative urban study by machine learning tools

Ziqi Cui, Qilin Wu, Baohua Wei, Sitong Guo
2022 Spring
Urban Study



Spatiotemporal Features of VAW incidents in Mumbai

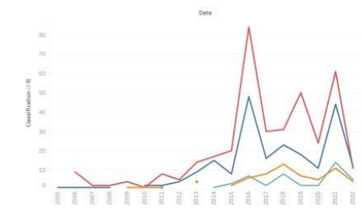


Physical Abuse

Chain Snatching
Petty Robbery
Rape /(Sexual) Assault
Touching / Groping

Verbal Abuse

Catcalls/Whistles
Commenting
Sexual Invites
Stalking



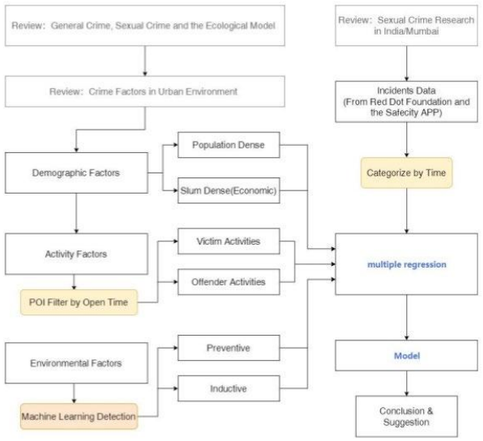
Non-Verbal Abuse

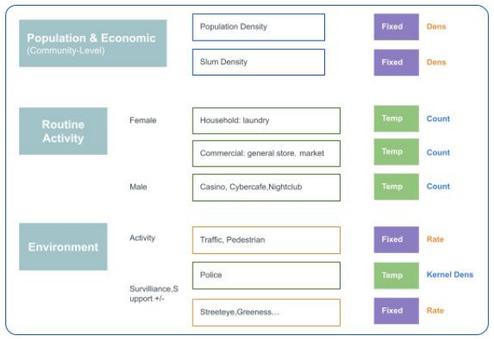
Indecent exposure
Indecent Exposure/Masturbation in public
Ogling/Lewd Facial Expressions/Staring
Taking pictures without permission

Other

In Mumbai, the incidence of sexual violence and the spatial patterns of communities may have a high degree of similarity due to the high urban density and geographic proximity between communities. VAW events are not prevalent on a natural day/week and related scale, illustrating the pattern of daily activity.

Methodology & Workflow

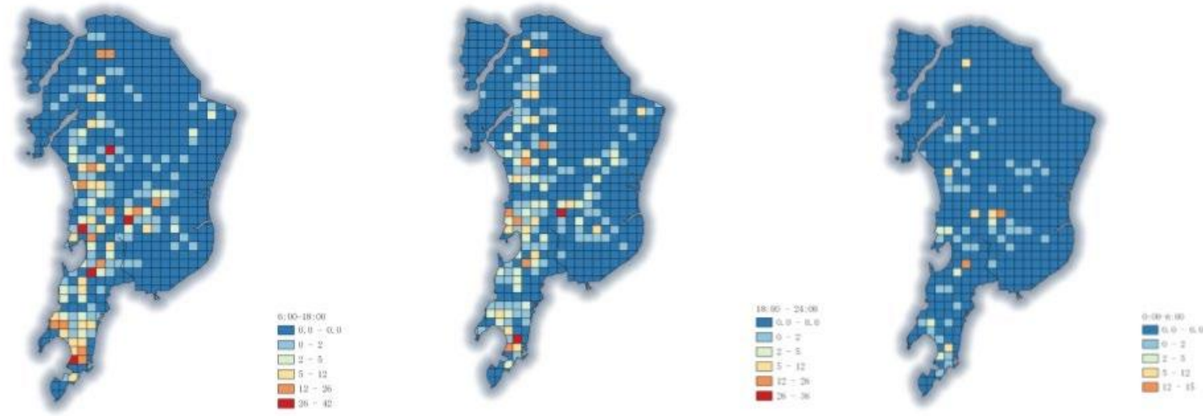




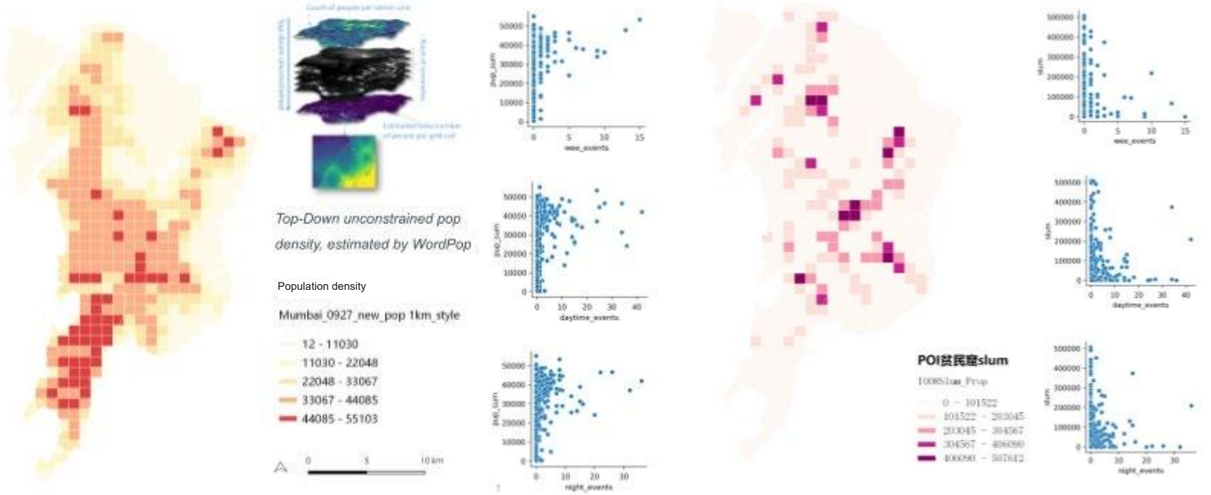
VAW Incident Data in Mumbai from Safecity APP

Title	Description	Address	Date	Clock	Classification
Online Harassment	Online harassment _S01589	Kurla	2009/7/4	20:55	Physical Abuse
Touching /Groping	Groping : 7 yrs ago I was returning from college by bus when a man on th	Friends Colony	2009/8/9	15:14	Physical Abuse
Showing Pornography \	In my school days my maths teacher told me to wait after school he told I	Naigaon	2010/3/22	13:50	Non-Verbal Abuse
Showing Pornography \	My friends showed me porn without my concern by saying its a funny vide	Colaba	2010/4/21	14:42	Others
Commenting/Sexual In	I was working at a reputed company as in intern during my college days. \	BKC	2010/8/5	12:19	Non-Verbal Abuse
Touching /Groping	_S03206 She was walking towards her local train on the station._S03206 s	Dadar	2010/8/17	15:00	Physical Abuse
Touching /Groping | Ra	On my way out of my college, a couple of Senior Students started bullying	Malad	2011/2/26	18:30	Physical Abuse
Indecent Exposure/Mas	Sexual advances S01525	Vikhroli	2011/3/15	21:15	Non-Verbal Abuse
Commenting/Sexual In	I was in 9th std was going to classes suddenly some men's started coming	Naigaon	2011/3/18	13:37	Non-Verbal Abuse

VAW Incidents Data

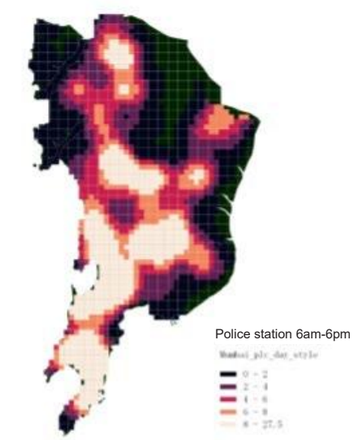


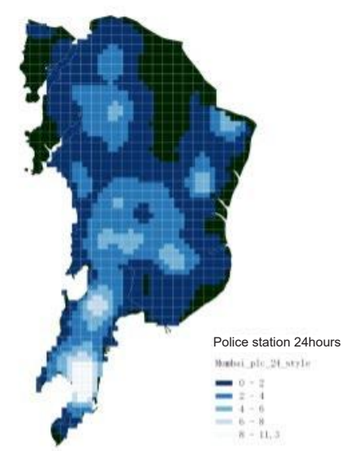
Population/Economy Data

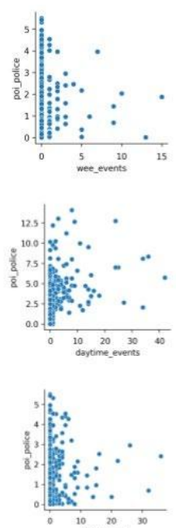


POI data & Open hours

police station(Kernel Density)

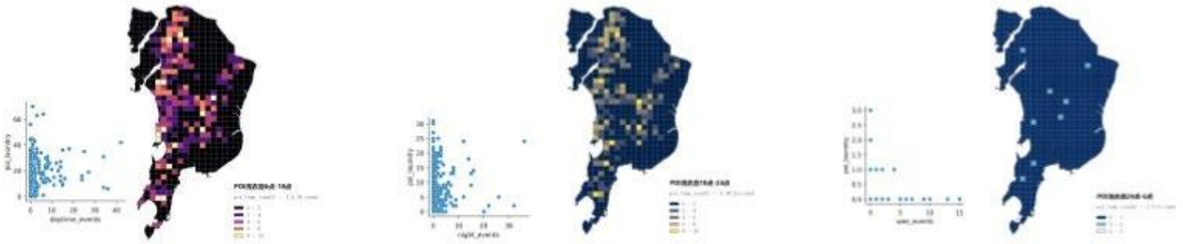






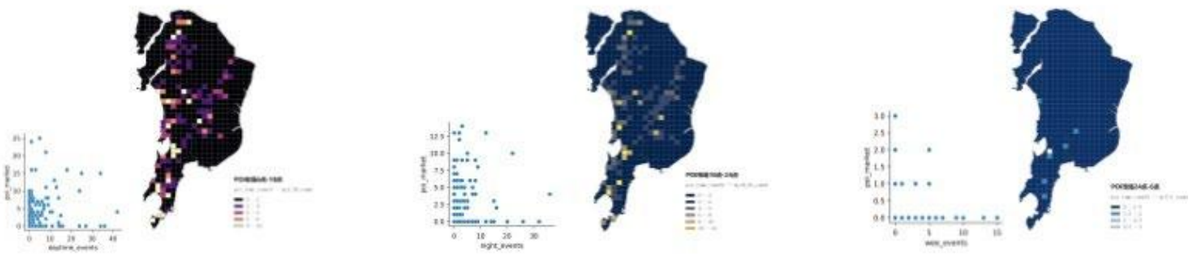
POI data & Open hours _ female label POI

laundry



grocery store

market



POI data & Open hours _ male label POI

casino

(citation1)

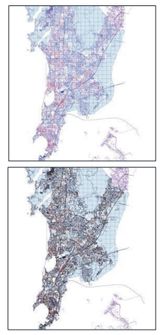
cybercafe

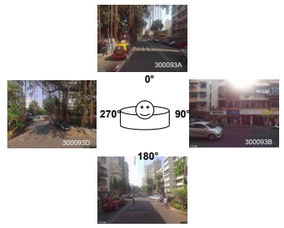
(citation1)

night club

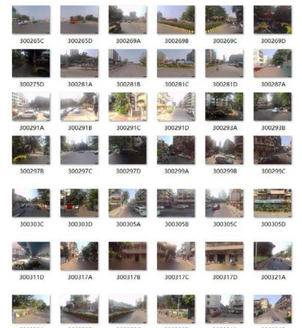
(citation1)

Google Street View crawling





Sample size: 1382*4=55244

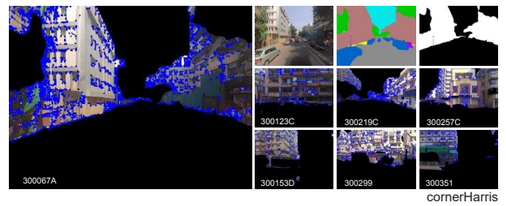


Images were collected along four directions of the road (i.e., 0 degrees, 90 degrees, 180 degrees, and 270 degrees) (Figure 2). Each image has a size of 640 × 480 pixels and a vertical angle of 0 degrees (spacing=0).

Sampling method: sampling every 200 meters. Select the point whose ID is an even number
Total: 13822 (average 21.26 points/ square)

From eyes on the street to safe cities: *"There must be eyes upon the street, eyes belonging to those we might call the natural proprietors of the street. The buildings on a street equipped to handle strangers and to insure the safety of both residents and strangers, must be oriented to the street. "*

— Jane Jacobs , The 'Death and Life of Great American Cities'



cornerHarris

Streetview Data and Indicators

Territorial-enclosure

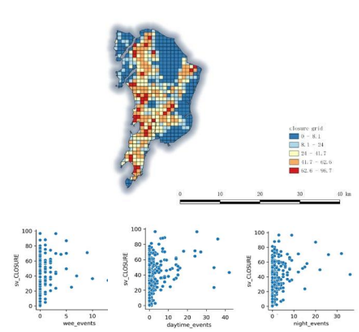
$$E_i = \frac{\sum_{j=1}^{n} Wall_j + \sum_{j=1}^{n} Fc_j + \sum_{j=1}^{n} Bd_j}{4N}$$

Surveillance-streeteyes

$$Se_i = \frac{\sum_{j=1}^{n} Feature_j}{4N}$$

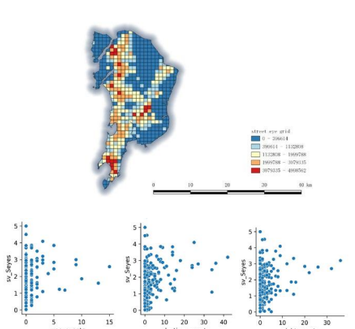
Activity-traffic flow

$$Tf_i = \frac{\sum_{j=1}^{n} C_j + \sum_{j=1}^{n} B_j + \sum_{j=1}^{n} Tk_j}{4N}$$



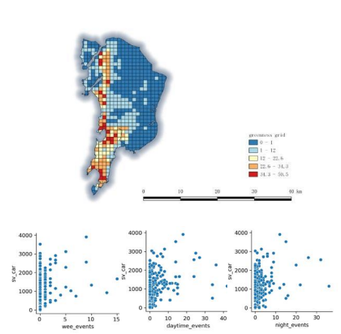
Activity-perdestrian

$$P_i = \frac{\sum_{j=1}^{n} Pe_j}{4N}$$



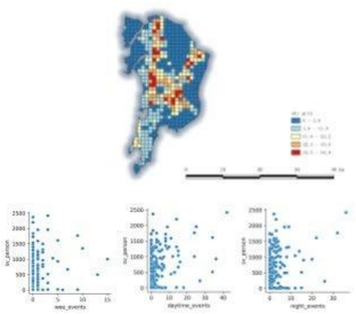
Environmental Image-greenness

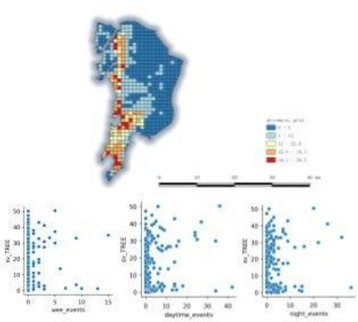
$$G_i = \frac{\sum_{j=1}^{n} Tr_j + \sum_{j=1}^{n} Pl_j}{4N}$$

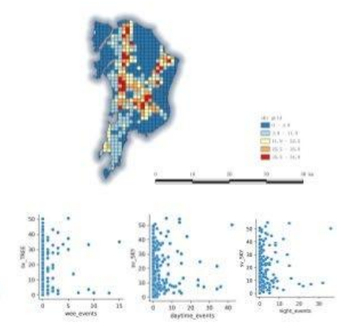


Environmental Image-sky openess

$$O_i = \frac{\sum_{j=1}^{n} Sky_j}{4N}$$







Crime Density (Y) = constant

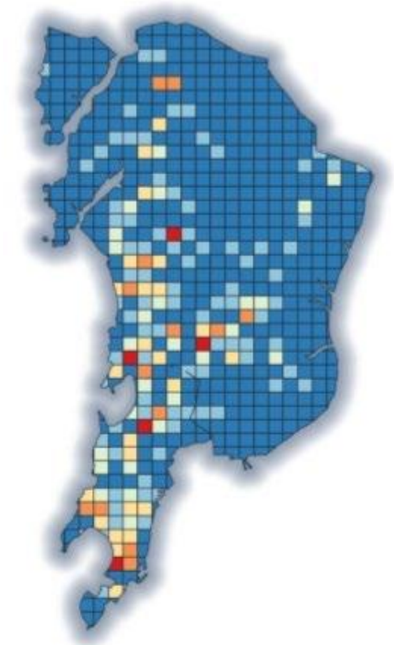
+ b1.1*population+b1.2*slum — Population/Economy

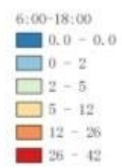
+b2*police+a3*store+a4*cybercafe+a5*casino+a6*nightclub+a7*laundry+a8*market — POI

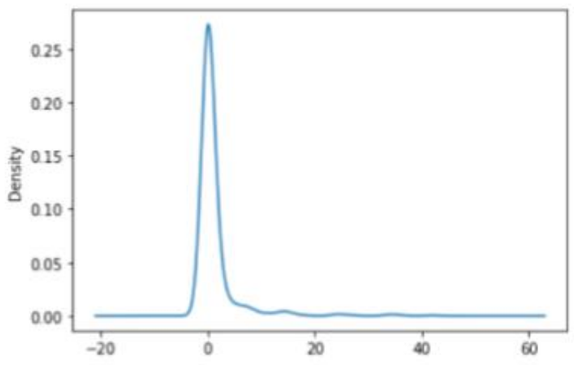
+a8*enclosure + a9*streeteyes+a10*tree+a11*sky +a12*car+a13*person — Street-view Related

+ error

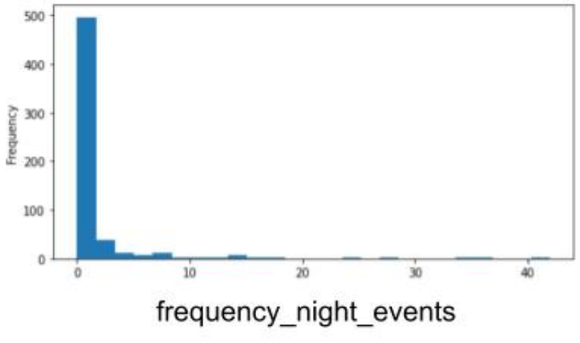
Model_daytime(06-18)





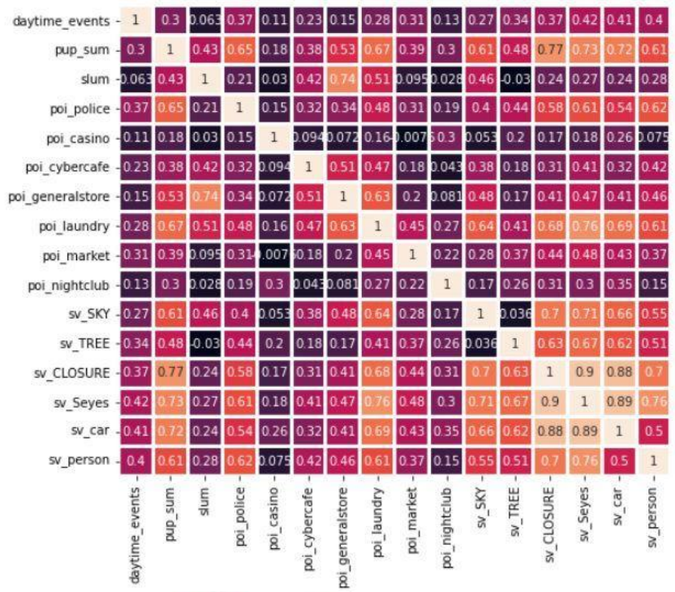


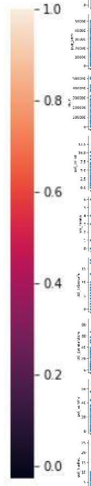
Density_night_events

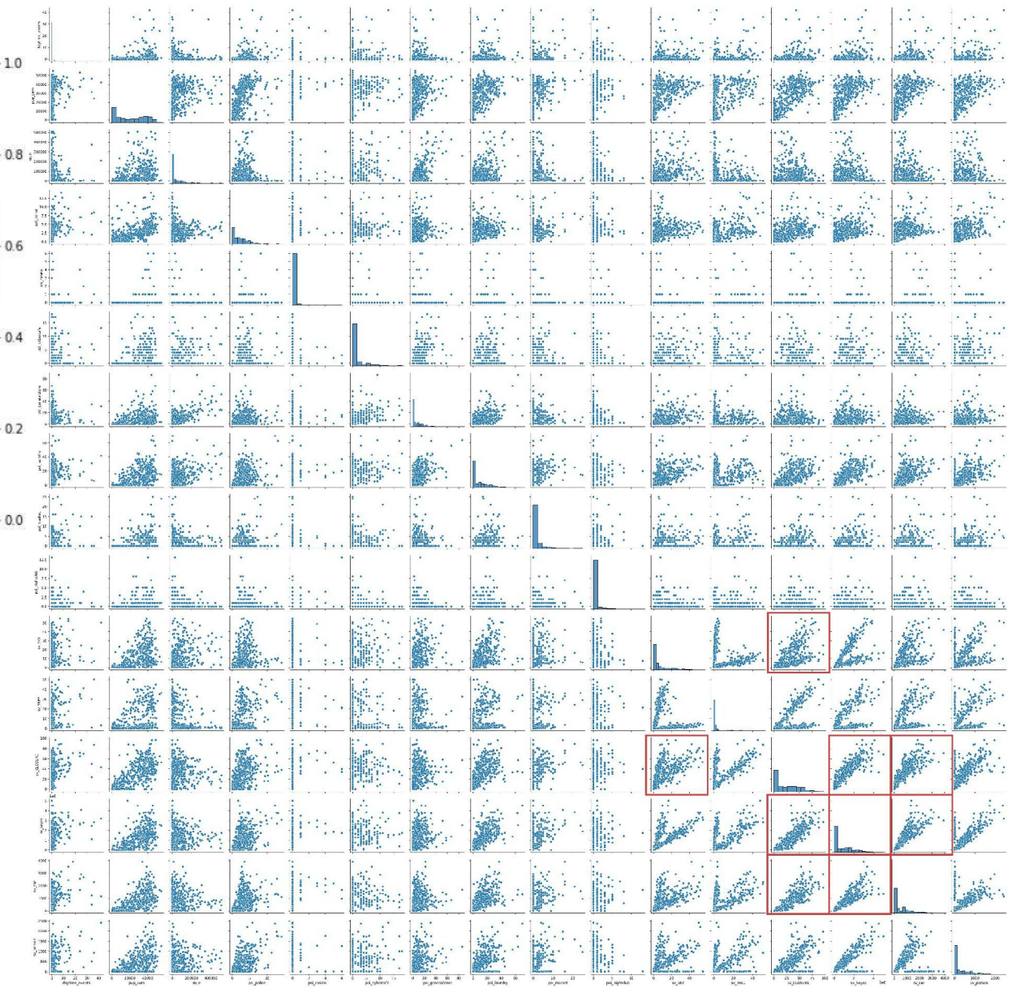


frequency_night_events

Model_daytime(06-18) correlation between X







	feature	VIF
0	const	2.863964
1	pup_sum	3.689918
2	slum	2.707834
3	poi_police	2.231257
4	poi_casino	1.227413
5	poi_cybercafe	1.534103
6	poi_generalstore	3.042794
7	poi_laundry	3.540773
8	poi_market	1.409574
9	poi_nightclub	1.258853
10	sv_SKY	9.804219
11	sv_TREE	7.785470
12	sv_CLOSURE	16.132887
13	sv_Seyes	20.407278
14	sv_car	13.261258
15	sv_person	5.909843

Model_daytime(07-18) remove some autocorrelation factors:

OLS Regression Results

Dep. Variable:	daytime_events	R-squared:	0.273
Model:	OLS	Adj. R-squared:	0.256
Method:	Least Squares	F-statistic:	16.60
Date:	Sat, 08 Oct 2022	Prob (F-statistic):	1.64e-32
Time:	10:27:58	Log-Likelihood:	596.44
No. Observations:	590	AIC:	-1165.
Df Residuals:	576	BIC:	-1104.
Df Model:	13		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.0053	0.006	-0.871	0.384	-0.017	0.007
pup_sum	-0.0520	0.022	-2.324	0.020	-0.096	-0.008
slum	0.0172	0.033	0.521	0.603	-0.048	0.082
poi_police	0.0774	0.033	2.364	0.018	0.013	0.142
poi_casino	0.0275	0.044	0.620	0.536	-0.060	0.115
poi_cybercafe	0.0647	0.028	2.293	0.022	0.009	0.120
poi_generalstore	-0.0577	0.052	-1.109	0.268	-0.160	0.044
poi_laundry	-0.1049	0.039	-2.686	0.007	-0.182	-0.028
poi_market	0.1238	0.036	3.402	0.001	0.052	0.195
poi_nightclub	-0.0082	0.047	-0.172	0.863	-0.101	0.085
sv_SKY	-0.0421	0.045	-0.928	0.354	-0.131	0.047
sv_TREE	-0.0272	0.039	-0.690	0.490	-0.104	0.050
sv_car	0.2417	0.057	4.221	0.000	0.129	0.354
sv_person	0.1711	0.039	4.352	0.000	0.094	0.248

Omnibus:	623.647	Durbin-Watson:	2.118
Prob(Omnibus):	0.000	Jarque-Bera (JB):	34195.437
Skew:	4.857	Prob(JB):	0.00
Kurtosis:	39.009	Cond. No.	26.6

Model Errors:

R-squared 0.273
MAE 0.041
MAPE inf
MSE 0.008
RMSE 0.088

	feature	VIF
0	const	2.767280
1	pup_sum	3.323563
2	slum	2.613892
3	poi_police	2.196124
4	poi_casino	1.211468
5	poi_cybercafe	1.488463
6	poi_generalstore	3.030559
7	poi_laundry	3.316404
8	poi_market	1.405809
9	poi_nightclub	1.258612
10	sv_SKY	6.951640
11	sv_TREE	5.654314
12	sv_car	8.209913
13	sv_person	4.111756

Positive Correlation: police(0.08)/ cyber_cafe(0.06)/ market(0.12)/ sv_car(0.24)/ sv_person(0.17)

Negative Correlation: pup_sum(-0.05)/ laundry(-0.10)

Comparison of the three models

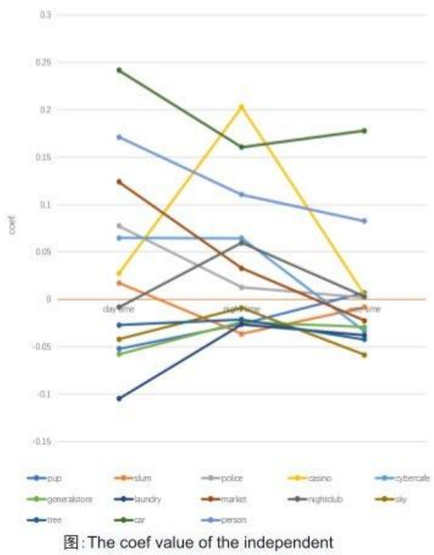


图: The coef value of the independent variable Y changes over time

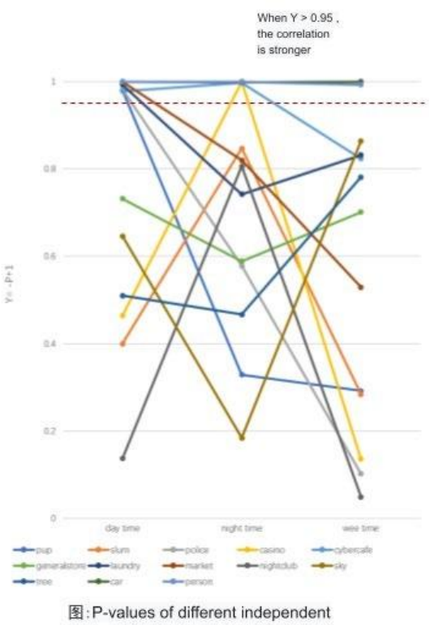


图: P-values of different independent variables Y change over time

What makes a park popular?

A quantitative urban study by machine learning tools

Ziqi Cui, Georgios Bekakos, Lora Fahmy
2023 Spring
AIA data-encoding course
Faculty: Gabriella Rossi, Salvador Calgua

picturesque buildings. Worth coming here for a calm walk. Dogs are not allowed unfortunately. Also a good spot fo g photos. Many loca spend time as well. The par su ny cafes so ca a place to sit a ck".

```
tags_to_fetch =
    {'natural': 'tree'},
    {'amenity': 'bench'},
    {'amenity': 'fountain'},
    {'amenity': 'drinking_water'},
    {'historic': 'memorial'},
    {'amenity': 'bicycle_parking',
    {'amenity': 'bicycle_rental'},
    {'amenity': 'waste_basket'},
    {'amenity': 'playground'},
    {'amenity': 'artwork'},
    {'amenity': 'pcnic_site'},
    {'amenity': 'restaurant'},
    {'amenity': 'bar'},
    {'amenity': 'toilets'},
    {'highway': 'footway'},
    {'landuse': 'grass'},
```

```
array(
['Firenze',
'Rotterdam',
'Utrecht',
'Lille',
'Venezia',
'Napoli',
'Toulouse',
'Bologna',
'Lisboa',
'Porto',
'Wien',
'Lyon',
'Stuttgart',
'Geneva',
'Kyiv',
'Frankfurt am Main',
'Marseille',
'Dublin',
'Strasbourg',
'Bordeaux',
'Nantes',
'Warszawa',
'Praha',
'Krakovia',
'Granada',
'A Coruña',
'Sevilla',
'Barcelona',
'Hamburg',
'Amsterdam',
'Luxembourg',
'Liverpool',
'Manchester',
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'Berlin',
'Paris',
'Bristol',
'Birmingham',
'Madrid',
'London',
'Milano',
'Roma']
```
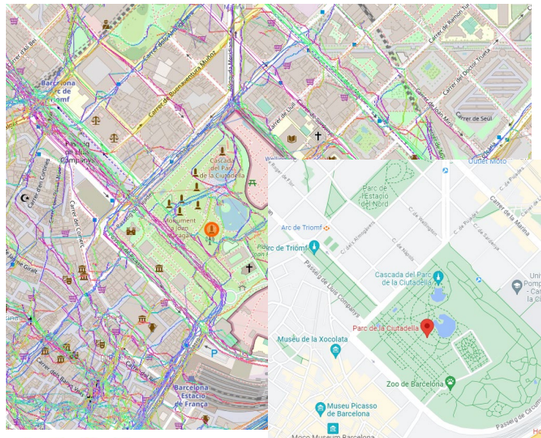
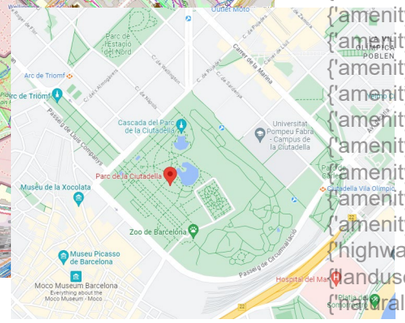
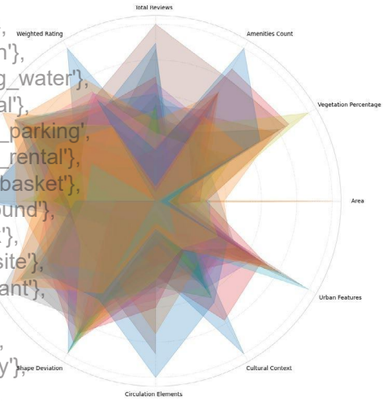







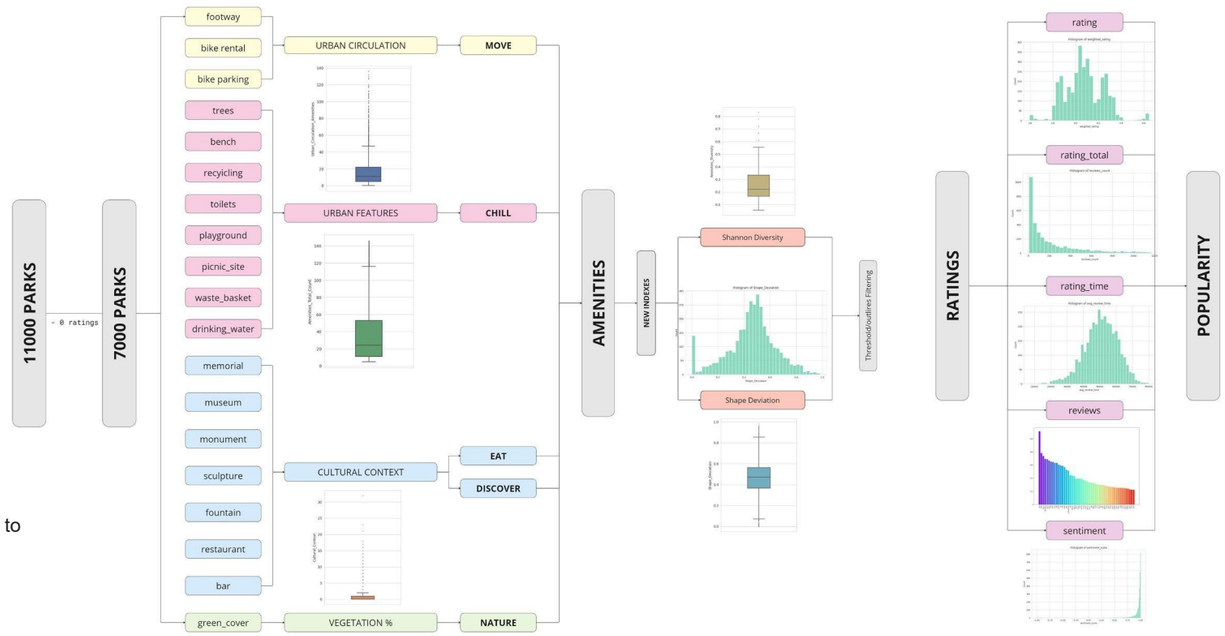
tags_to_fetch =
[{'natural': 'tree'},
{'amenity': 'bench'},
{'amenity': 'fountain'},
{'amenity': 'drinking_water'},
{'historic': 'memorial'},
{'amenity': 'bicycle_parking'},
{'amenity': 'bicycle_rental'},
{'amenity': 'waste_basket'},
{'amenity': 'playground'},
{'amenity': 'artwork'},
{'amenity': 'pcnic_site'},
{'amenity': 'restaurant'},
{'amenity': 'bar'},
{'amenity': 'toilets'},
{'highway': 'footway'},
{'landuse': 'grass'},
{'natural': 'wood'}]

★★★★☆

"Wonderful park free of charge. Beautiful monuments and picturesque buildings. Worth coming here for a calm walk. Dogs are not allowed unfortunately. Also a good spot for making photos. Many locals spend time here as well. The park is surrounded by many cafes so you can easy after find a place to sit and have a small snack".

Public parks are vital urban assets.
Their popularity depends on features like facilities,diversity, safety, and accessibility. Leveraging data on these features can help formulate strategies to improve lesser-used parks, increasing public engagement in urban spaces.

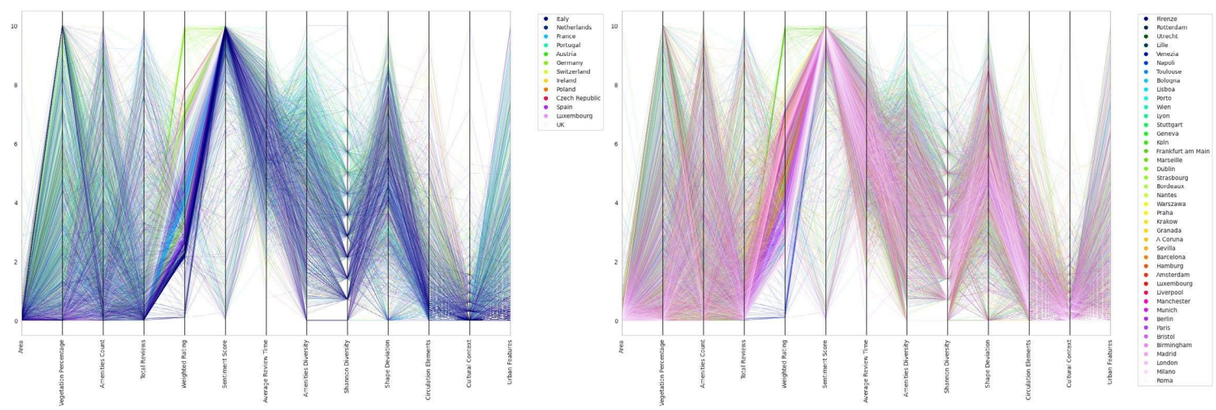
Dataset structure



Webscraping 42 cities in Europe

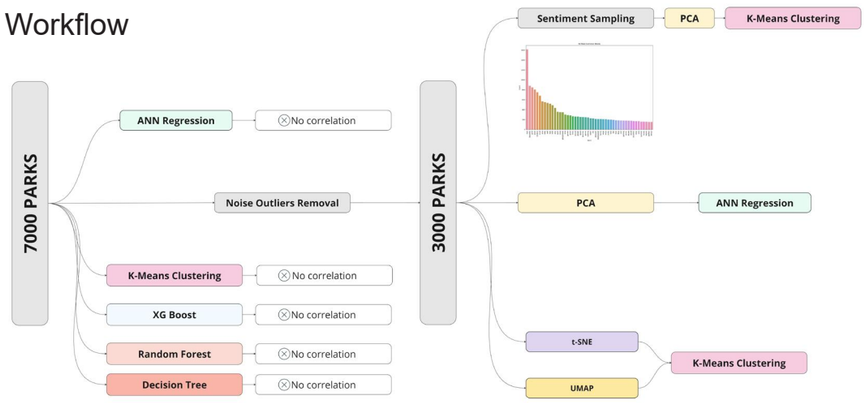
array(
['Firenze',
'Rotterdam',
'Utrecht',
'Lille',
'Venezia',
'Napoli',
'Toulouse',
'Bologna',
'Lisboa',
'Porto',
'Wien',
'Lyon',
'Stuttgart',
'Genève',
'Köln',
'Frankfurt am Main',
'Marseille',
'Dublin',
'Strasbourg',
'Bordeaux',
'Nantes',
'Warszawa',
'Praha',
'Krakovia',

After all the direct application that didnt emerge acceptable patterns, we ended up with a 3000 points dataset and those notable features. We begin with a Parallel Coordinates Plot overlaid with a Spider Chart, representing grouped parks per cities. Though seemingly chaotic, the overlay provides a multi-dimensional perspective while we can distinguish weighted rating and sentiment score colors giving an initial segmentation for Italy and Germany.

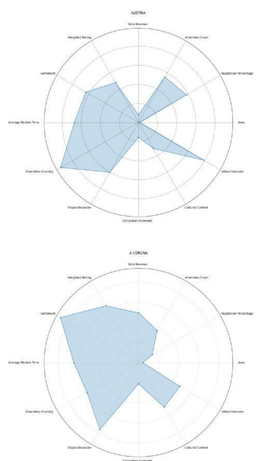


We now drill down to individual cities using the same plot structure. This city-centric view reveals local patterns that might be obscured in larger groups like the locally low ratings of Napoli and the high ratings of Stuttgart.

Workflow



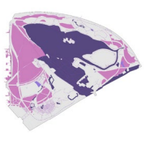
We started with a gut feeling that there was a direct and linear relation between the scores of a park and it's appreciation but we had to rethink that very quickly as we didn't find any relation. We removed the noise and reduced our dataset from ouliers to work on a smaller set that again wasn't promising, what gave us a promising result was grouping a subset of parks that has a general very good feedback and we started to look for relations there leaving behind the attempt of finding a relation between people's opinion an park's urban design. Following a guidethrow of that process.



Park samples

CHILL
shape Deviation : 0.8
amenities Diversity : 0.7
weighted_rating: 4.7

DISCOVER
shape Deviation : 0.9
amenities Diversity : 0.5
weighted_rating: 4.9

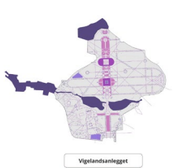


MOVE
shape Deviation : 0.6
amenities Diversity : 0.7
weighted_rating: 4.2

NATURE
shape Deviation : 0.4
amenities Diversity : 0.3
weighted_rating: 4.4

CHILL
shape Deviation : 0.8
amenities Diversity : 0.7
weighted_rating: 4.6

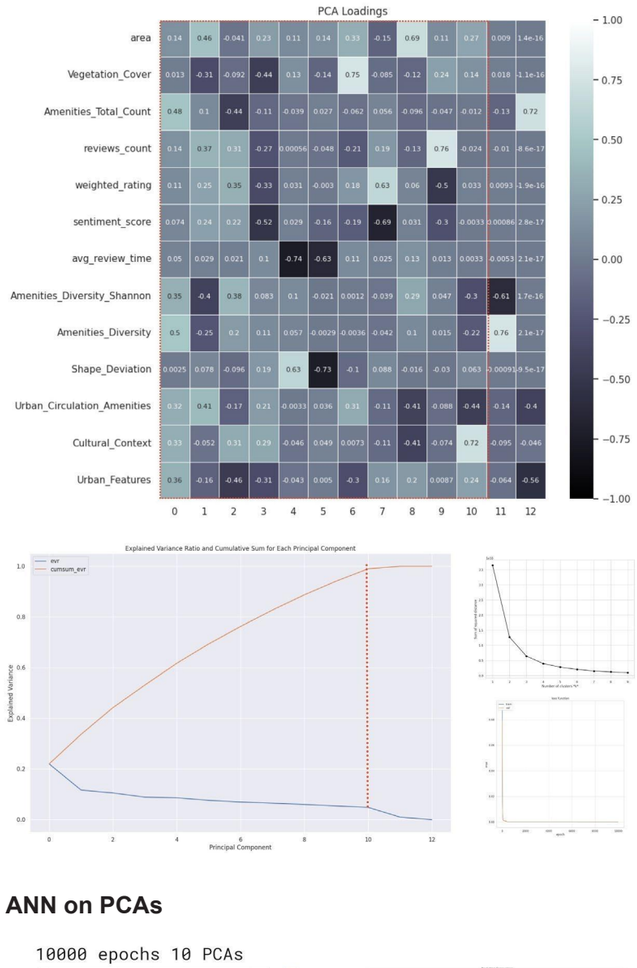
CHILL
shape Deviation : 0.8
amenities Diversity : 0.7
weighted_rating: 4.1



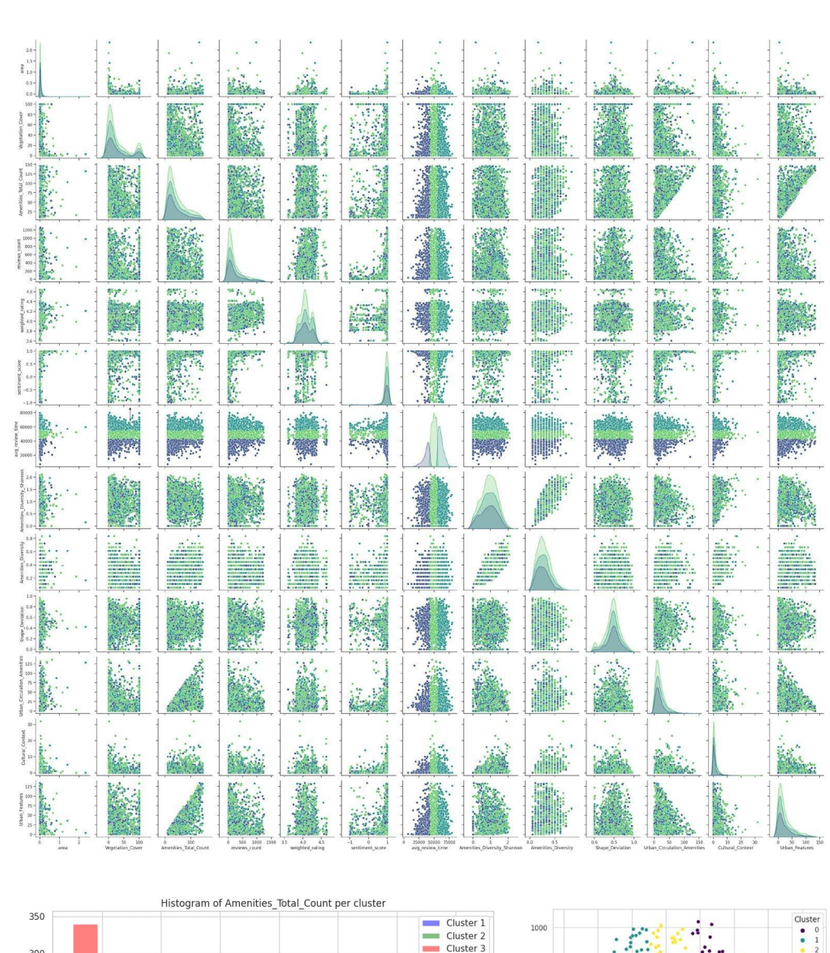
DISCOVER
shape Deviation : 0.7
amenities Diversity : 0.7
weighted_rating: 4.2

DISCOVER
shape Deviation : 0.8
amenities Diversity : 0.5
weighted_rating: 4.8

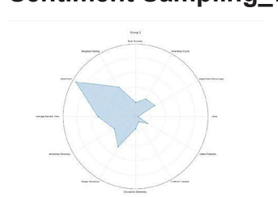
PCA

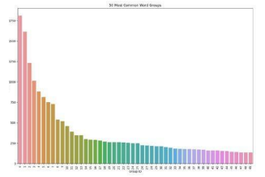


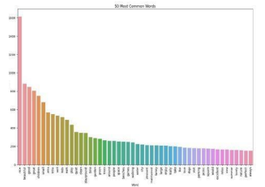
K-means

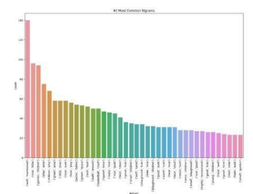


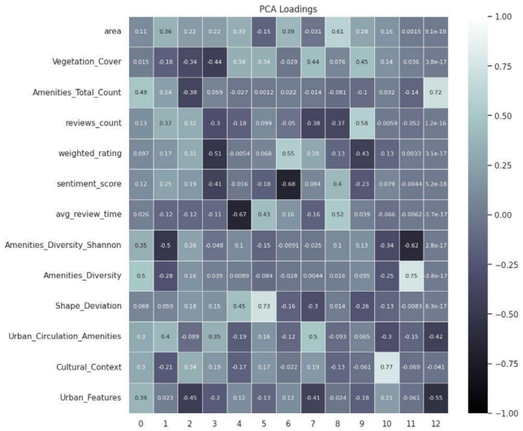
Sentiment Sampling_7000 parks

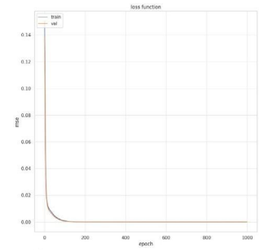


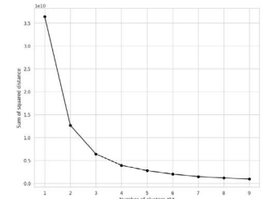




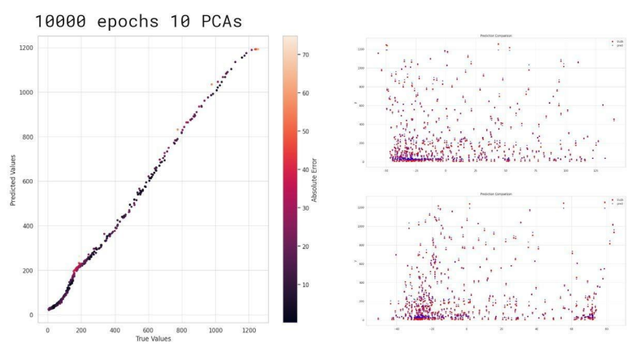


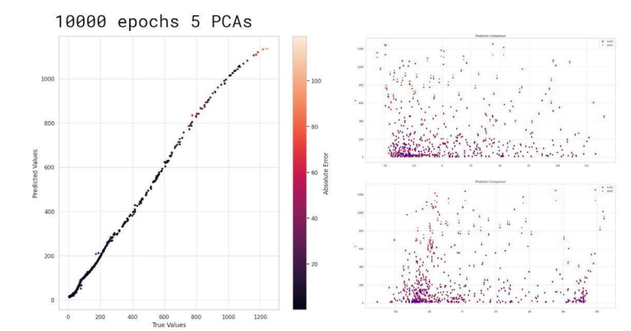


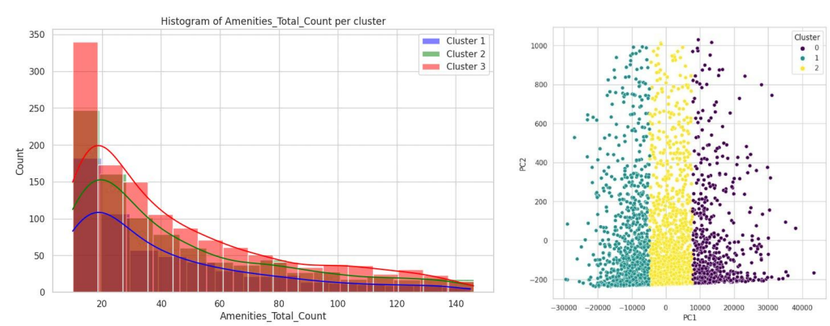




ANN on PCAs







We've downscaled our dataset via Principal Component Analysis (PCA), which can be visualized through histograms, boxplots, and the heatmap, with convergence at PCA 10. This means our features almost equally describe our model.
After an initial ANN failed, we implemented an ANN based on the PCA's matrix, yielding positive prediction results as you can view from the diagonals one note we suspect that due to the many filtering + outlier based on quantiles iterations the data somehow are formed in our intent and thats why the diagonals are so proper. We attempted K-means clustering on the pca clouds, but found distinct segmentation by review times - noon, afternoon, evening. The data structures can be seen here.

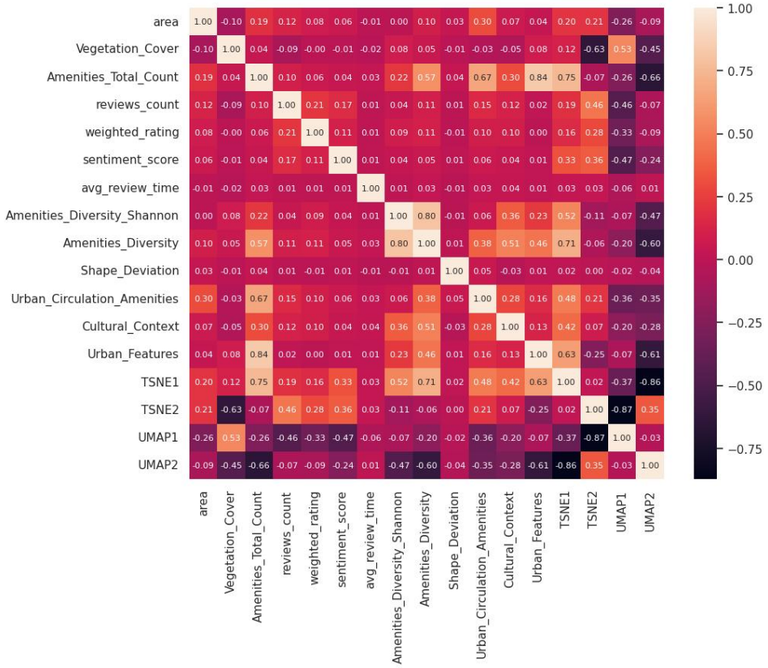
K-means Clustering

Then, we've sampled a smaller data group for experimentation. Following sentiment analysis, we analyzed word counts from reviews, grouped synonyms based on the top 100 most use words, and then applied PCA on the data attributes that contained those filters. Similar results on pca heatmaps and the kmeans led us to explore another method for tackling non-linearity."

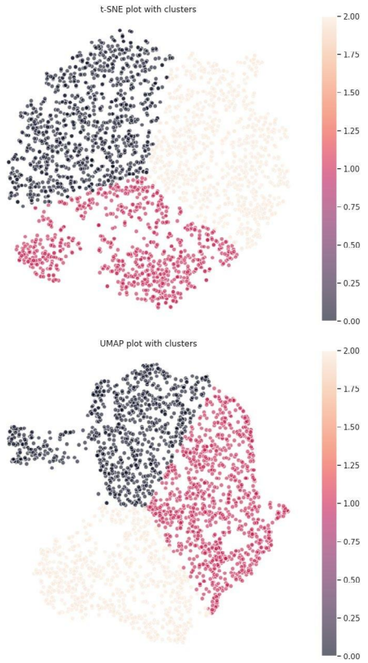
7000 PARKS

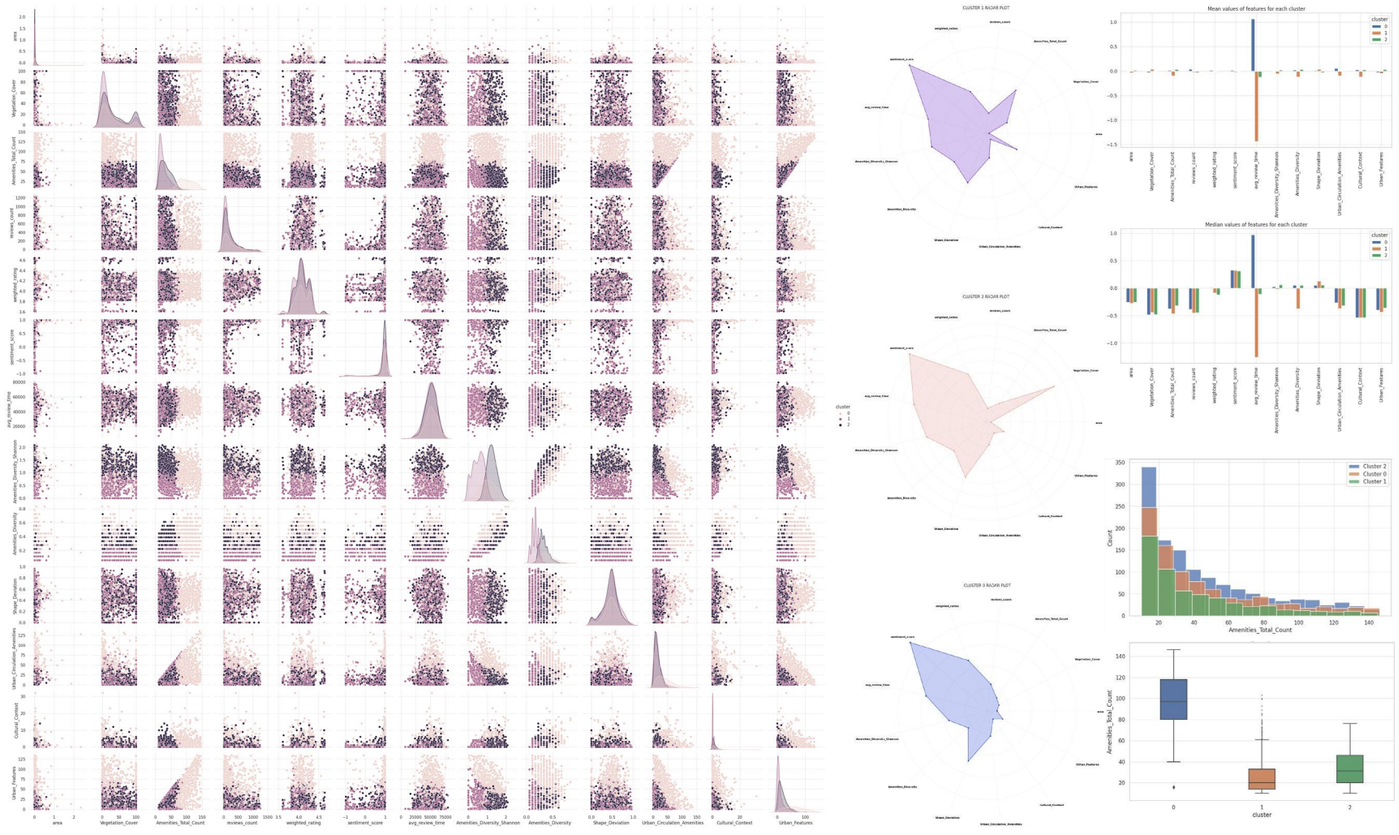
Our key finding: using t-SNE and UMAP. t-SNE, excels at preserving local structure and forming distinct clusters. UMAP, or Uniform Manifold Approximation and Projection, balances preserving global and local structure.

Initial visualizations show segmentations by feature color. However, after applying k means clustering on the output of those algorithms, three distinct groups emerged, a critical step forward in understanding our complex, multi-dimensional data."



K-means Clustering

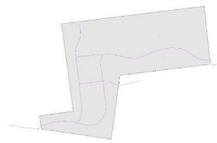




Moving further into K-means clustering, we can now better understand the structure of these data clusters by examining medians, averages, data distributions, and box plots. The pair plot we believe is successful into segmenting the features properly while of course some of them remain totally random due to the nature of our project and the webscrapping.

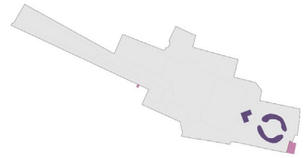
Parc Dalamau

MOVE
shape Deviation : 0.5
amenities Diversity : 0.3
year of construction: 1943
sentiment: 0.99
weighted_rating: 4.1
avg_review_time: 4:29:04 PM



Parco Aldo Moro

NATURE
shape Deviation : 0.53
amenities Diversity : 0.5
year of construction: 1949
sentiment: 0.95
weighted_rating: 4.1
avg_review_time: 2:27:53 PM

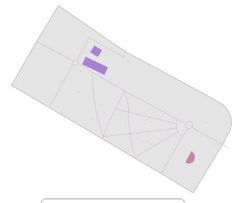


Parco Sandro Pertini

CHILL
shape Deviation : 0.36
amenities Diversity : 0.7
year of construction: 2010
sentiment: 0.98
weighted_rating: 3.9
avg_review_time: 15:38:40 PM

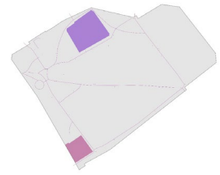
Greenbank Park

MOVE
shape Deviation : 0.37
amenities Diversity : 0.77
year of construction: 1997
sentiment: 0.99
weighted_rating: 3.9
avg_review_time: 14:41:01 PM



Parco 25 Novembre

CHILL
shape Deviation : 0.5
amenities Diversity : 0.55
year of construction: 1884
sentiment: 0.97
weighted_rating: 3.7
avg_review_time: 11:24:46 AM

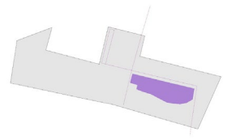


Bruce Castle Park

CHILL
shape Deviation : 0.5
amenities Diversity : 0.55
year of construction: 1892
sentiment: 0.97
weighted_rating: 4.3
avg_review_time: 18:15:56 PM

Garenne Lemot

CHILL
shape Deviation : 0.38
amenities Diversity : 0.55
year of construction: 1827
sentiment: 0.98
weighted_rating: 4.3
avg_review_time: 5:20:08 PM



Parco 4 Aprile

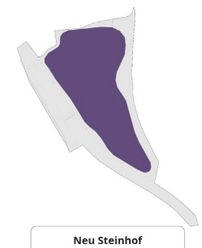
NATURE
shape Deviation : 0.52
amenities Diversity : 0.52
year of construction: 1965
sentiment: 0.99
weighted_rating: 4.1
avg_review_time: 2:27:53 PM

Pauluspark

CHILL
shape Deviation : 0.44
amenities Diversity : 0.52
year of construction: 1905
sentiment: 0.99
weighted_rating: 4.2
avg_review_time: 2:14:19 PM

Park Sielecki

MOVE
shape Deviation : 0.52
amenities Diversity : 0.22
year of construction: 1965
sentiment: 0.98
weighted_rating: 4.3
avg_review_time: 2:27:53 PM



Neu Steinhof

CHILL
shape Deviation : 0.62
amenities Diversity : 0.44
year of construction: 1904
sentiment: 0.99
weighted_rating: 4.3
avg_review_time: 12:52:29 PM

Eugenia de Montijo

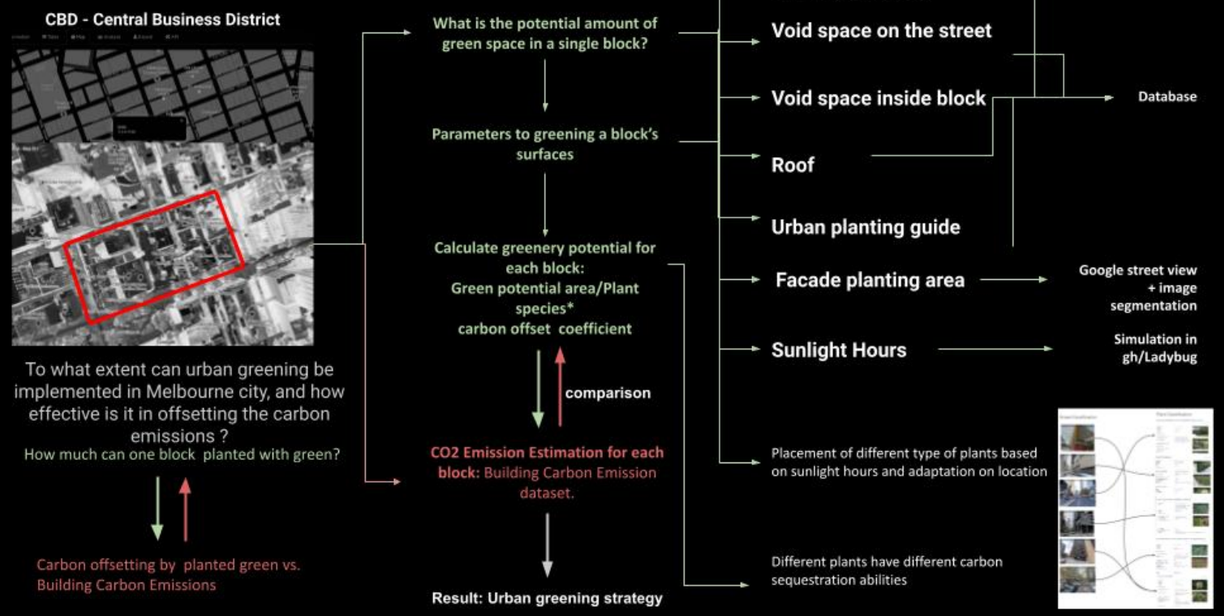
NATURE
shape Deviation : 0.52
amenities Diversity : 0.52
year of construction: 1965
sentiment: 0.99
weighted_rating: 4.1
avg_review_time: 2:27:53 PM

From our spider plots, we can discern diversity in vegetation cover and shape deviation among the three clusters.

Go to Green

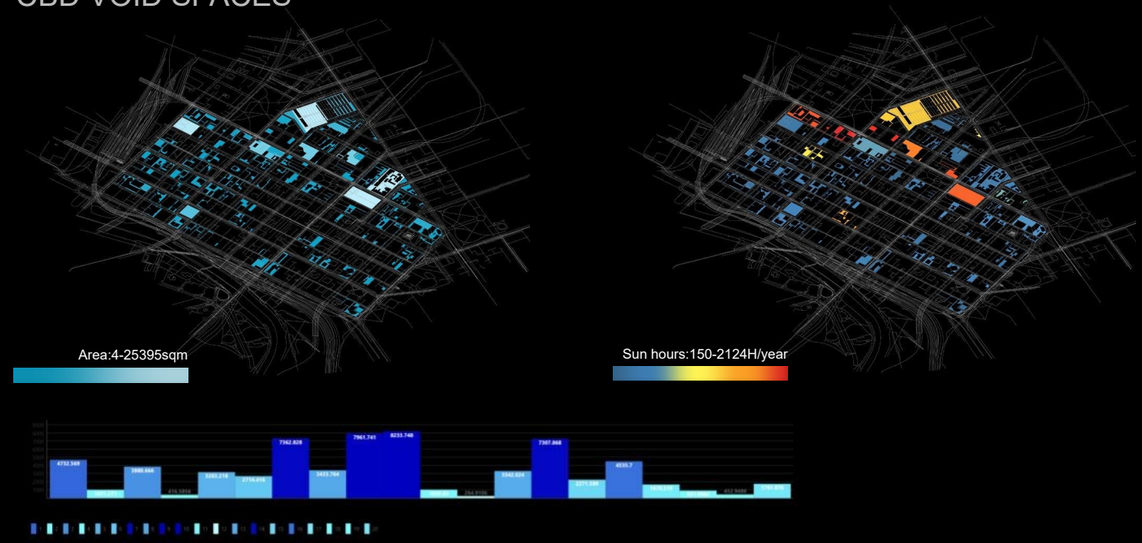
Unveiling the potential of urban blocks for greenery insertion in Melbourne:
Offsetting carbon emission of the city blocks by implementing a comprehensive greening plan

Ziqi Cui, Ezgi Nalci, Littieri Machado Lamb
2023 Spring
AI Urban Stuido
Faculty: Angelos Chronis, Ebrar Eke, Ondrej Vessely

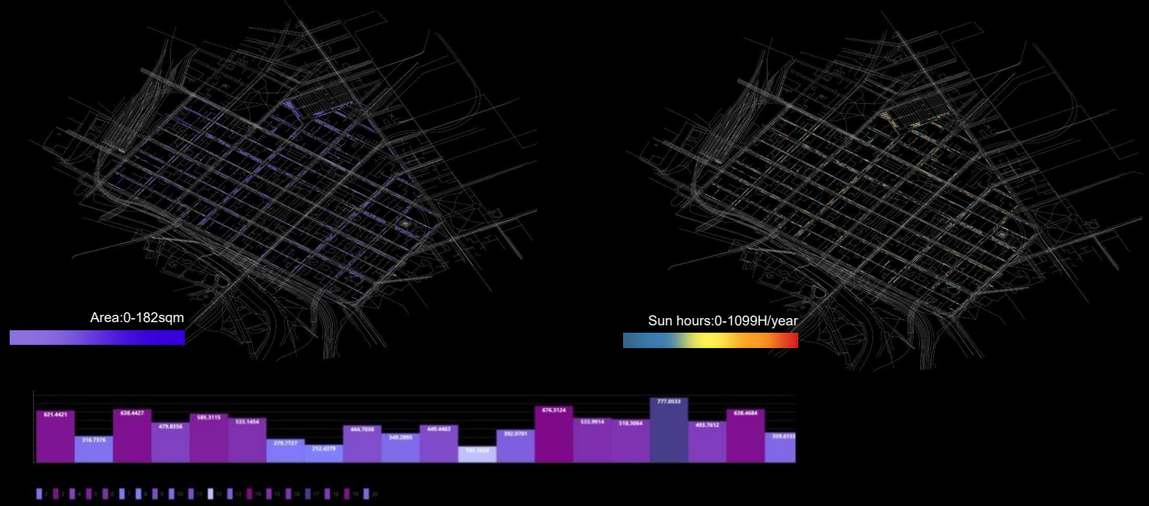


This proposal aims to uncover the untapped potential of surfaces within Melbourne city blocks for the implementation of greenery. It aims to identify suitable locations for the insertion of greenery, as well as quantify the CO2 absorption capacity resulting from the greenery. The objective` is to provide valuable insights for enhancing urban greening strategies and incentivizing carbon offsetting in the built environment to reinforce a sustainable city approach.

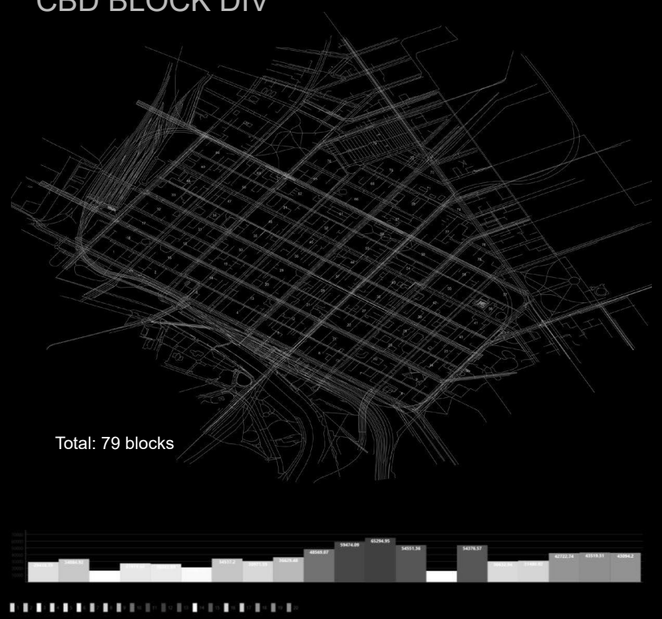
CBD VOID SPACES



CBD STREET PARKINGS

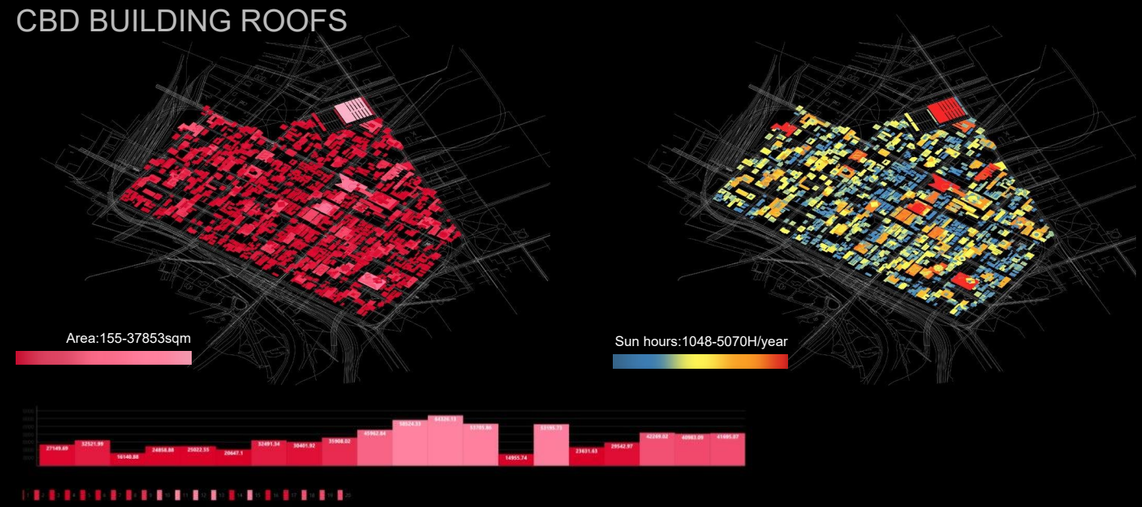


CBD BLOCK DIV



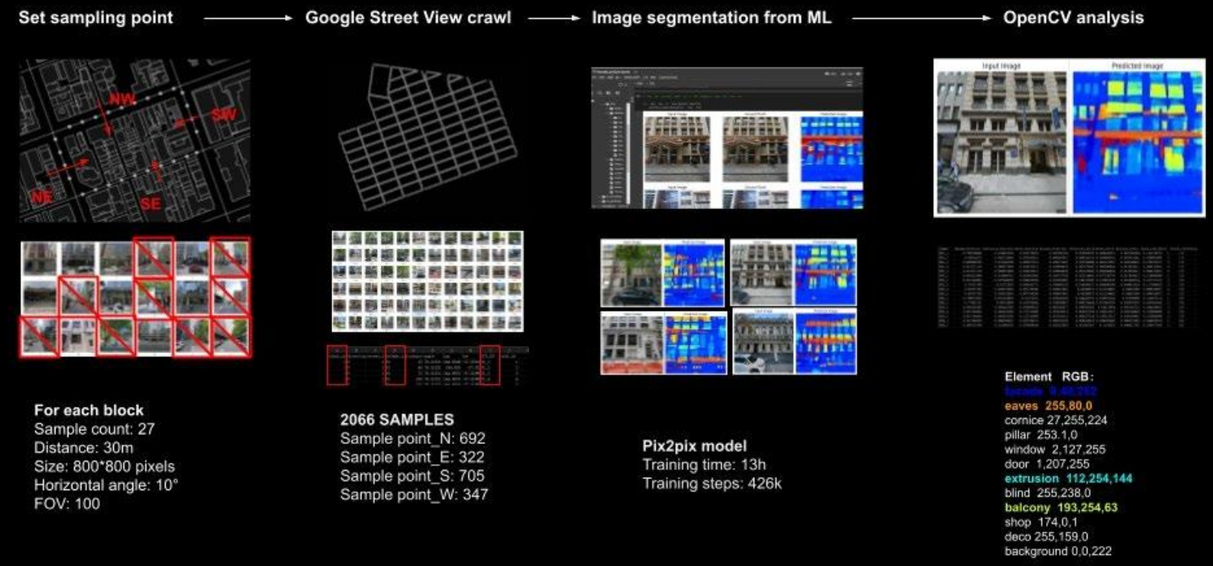
CBD STREET GREENERY

CBD BUILDING ROOFS

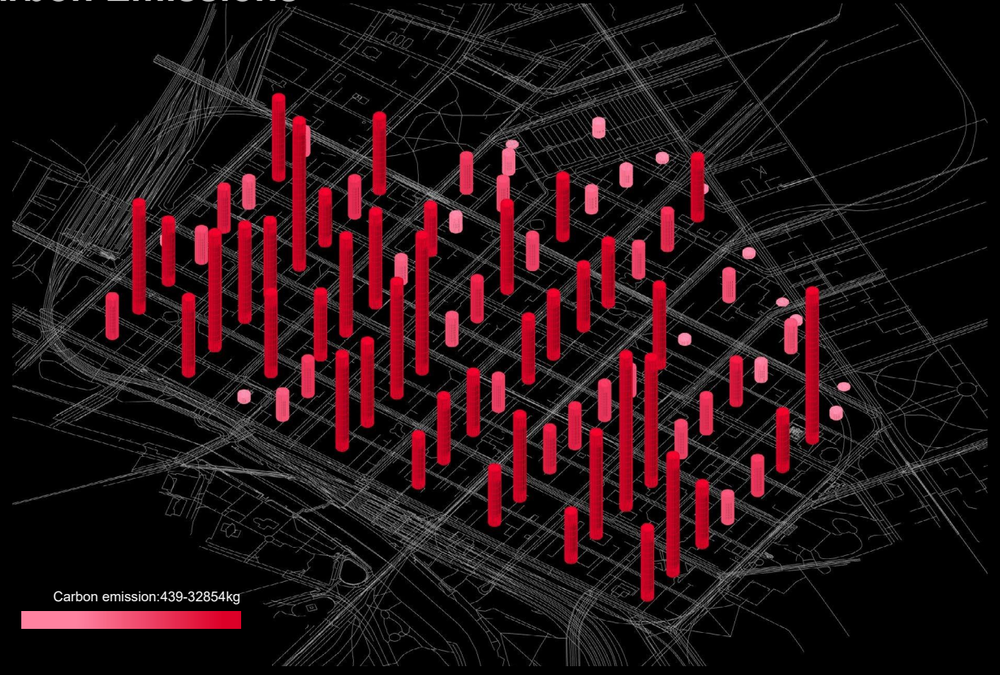


Data analysis: abailable planting area

Workflow: how to get availabe facade data



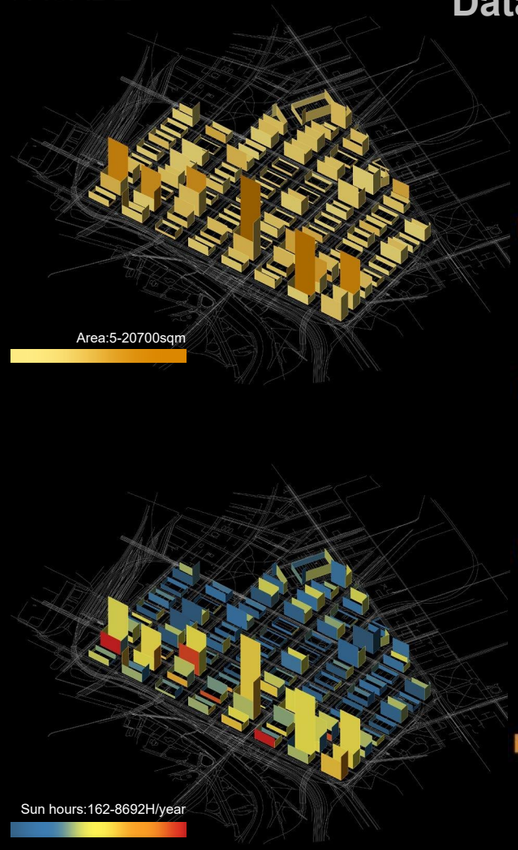
Carbon Emissions



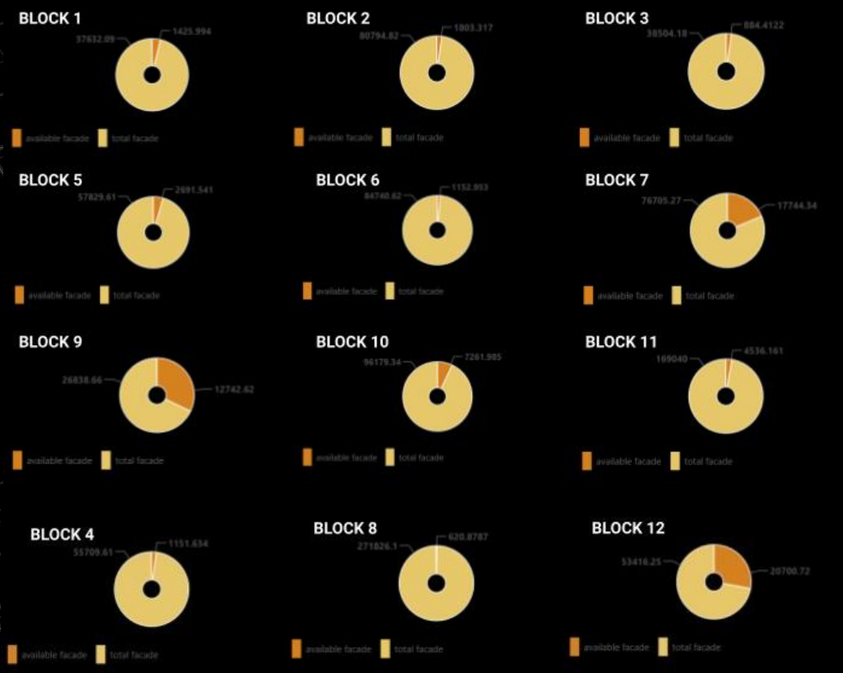
Carbon Emissions (kgCO2) = Electricity Consumption (kWh) x Carbon Intensity (kgCO2/kWh)

Carbon Intensity can vary depending on the region and the mix of energy sources (e.g., coal, natural gas, renewables) used to generate electricity.
On a life-cycle basis, coal emits 820 grams of carbon dioxide (CO2) equivalent per kWh

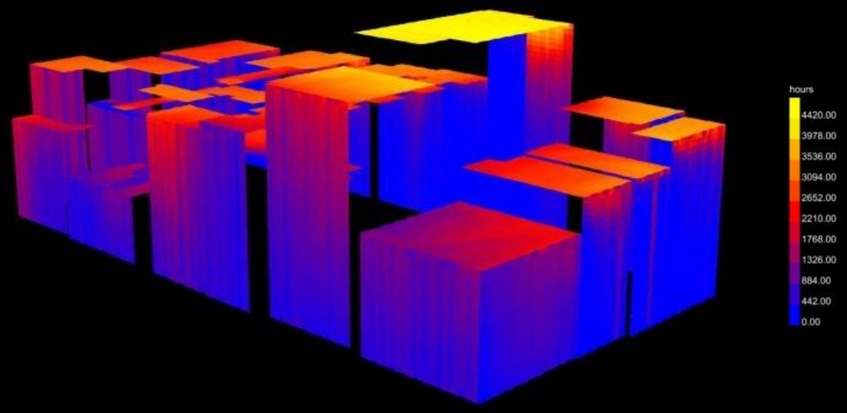
FACADE

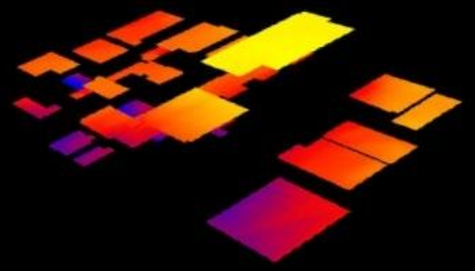


Data analysis: abailable planting area on facade

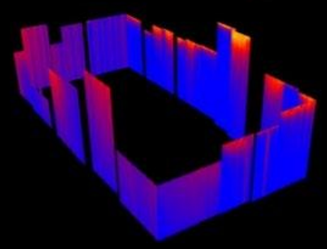


Block 30 Plant type distribution





Roof space + full sun plants



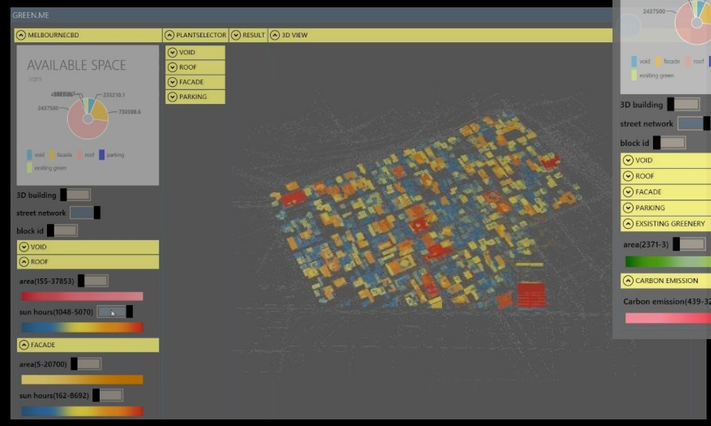
Sun hours < 1700 facade space
+ shade plants

Sun hours > 1700 facade space
+ partial sun plants

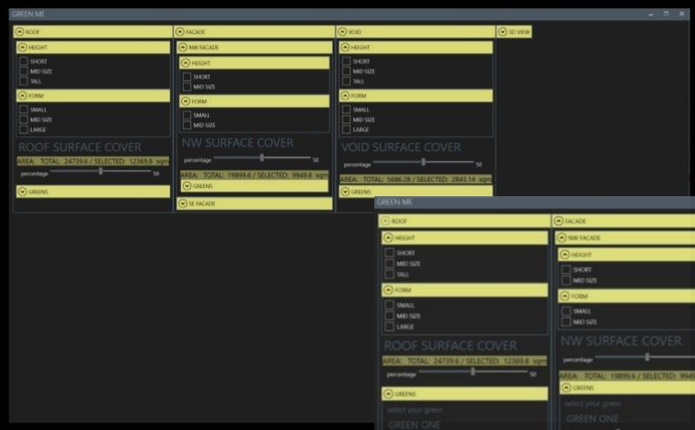
void space + shade plants

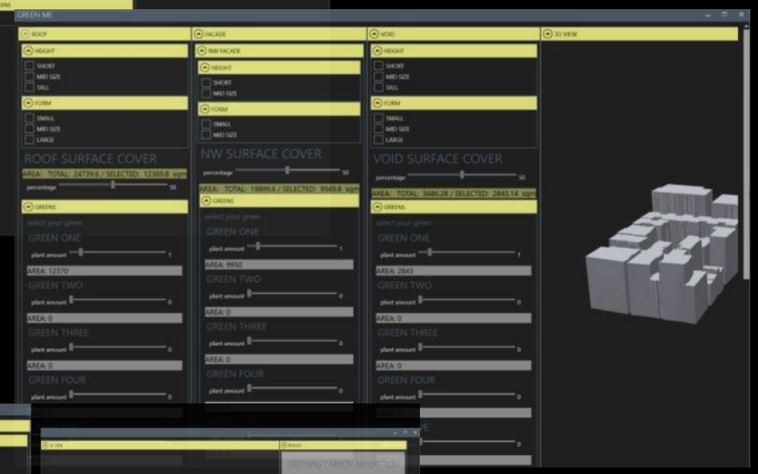
"GREEN.ME" Application Interface

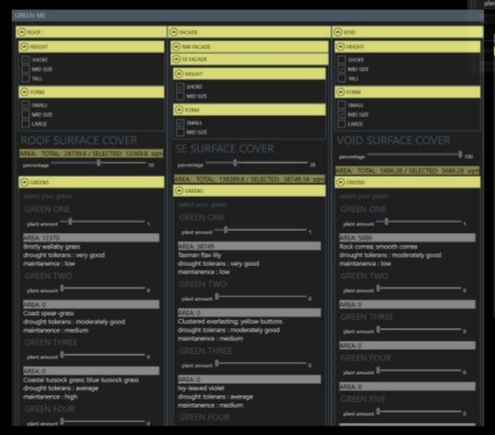
1. Overview of CBD

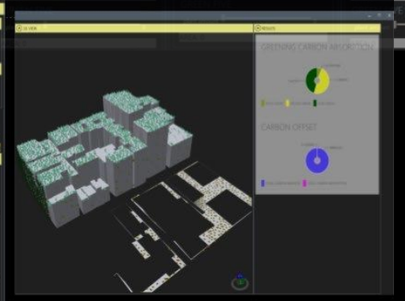


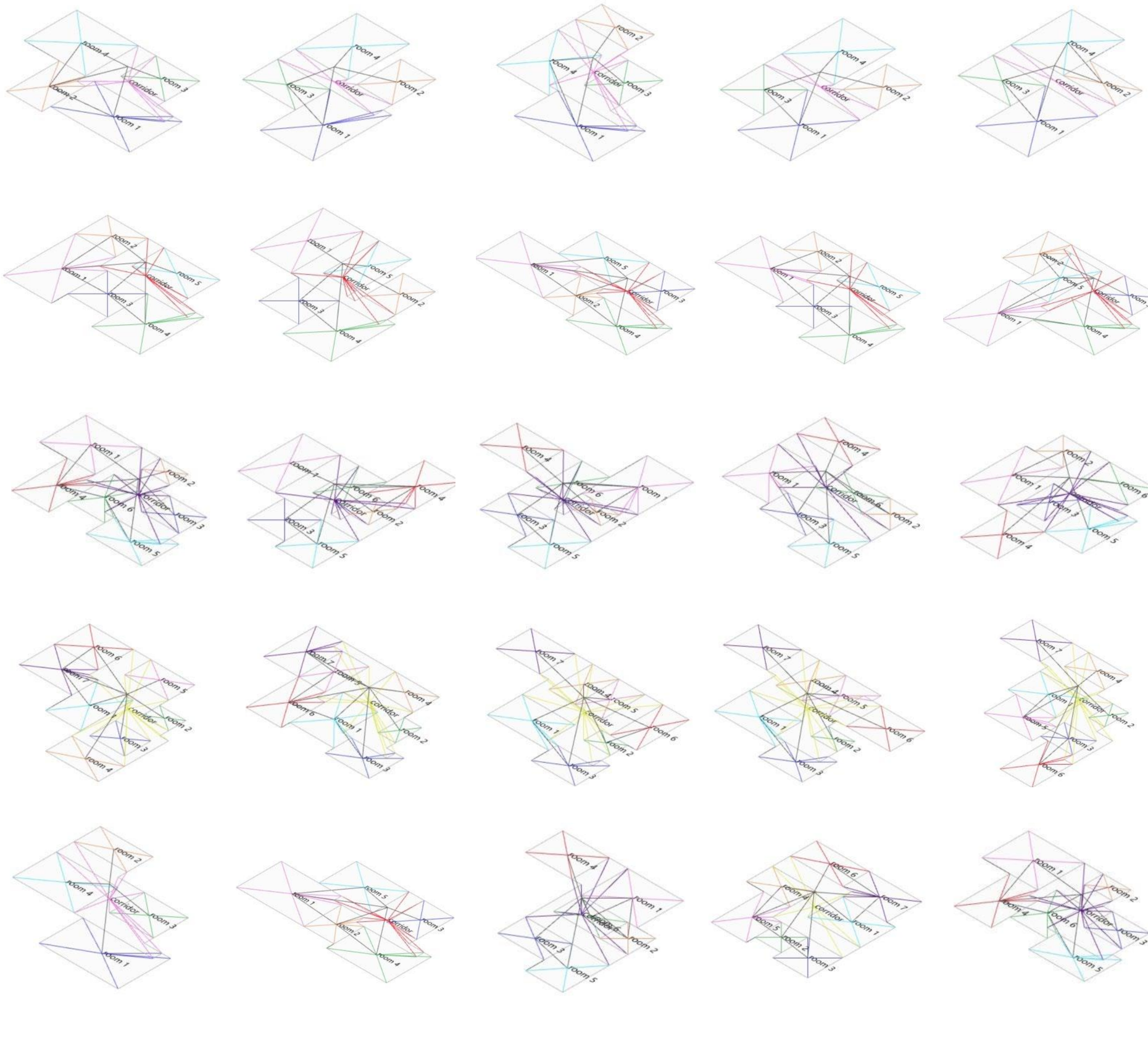
2. Zoom into specific block









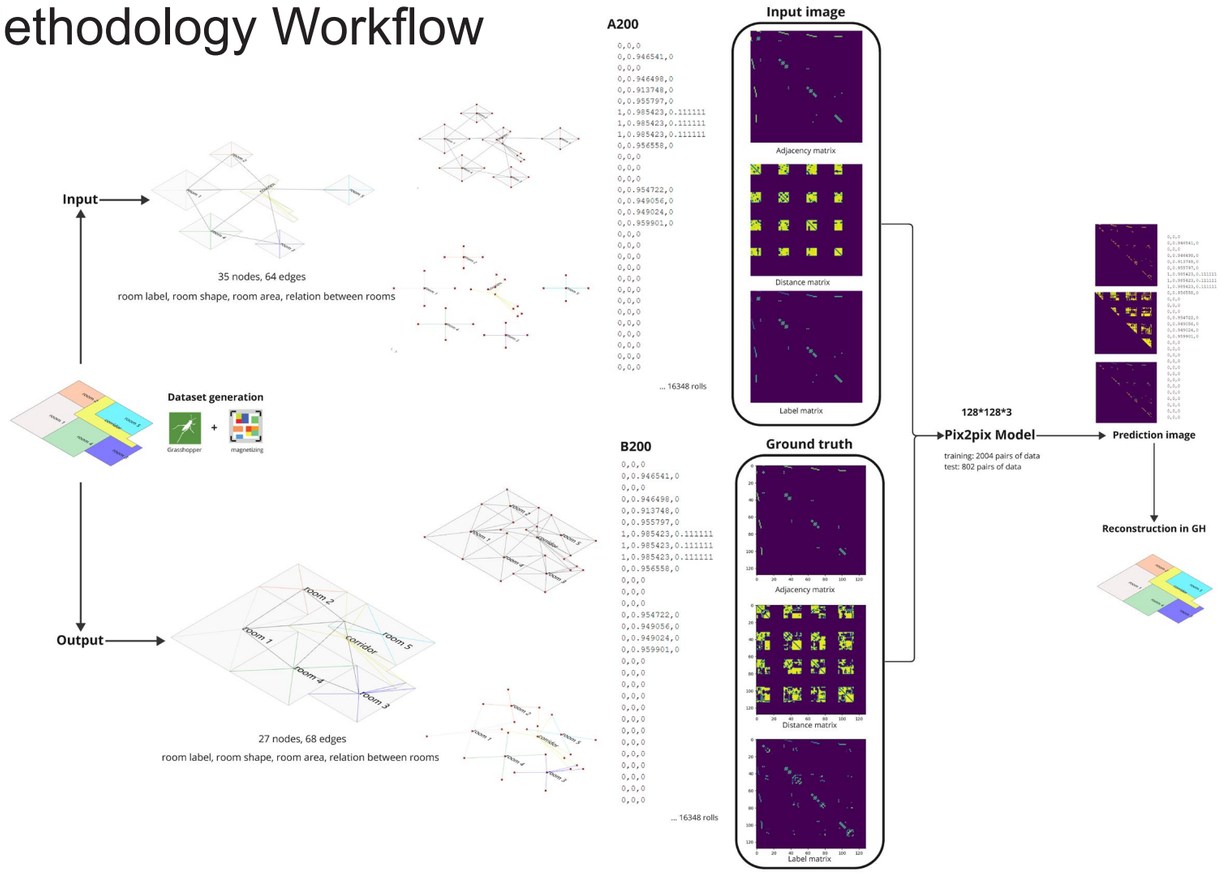


"Follow with corridor"

A GAN-based floor plan generator

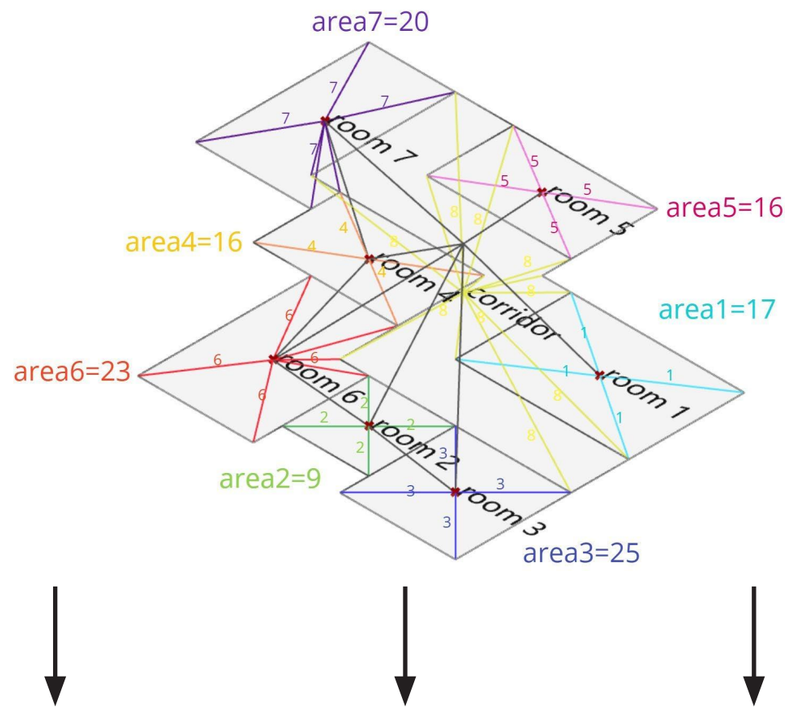
Ziqi Cui
2023 Fall
IAAC thesis studio
Faculty: Oana Taut

Methodology Workflow



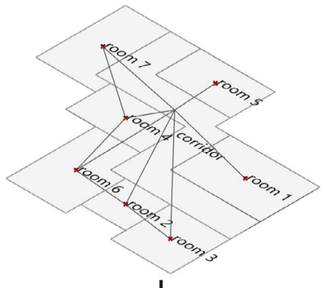
Embed Feature

It includes: room label, room shape, room area, relation between rooms.

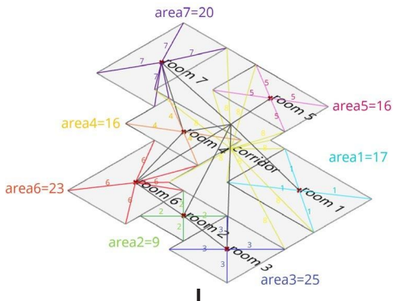


Got input dataset

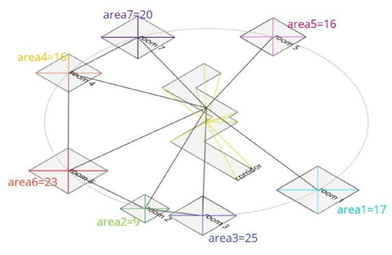
Room relationship (Output)



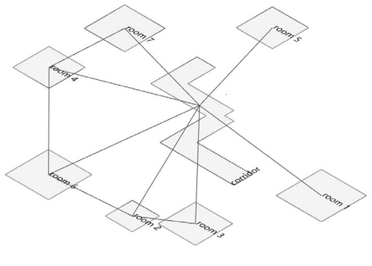
Room feature (Output)



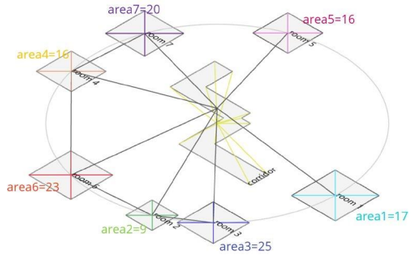
Room feature (Input)



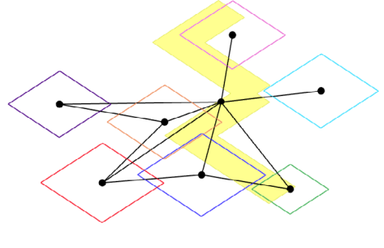
Room relationship (Input)



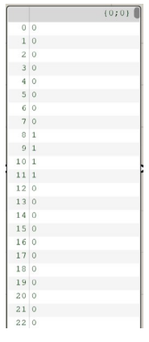
Room feature (Input)



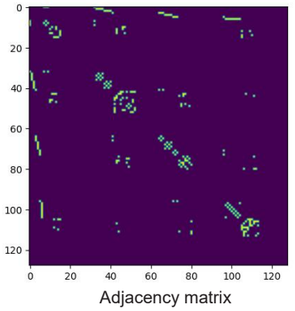
Room generation result (Kangaroo)



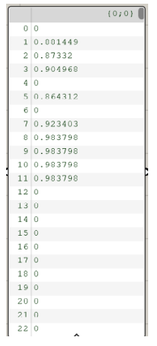
Room relation/room geometry
Nodes connections(yes/no)



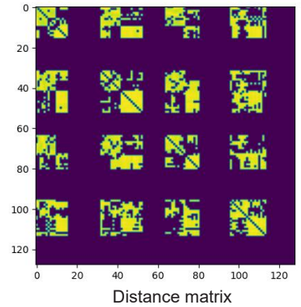
Adjacency matrix



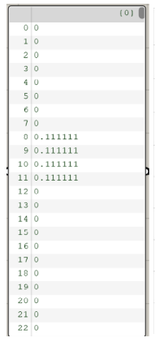
Room relation/room geometry
Nodes distance/ edge length



Distance matrix

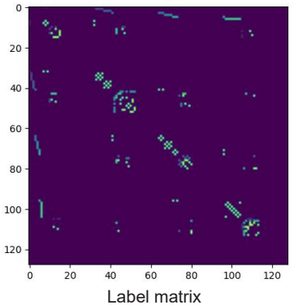


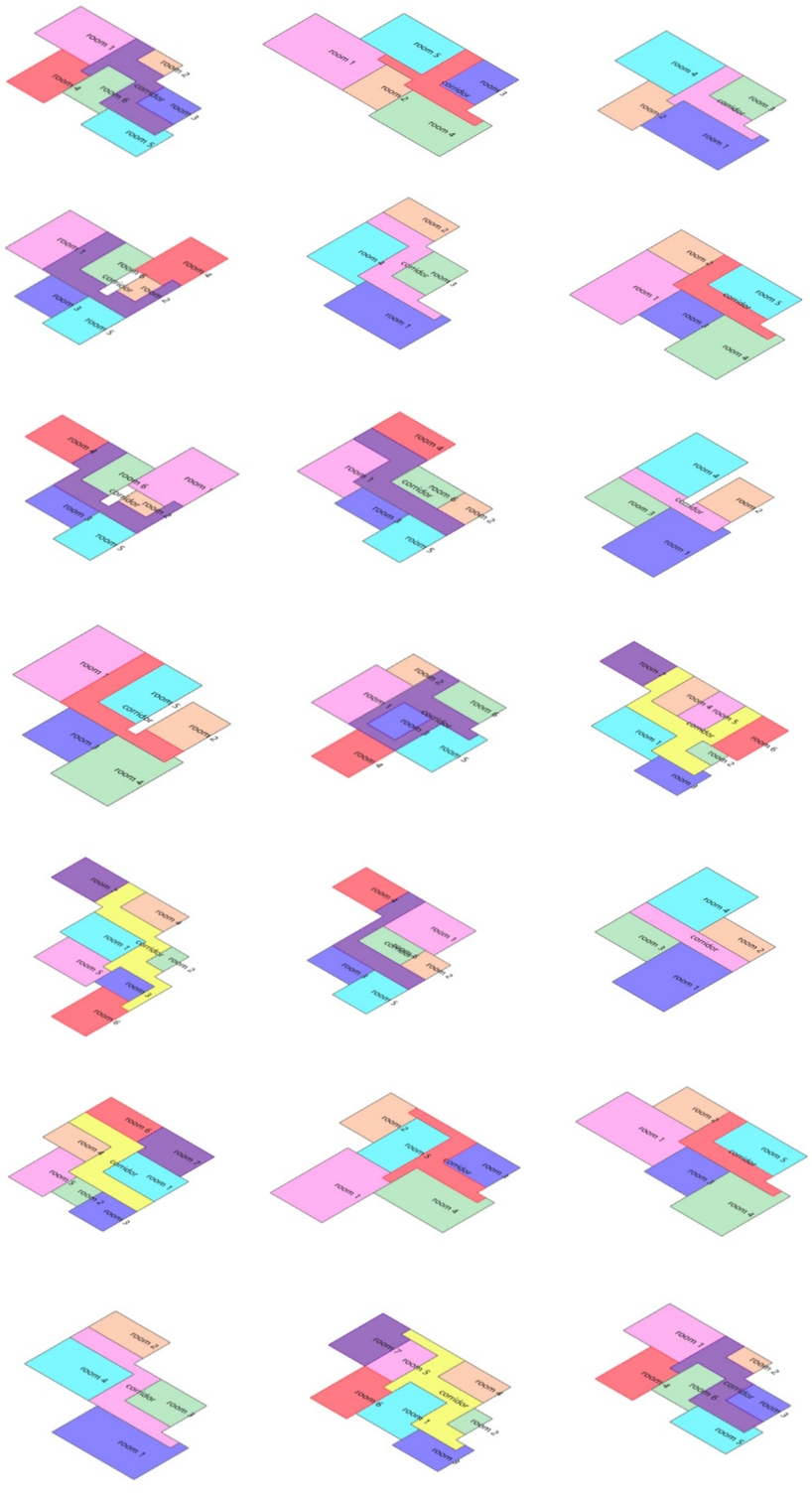
Room label
edge label



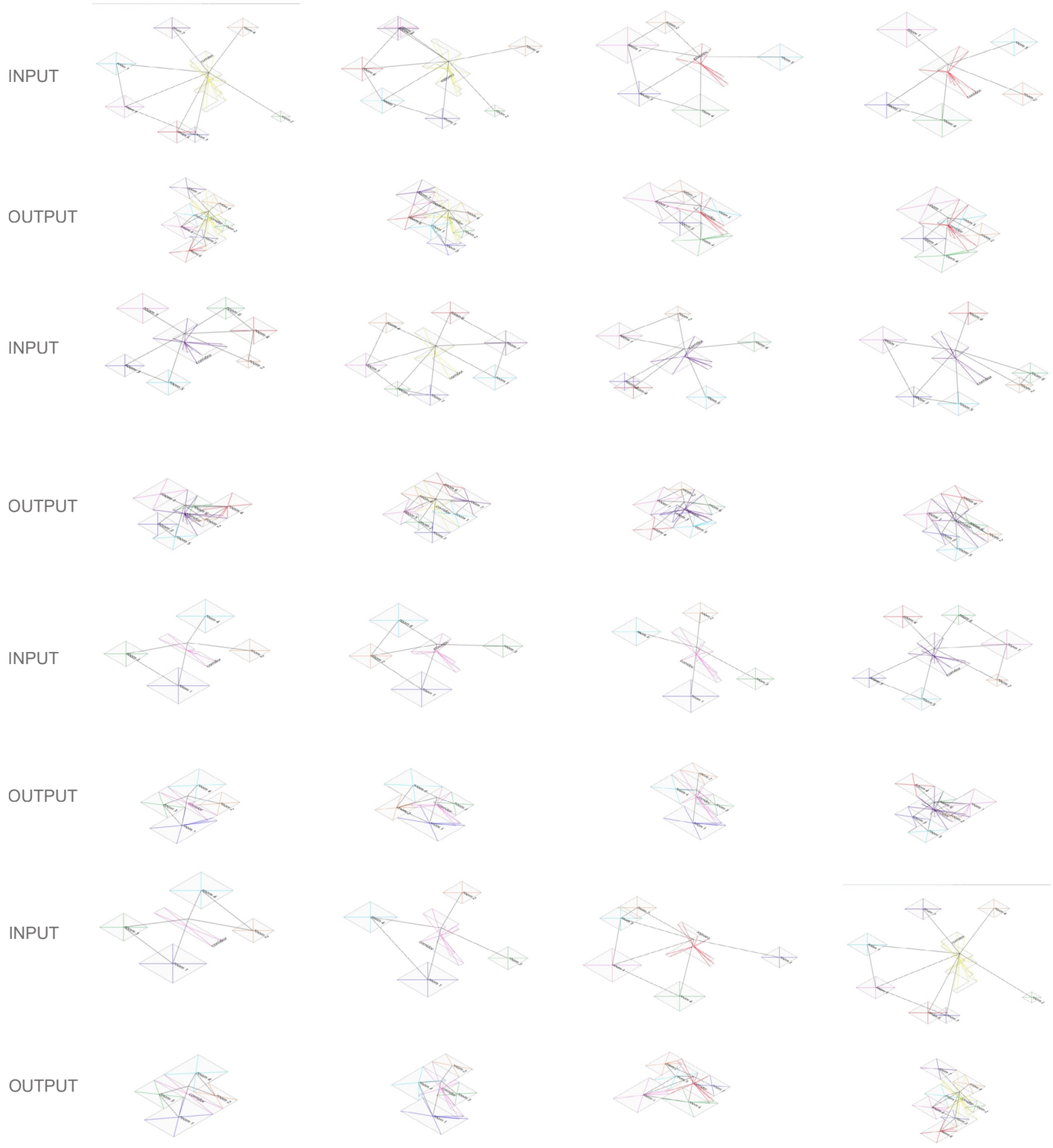
Room type	Label	Label in pixel matrix(0-1)
Room01	1	0.11111
Room02	2	0.22222
Room03	3	0.33333
Room04	4	0.44444
Room05	5	0.555556
Room06	6	0.666667
Room07	7	0.777778
Corridor	8	0.888889
Others	9	1

Label matrix





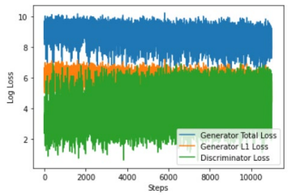
Grasshopper Generate Dataset



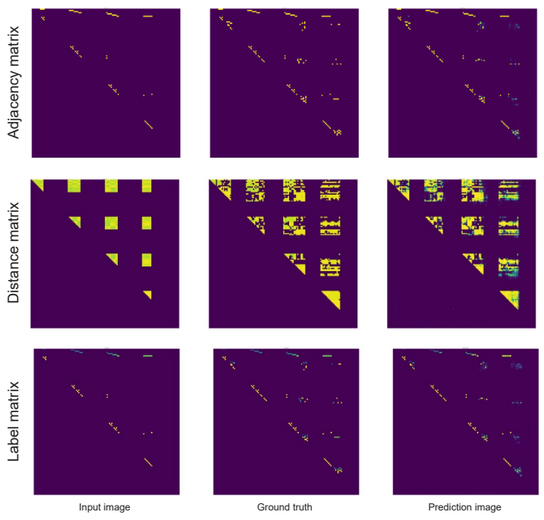
Dataset Variation

Around 2000 pairs of data with different area, rooms and configuration.

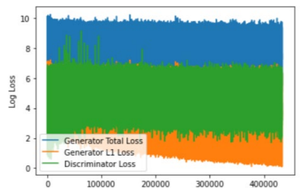
Training Start



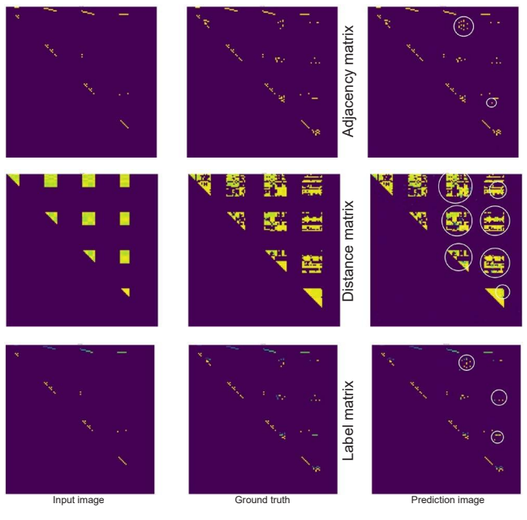
Step:8999
Generator loss: 8.926547394154905
Generator l1 loss: 5.807867540837871
Discriminator loss: 3.611856884098233



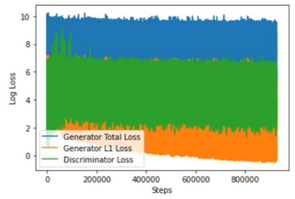
1st Training result_433,999 steps



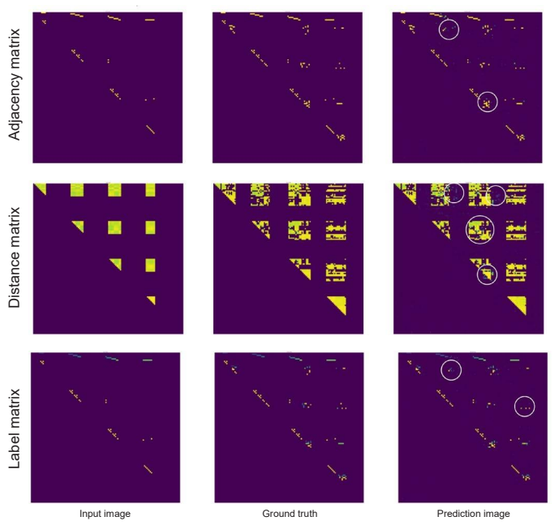
Step: 433999
Generator loss: 7.69580344655172
Generator l1 loss: 4.187161721570787
Discriminator loss: 4.823407427206053



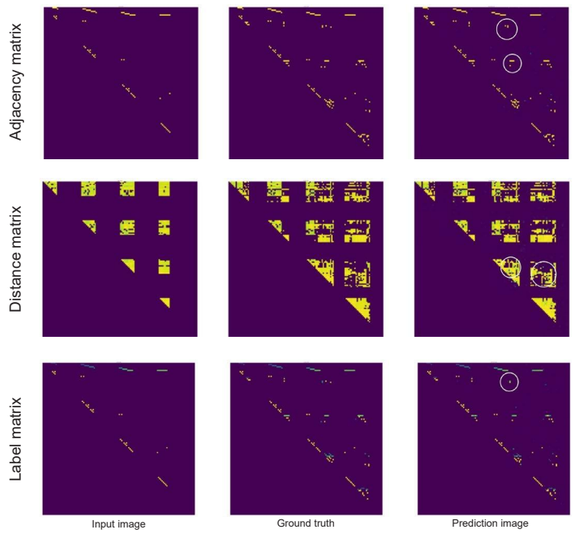
2nd Training result_988,999 steps



Step:988999
Generator loss: 7.494118938771732
Generator l1 loss: 3.8295014311102595
Discriminator loss: 4.693514201845212



Prediction Result

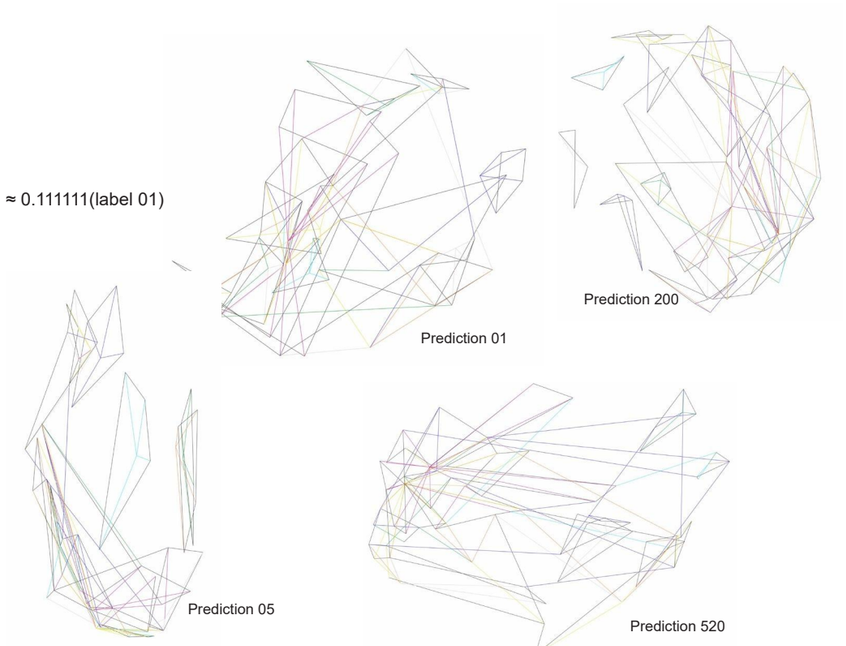


1st Reconstruction Result

Most of generation result has distortion.

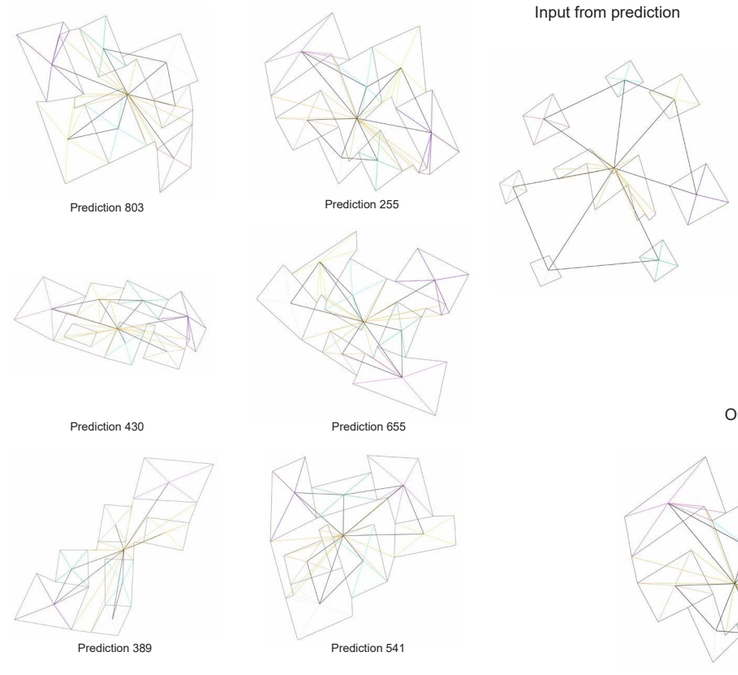
0.000000,0.000500,0.000019
0.000000,0.864024,0.000017
0.000000,0.860430,0.000000
0.000000,0.904417,0.000014
0.000000,0.943931,0.000000
0.000000,0.870392,0.000008
0.000000,0.000000,0.000000
1.000000,0.983873,0.087308
1.000000,0.986280,0.084551
1.000000,0.986758,0.099570
1.000000,0.991310,0.119562
0.000000,0.952508,0.000005
0.000000,0.967902,0.001508
0.000000,0.000533,0.000187
0.000000,0.861662,0.000000
0.000000,0.923679,0.000164
0.000000,0.000018,0.000002
0.000000,0.977263,0.000073
0.000000,0.952933,0.000032
0.000000,0.913780,0.000843
0.000000,0.000018,0.000074
0.000000,0.839390,0.000853
0.000000,0.881369,0.000038
0.000000,0.000053,0.000471

≈ 0.111111(label 01)



2nd Reconstruction Result

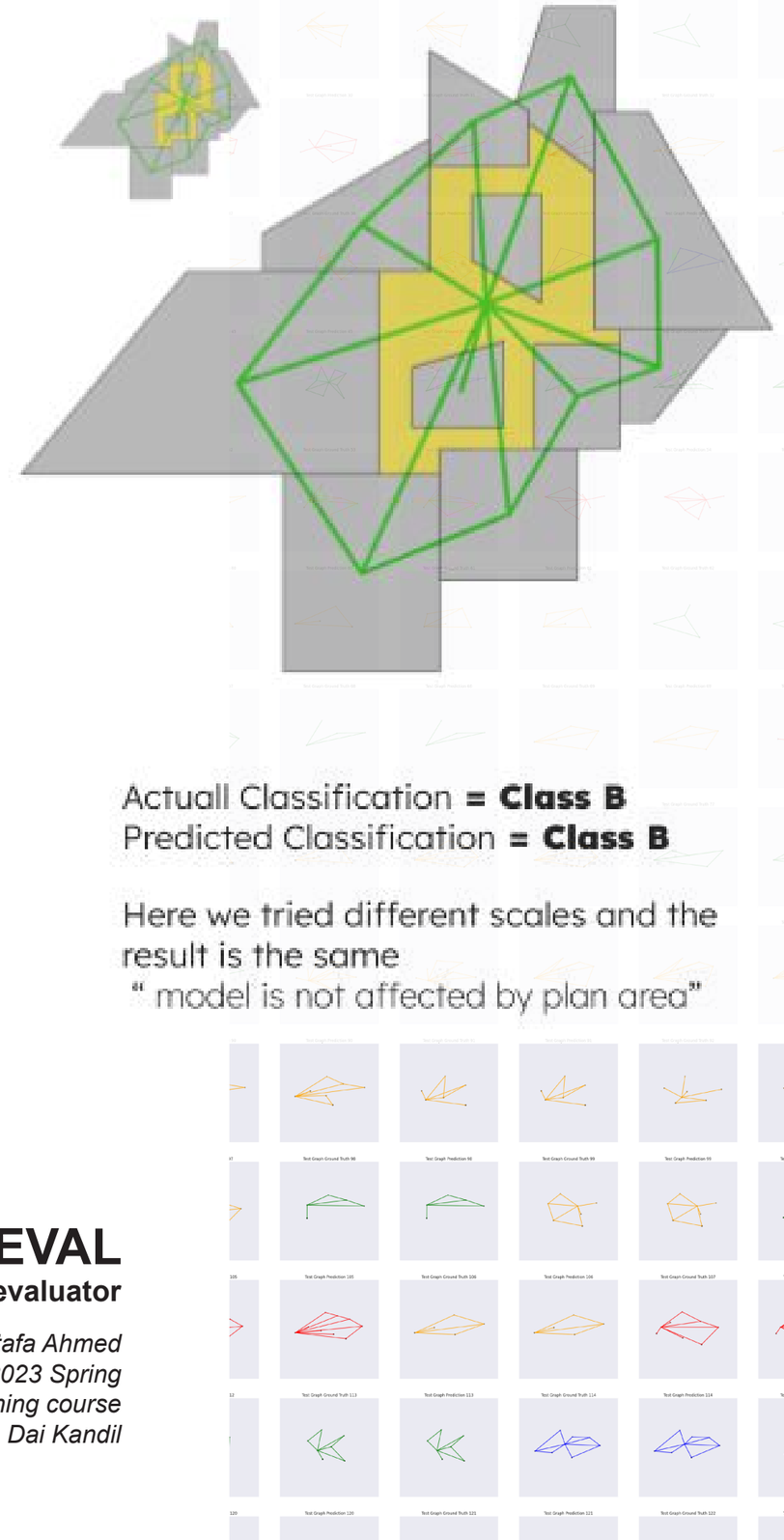
Generation result still has distortion, but work better than before.



FLOOR _PEVAL

A GML based floor plan evaluator

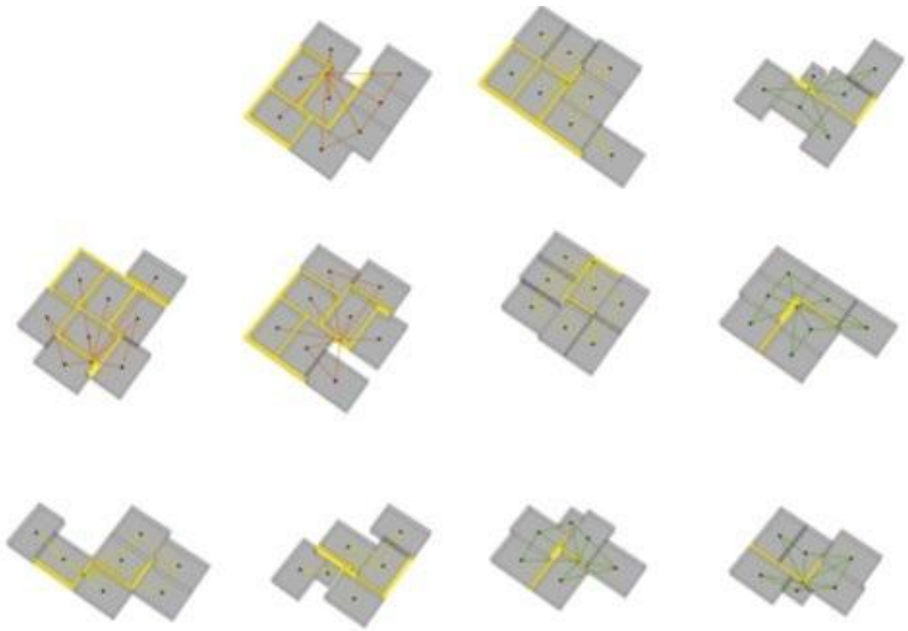
Ziqi Cui, Mostafa Ahmed
2023 Spring
AIA Graph Machine learning course
Faculty: David Andres Leon, Dai Kandil



Actuall Classification = **Class B**
Predicted Classification = **Class B**

Here we tried different scales and the result is the same
" model is not affected by plan area"

Dataset generation

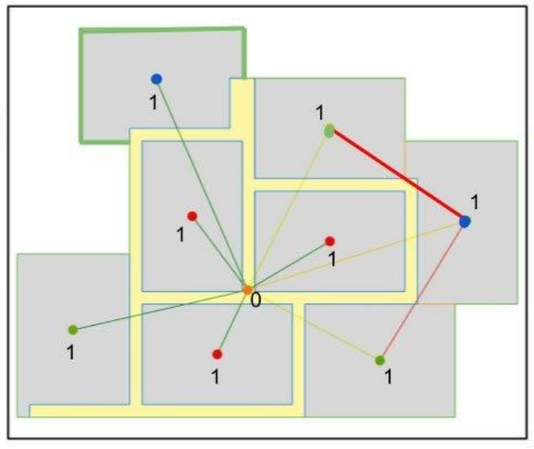


- Generating Random Floor plans
- Classifying the spaces into Rooms + Corridor
- Classifying space edges into " To corridor , To Exterior & To neighbour"
- Calculating each room space quality index = (1- to corridor %) +to exterior %
- Build a graph Nodes = rooms centers Edges = connections between nodes
- Nodes encoded data is:
  Room type
  Room Quality index
  Room Area
  Room perimeter
  Edges encoded data is:
  Edges Vertices
  Edges Vertices to neighbour
  % average
- Graph encoded data is:
  Plan Quality
  Plan efficiency "Corridor% to plan area"
  Floor plan Area

Embed feature

Nodes encoded data is:
Room type: 1
Room Quality index: length to outside/room perimeter
Room Area
Room perimeter

x	y	z	RoomType	RoomQualit	RoomArea	RoomPrem
4.99	-1.9	1.75	1	1.4309001	112.7376	45.18459



Edges encoded data is:
Edges Vertices: 2
Edges Vertices to neighbour %
average: (length to neighbour 1/room 1 perimeter + length to neighbour 2/room 2 perimeter)/2

vi	vj	adjacency
1	3	0.035823

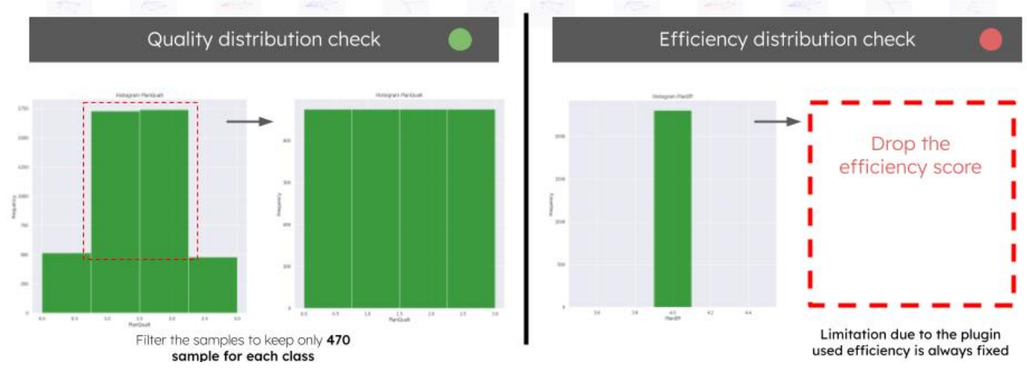
4 Classes(A,B,C,D)

Graph encoded data is:
Plan Quality: average room quality
Plan efficiency: corridor% to plan area
Floor plan Area

PlanQualt	PlanEff	PlanArea
0.821710133	4	369.8214

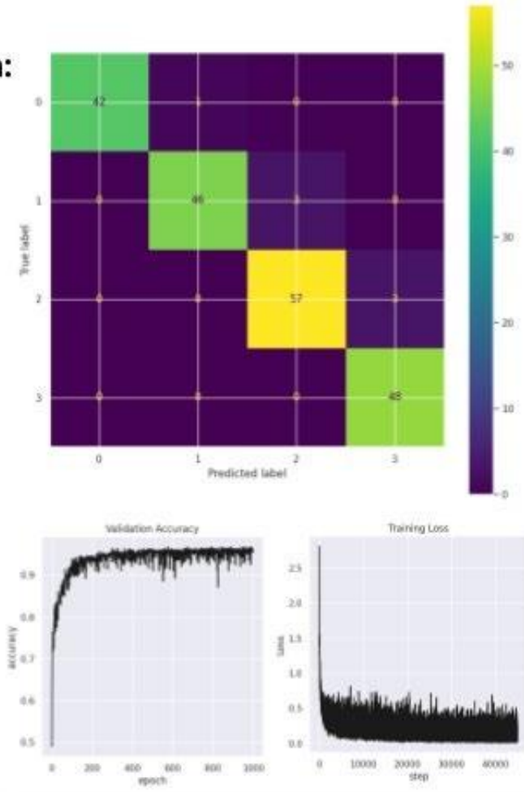
Corridor= 0
rooms= 1

We decided to drop room area and room perimeter because they are already influencing room quality index and this will result in more generic model.

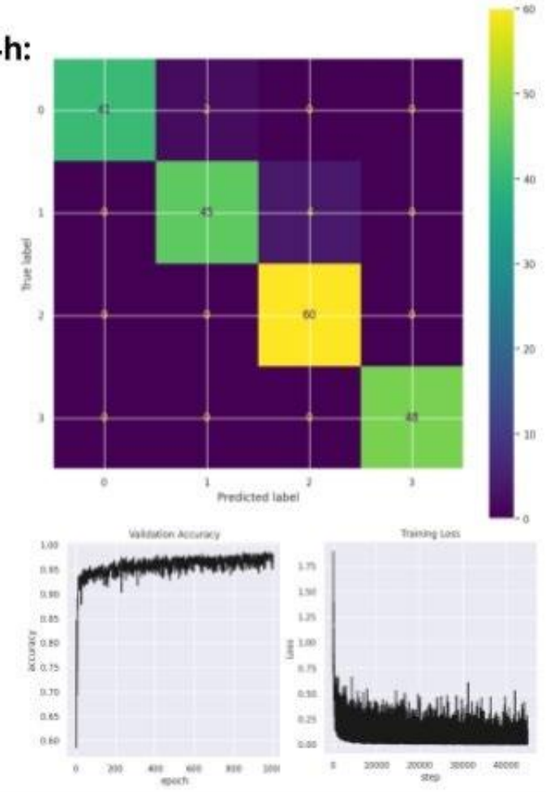


Training model

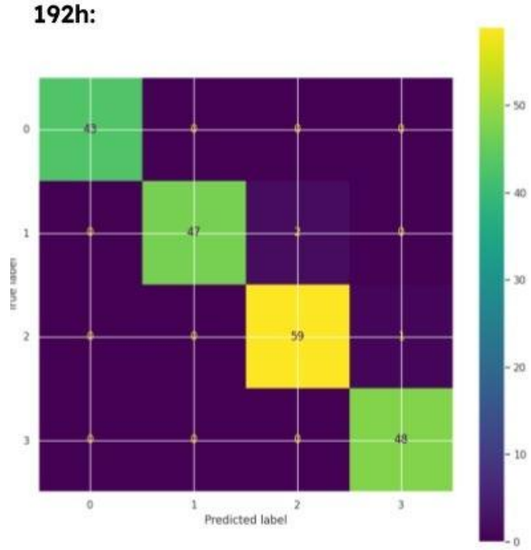
16h:

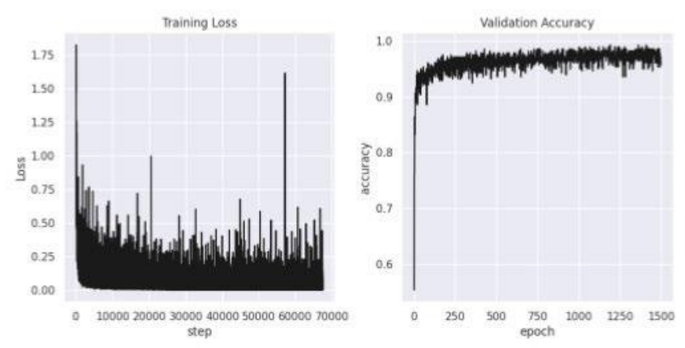


64h:



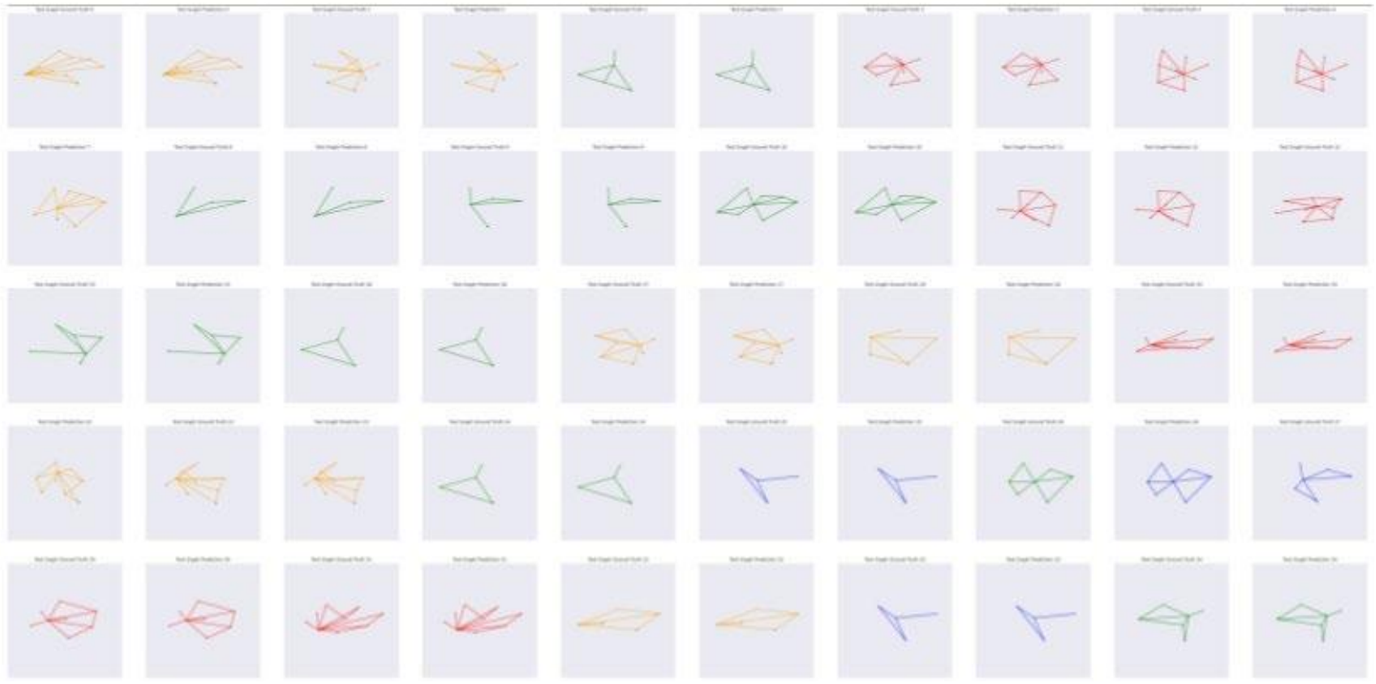
192h:



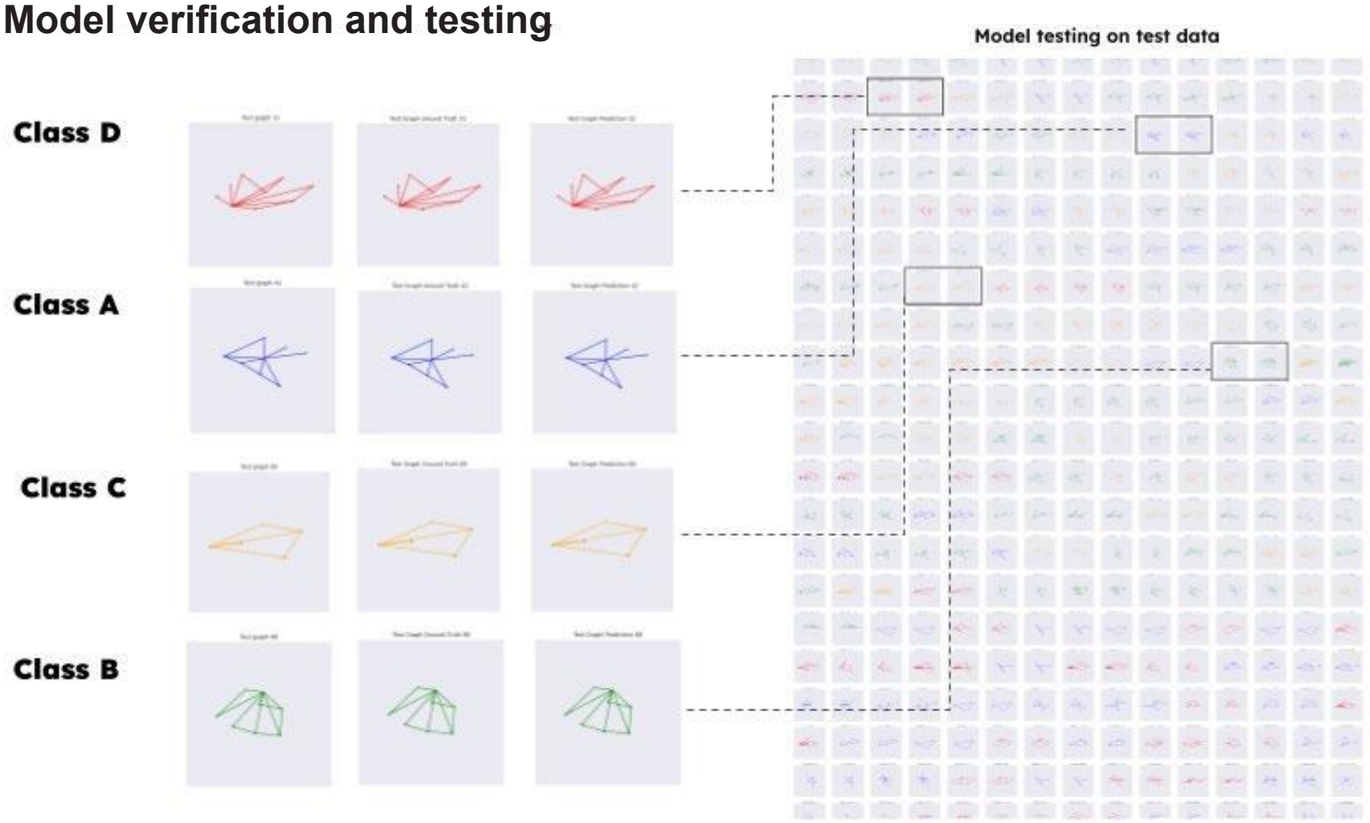


Model Accuracy = **98%**

Graph Samples “ Classified by quality

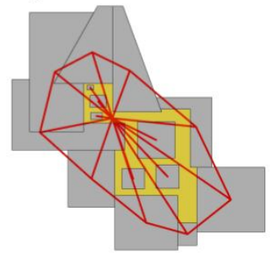


Model verification and testing

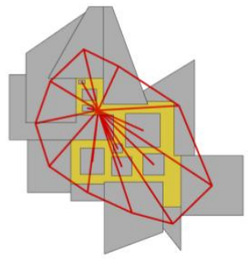


Prediction Result

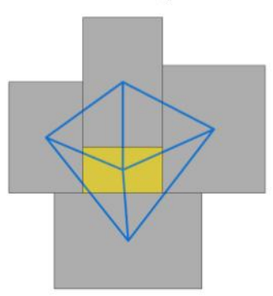
The model performed in a very goodway regarding predicting plans quality within training domain, also it’s performance was good for predicting unseen out of training domain data with a minor margin of error (only one class up or down in some plans), overall graph SAGE classification showed a very good potential for our particular problem with acceptable ability to generalize.



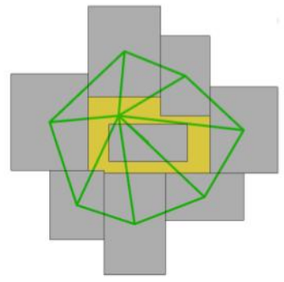
Actuall Classification = **Class D**
Predicted Classification = **Class C**



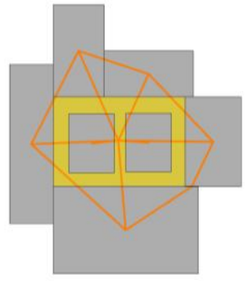
Actuall Classification = **Class D**
Predicted Classification = **Class C**



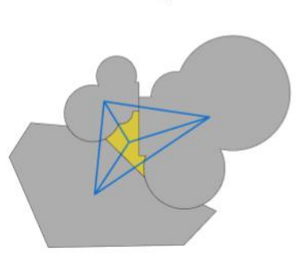
Actuall Classification = **Class A**
Predicted Classification = **Class A**



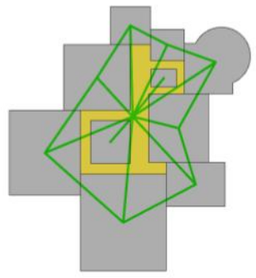
Actuall Classification = **Class B**
Predicted Classification = **Class B**



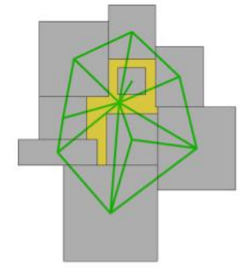
Actuall Classification = **Class C**
Predicted Classification = **Class C**



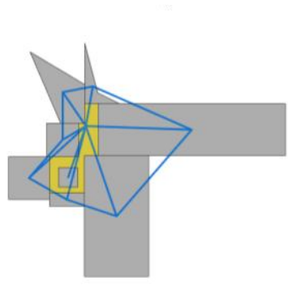
Actuall Classification = **Class A**
Predicted Classification = **Class A**



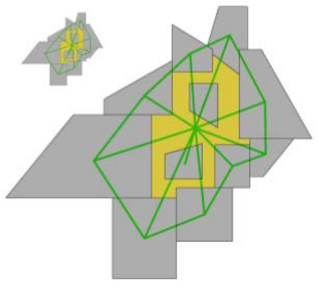
Actuall Classification = **Class B**
Predicted Classification = **Class B**



Actuall Classification = **Class B**
Predicted Classification = **Class B**



Actuall Classification = **Class A**
Predicted Classification = **Class B**



Actuall Classification = **Class B**
Predicted Classification = **Class B**

Actuall Classification = **Class B**
Predicted Classification = **Class A**

Generative AI in 3D

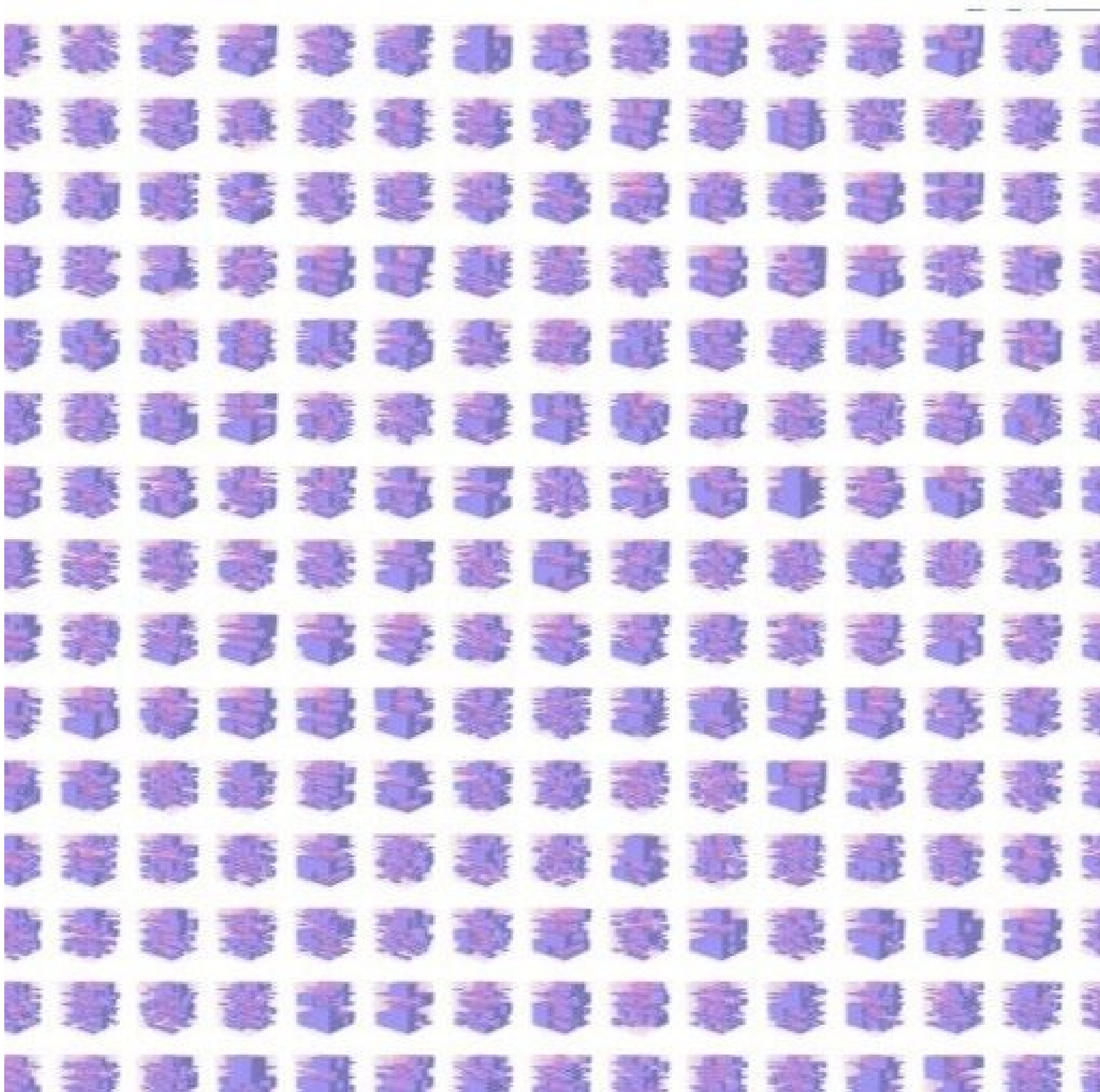
Two ways to use generative AI to reconstruct 3D building volumes

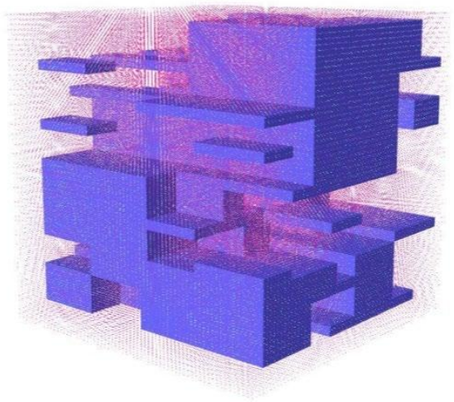
Ziqi Cui, Eugenia Raigada, Littieri Machado Lamb

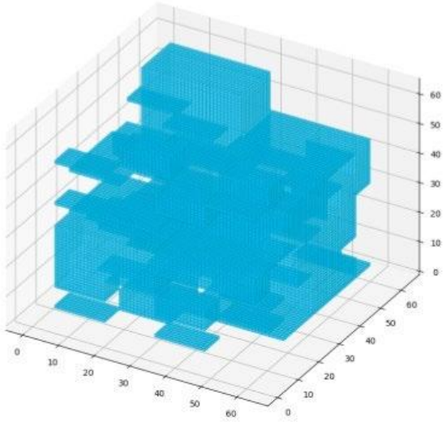
2023 Spring

AIA generative course

Faculty: Oana Taut, Dami Akinniyi

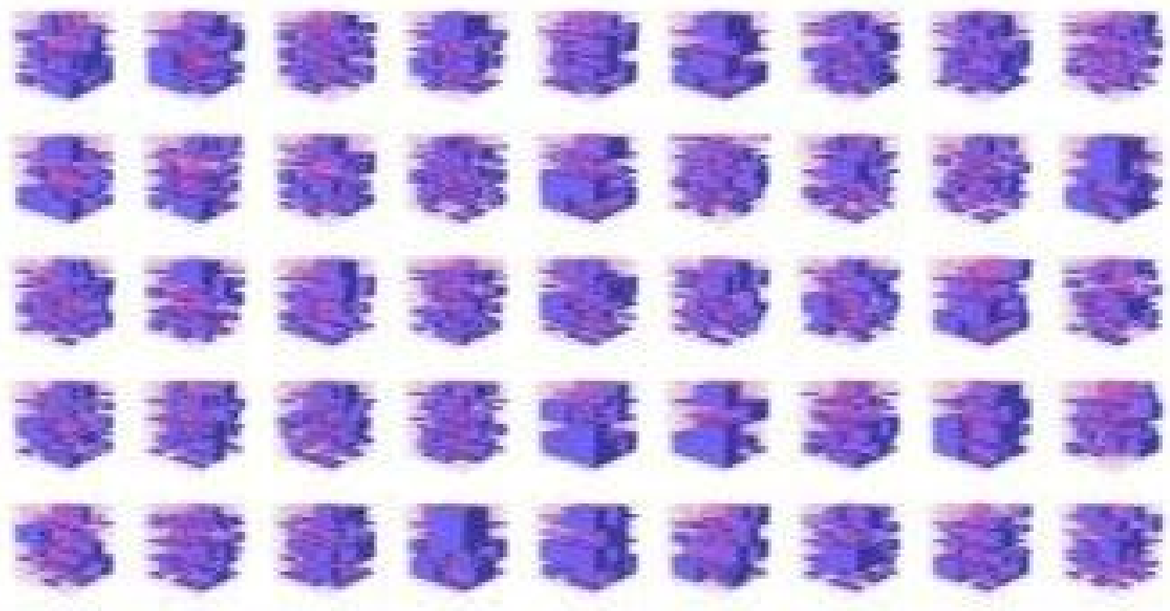




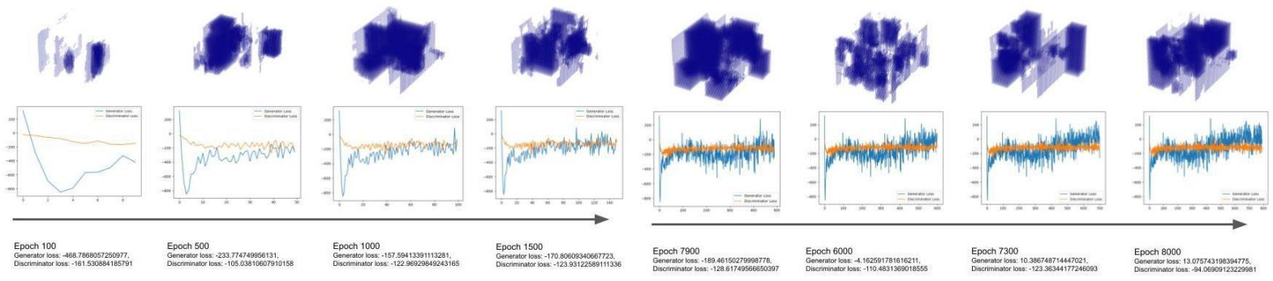


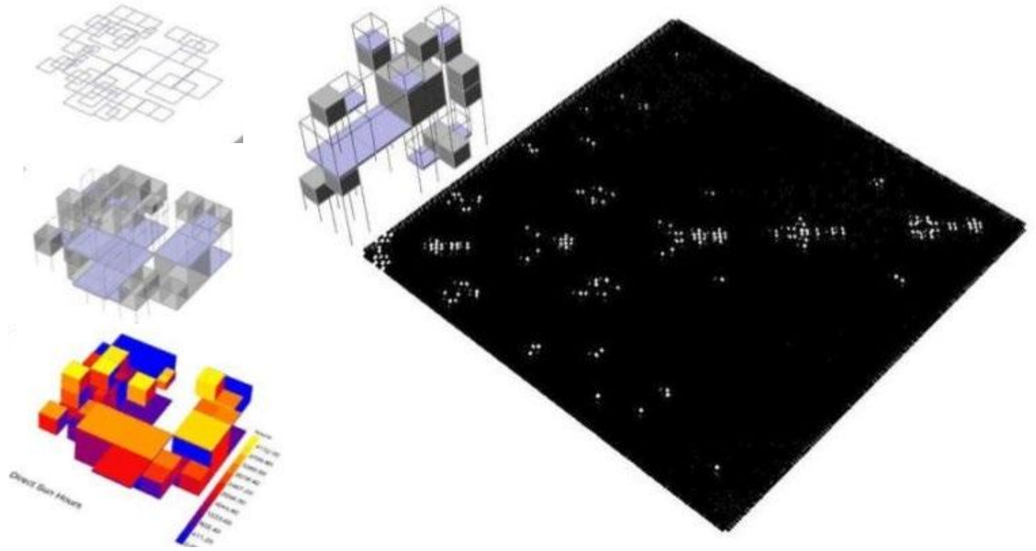
3D binary voxels + 3D convolutional network

Dataset generation: around 2000 samples



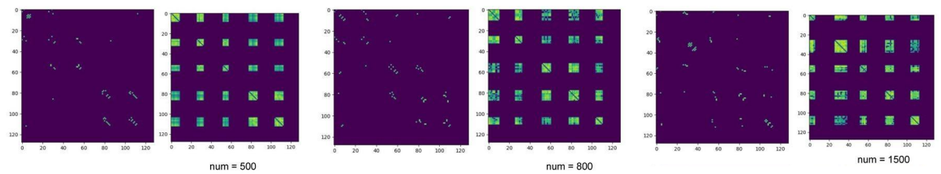
Training evolution



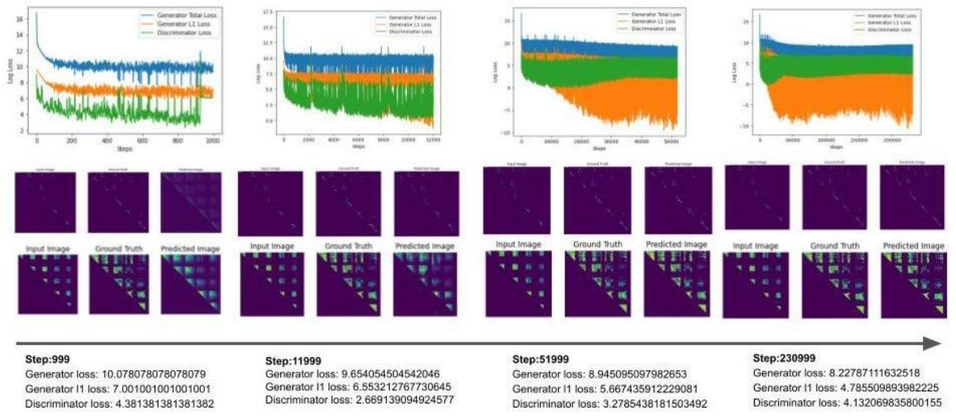


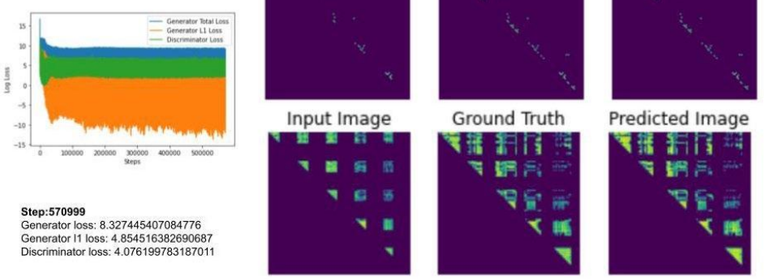
2d adjacency matrix + pix2pix model

Dataset generation: 1965 pairs of training and test dataset

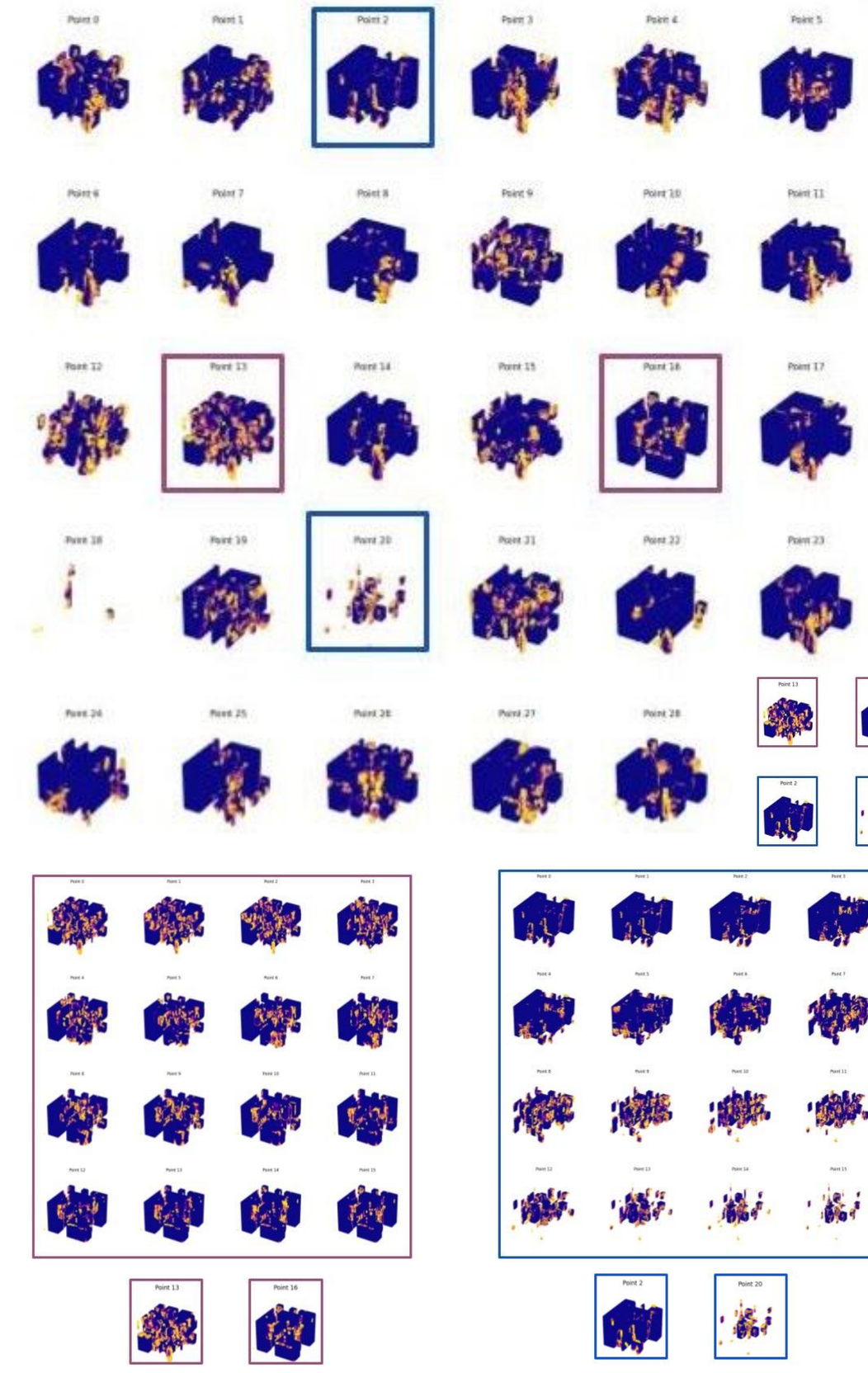


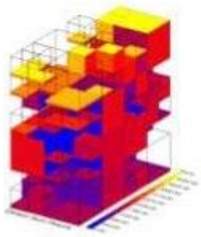
Training evolution

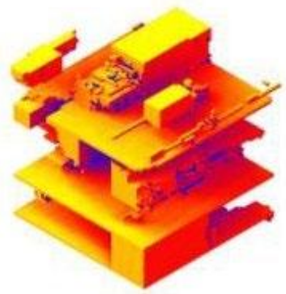


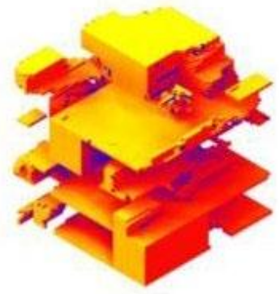


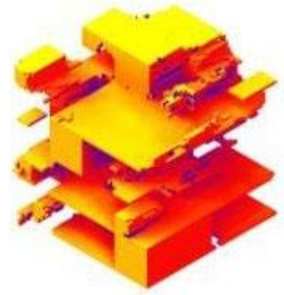
Latent space exploration

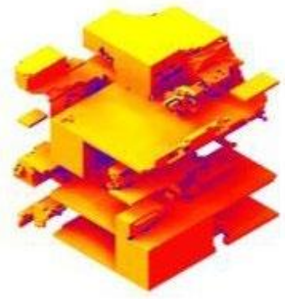


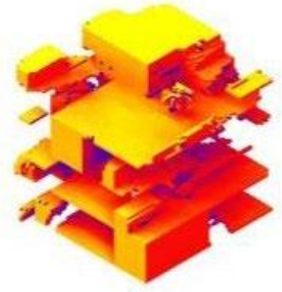


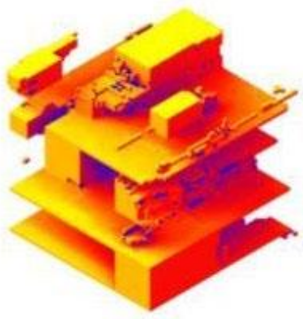


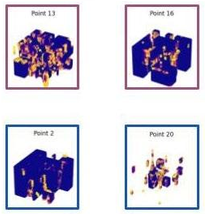




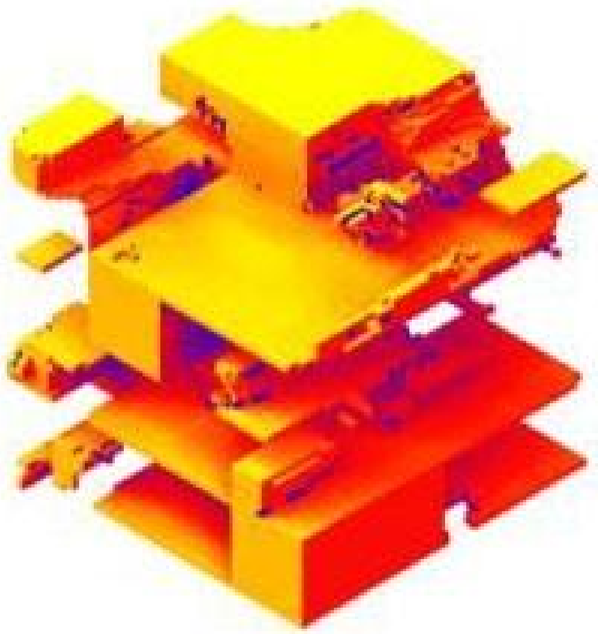




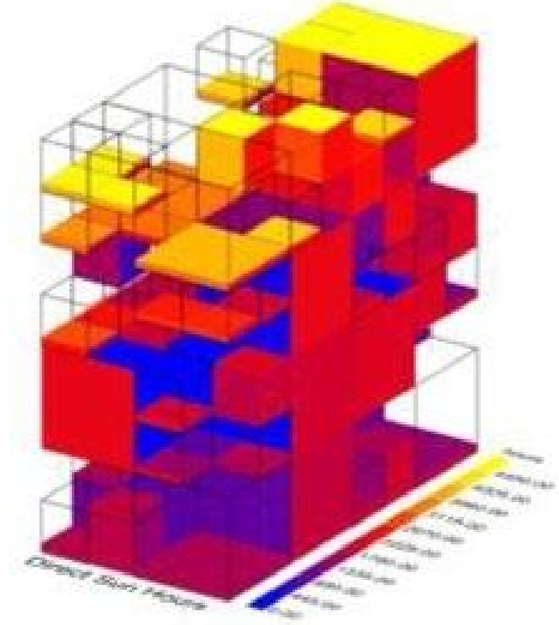




3d binary voxels



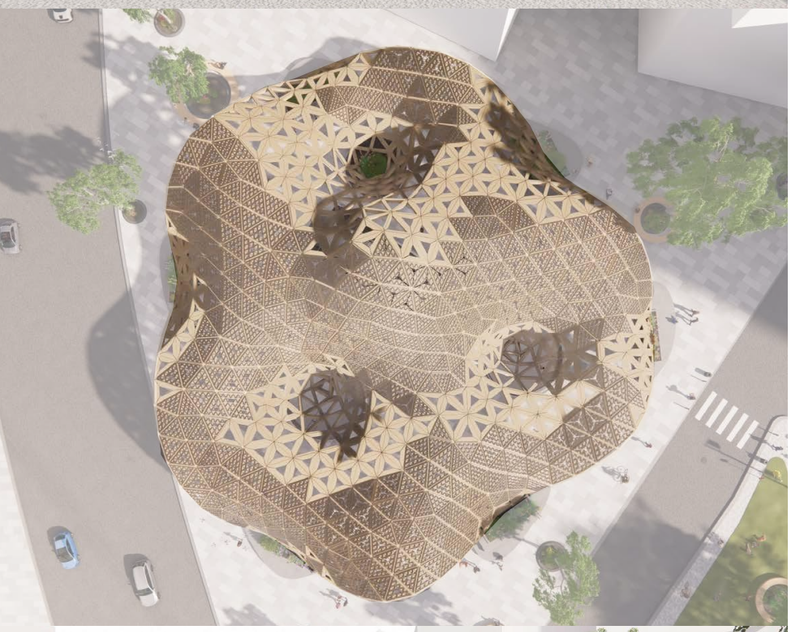
2d adjacency matrix



Climate-friendly Transport Hub

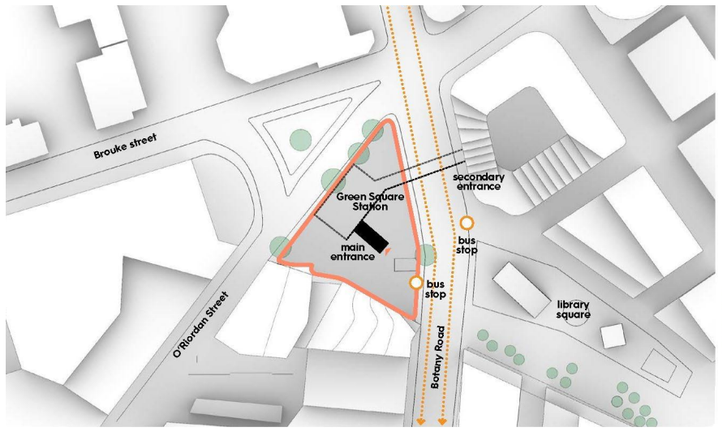
Building performance simulation and responsive design

Laurence Antelme, Zigi Cui, Littieri Lamb, Georgios Bekakos
2022 Fall
Faculty: Rodrigo Aguirre, Irene Martin Luque



N

site boundary



bus stop

bus lines

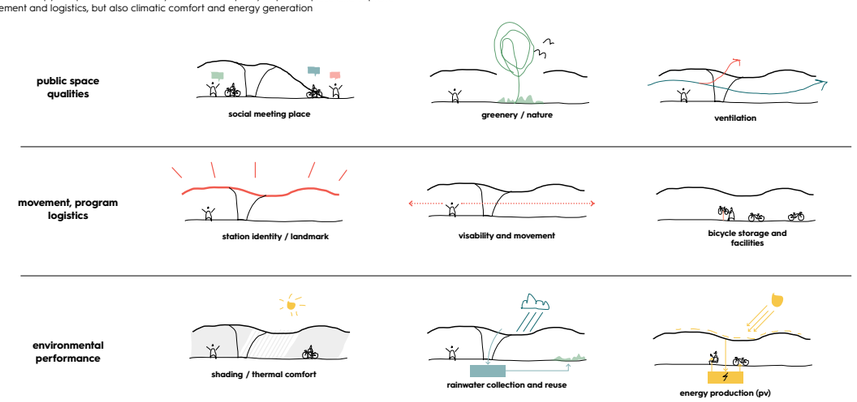
site boundary

N

0 20m 50m

Roof canopy qualities

The roof canopy has potential to not only enhance the quality of public space and improve movement and logistics, but also climatic comfort and energy generation



Existing site context

Today the site is an open paved surface with some existing trees. The entrance staircase and escalators to Station below are housed in a small pavilion

Movement lines across the site

Site programming is organised around pedestrian movement, insuring that wayfinding and convienence are given priority

Remaining space for program

The remaining spaces are available for program such as bicycle parking, a cafe area and greenery.

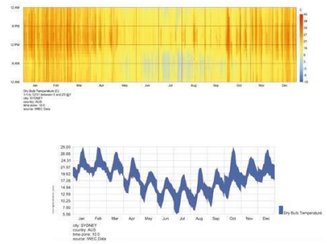
Structure support points

Areas are identified where structural support can be placed that do not interfere with movement and visibility

Connected roof canopy

To strengthen Green Square as a meeting place, a roof canopy covers most of the site providing shade, protection from rain and providing a human scale from the neighbouring tall buildings.

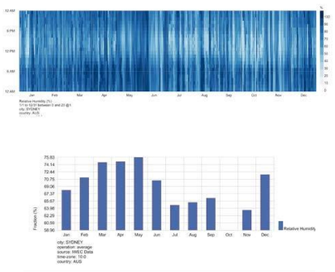
Hourly plot and Monthly Chart: Dry bulb temperature



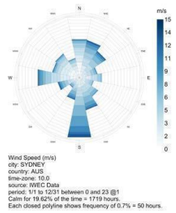
Sydney is generally warm throughout the year with many days above 15°C in all months. Some cooler temperatures in July around 10°C and over 30°C during the summer months.

Climate change predictions show a potential increase in temperatures, making cooling the city and reducing heat island effect important design considerations.

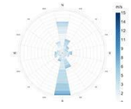
Hourly plot and Monthly Chart: Relative Humidity



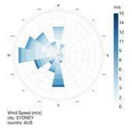
Sydney a relative humidity above 60% in all months of the year. Rainfall occurs in all months with the autumn months are the wettest and spring the driest.



Wind Speed (m/s)
city: SYDNEY
country: AUS
time-zone: 10.0
source: WEC Data
period: 1/1 to 12/31 between 0 and 23 @1
Calm for 19.62% of the time = 1719 hours.
Each closed polyline shows frequency of 0.7% = 50 hours.

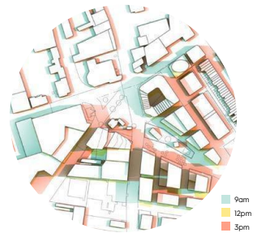


Summer months
Wind predominantly from the south and northeast

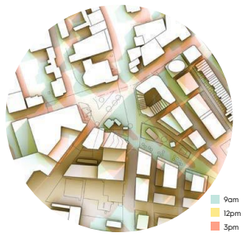


Winter months
Wind predominantly from the south and west

Sun and shadow analysis



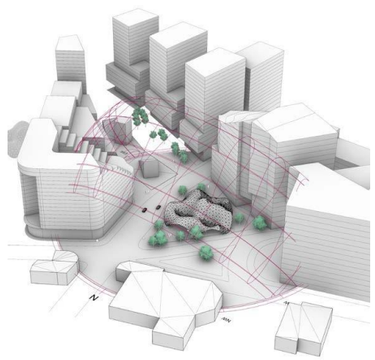
SUMMER SOLSTICE
21. December



WINTER SOLSTICE
21. June

SHADOW ANALYSIS

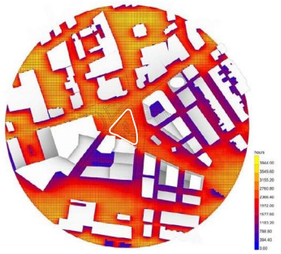
Sydney has a high global heat index (GHI) throughout the year indicating the importance of shading indoor and outdoor areas to create comfortable spaces for people and reducing the effects of heat islanding. Higher annual GHI shows a strong potential for the use of photovolatic and solar thermal strategies.



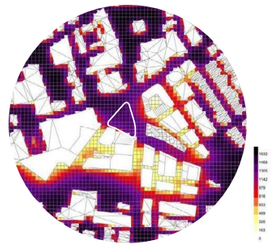
SUN PATH IN CONTEXT

The site faces north and despite the tall buildings, the site is very exposed to direct sun throughout the day

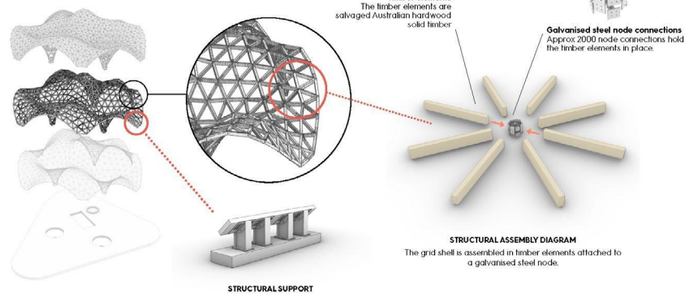
Sun hours and irradiation analysis



ANNUAL DIRECT SUN HOURS
High solar exposure indicates that shading is important for user comfort on the square.

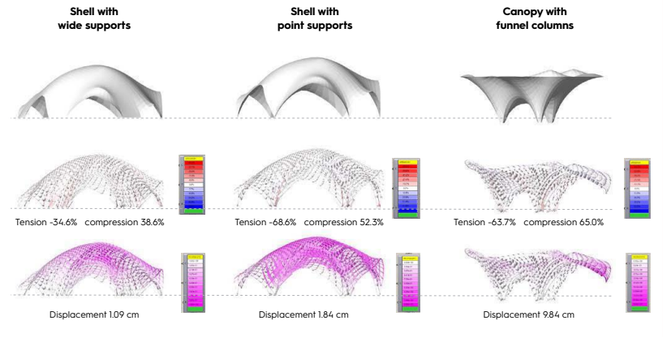


ANNUAL IRRADIATION
There is strong potential for year-round solar energy production on the site.



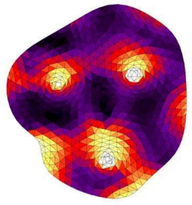
STRUCTURAL SUPPORT

STRUCTURAL ASSEMBLY DIAGRAM
The grid shell is assembled in timber elements attached to a galvanised steel node.

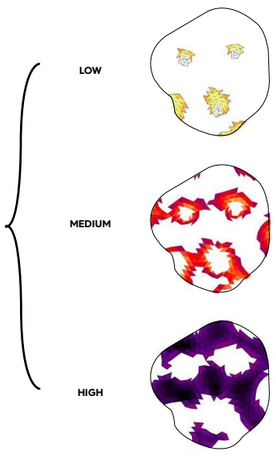


Climatic panel concept

Cross-laminated timber panel system fits within the triangular structural grid.



IRRADIATION ANALYSIS
We took an irradiation analysis of the canopy and based on the results, created 3 panel types the respond to different climatic conditions

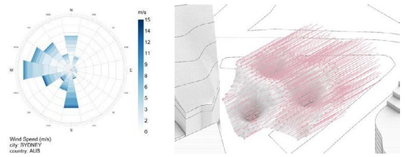


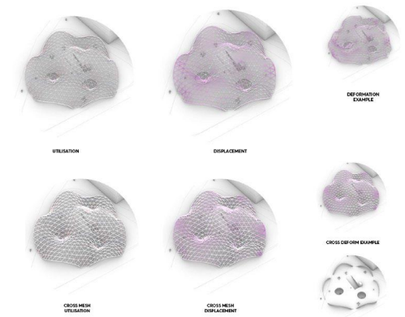
1. OPEN
The panels with the lowest radiation explosure are left open. These are also facing south which allow in diffused light in for daylighting the space below. They are also mostly located lower in the mesh, which increase visablity around the site

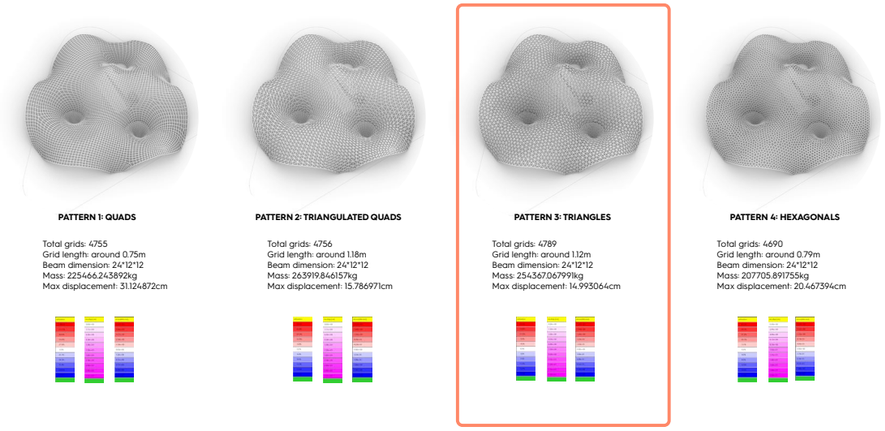
2. SEMI-OPEN
These panels exposed to much less irradiation and are semi open to allow some shading and a little light to the space below

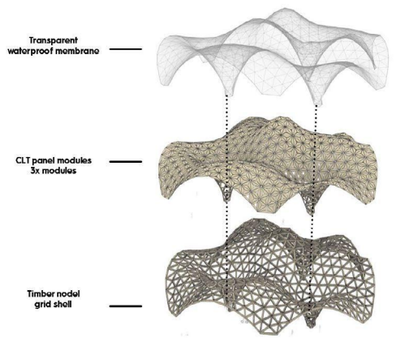
3. CLOSED
Where radiation was the highest the panel is fully closed to shade the space below

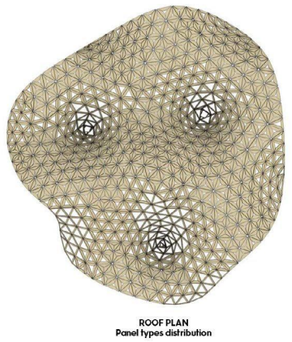
sation:
ə - North west



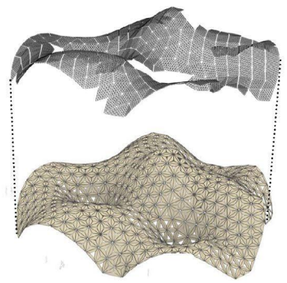


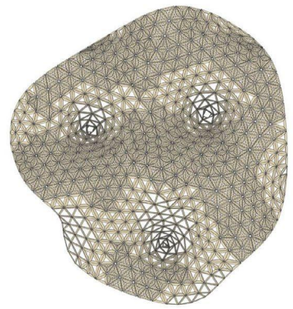




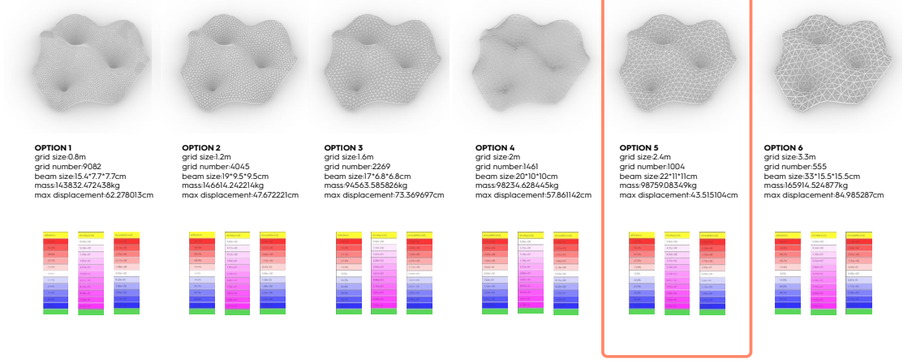


ROOF PLAN
Panel types distribution



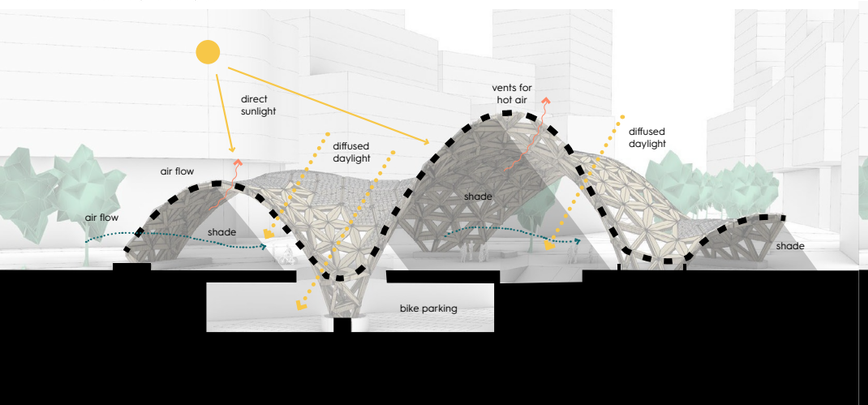


PHOTOVOLTAIC PANELS (PV)
PV are attached to the outer layer of the canopy in areas



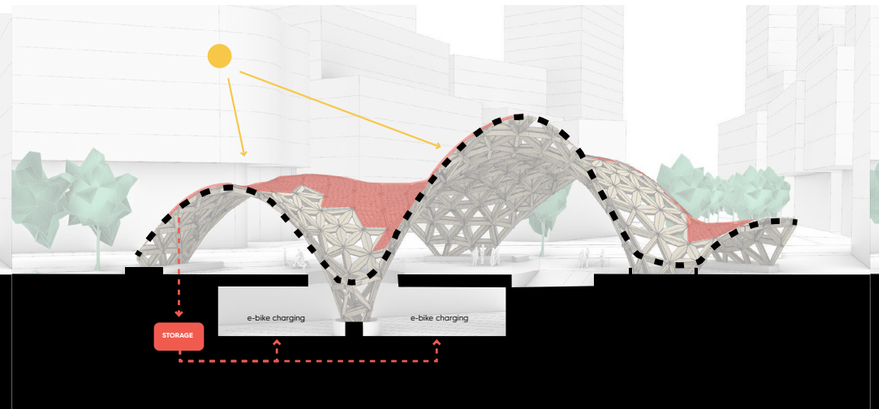
Passive design

The canopy shades the interior by blocking direct sunlight with closed panels, while open panels allow diffused daylight in from the south. The open air canopy also allows air to pass through vents at high points in the canopy.



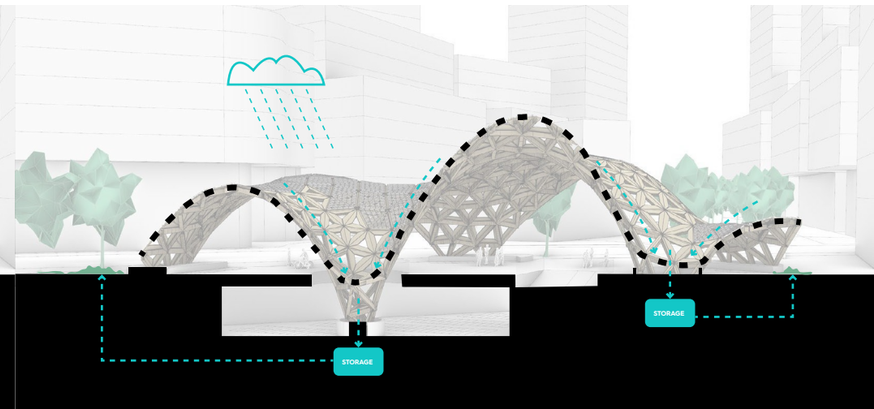
E-bike charging system

Solar energy is collected and storged in battery, then distributed to charge e-bikes and outdoor lighting. The **2735m2** pv system generates approx. 976,590kw/m² per year.



Rainwater catchment and reuse

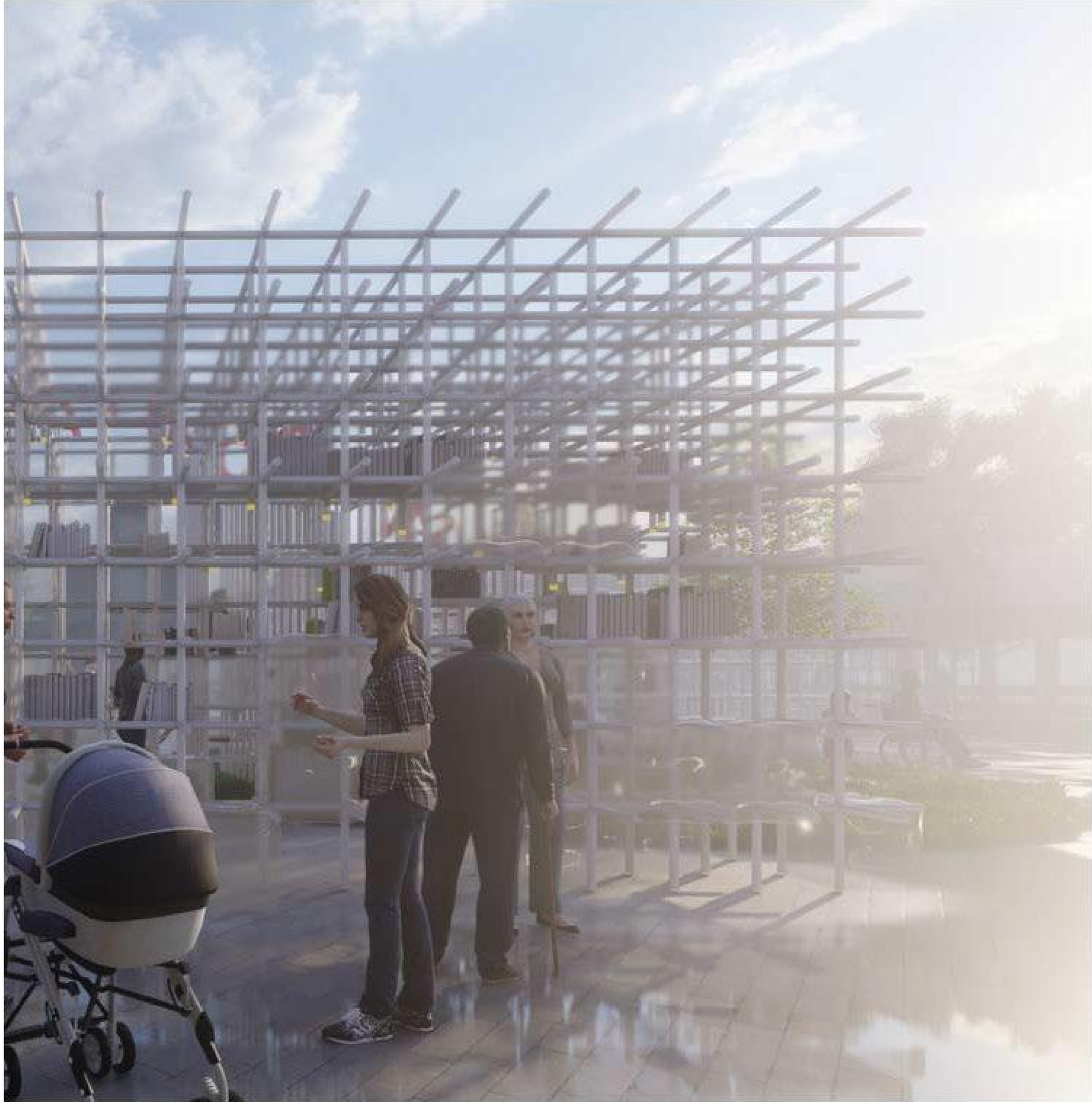
Rainwater is collected on the roof and channeled down the funnel columns and stored in cisterns. Water is then pumped with energy from the pv system to irrigate the site.



Modular Inflatable Interactive Pavilion

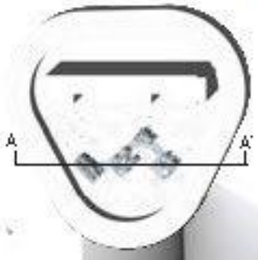
How will urban facilities meet the diverse and dynamic needs of the public?

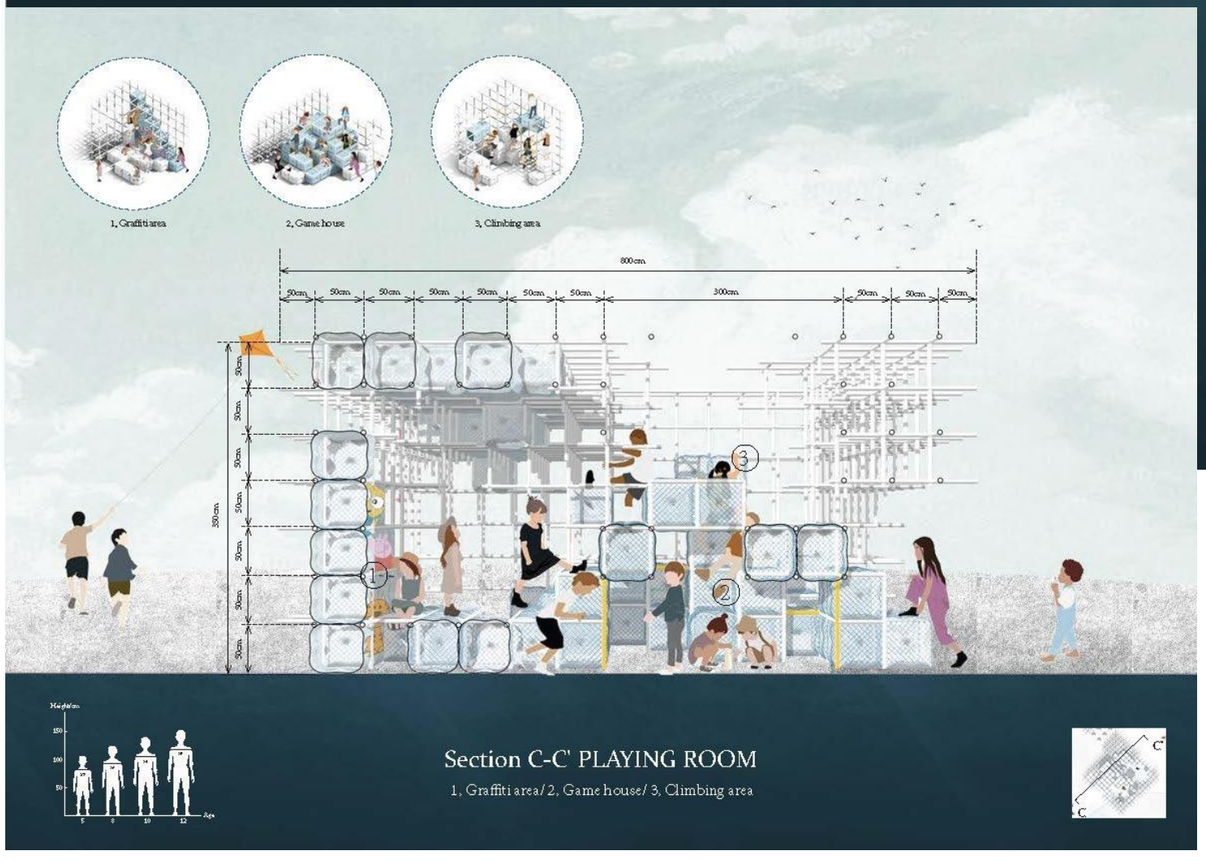
Zigi Cui, Boshi Chen, Yihao Tong, Xinjie Zhang
2021 Fall
Design and construction Studio
Faculty: Prof.Ingrid Paolatti, Prof.Elena Meh



Section A-A'

1, Reading Room
2, Meeting Room
3, Rest Room
4, Play Room

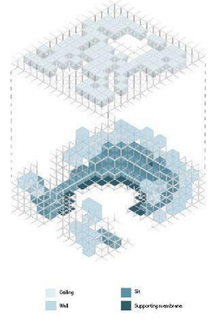


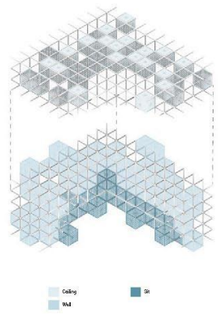


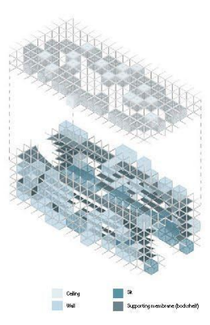
Section C-C' PLAYING ROOM

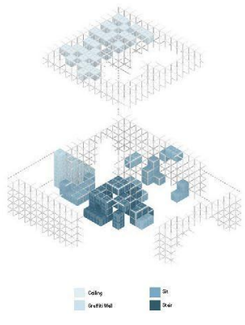
1, Graffiti area/ 2, Game house/ 3, Climbing area

DISTRIBUTION OF INFLATABLE BLOCKS WITH DIFFERENT FUNCTIONS

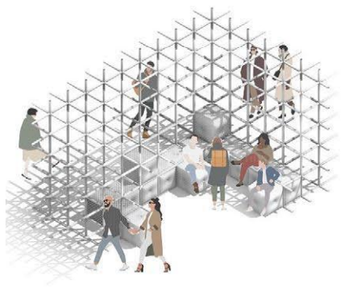

Meeting Room


Relax Room


Reading Room


Play Room

NCTION OF INFLATION BLOCK

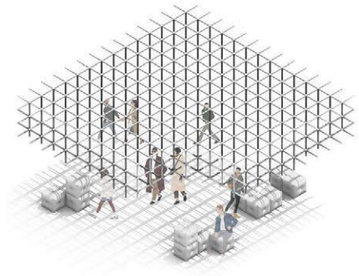


Deflation status

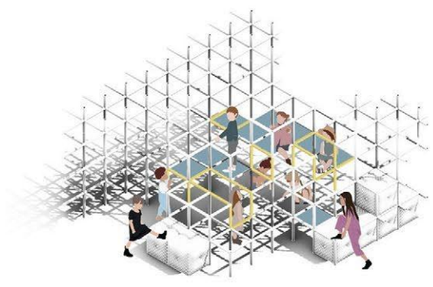
Deflation status

Inflation status
(It can be a wall to divide a space.)

Inflation status
(It can be used as a seat.)

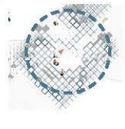


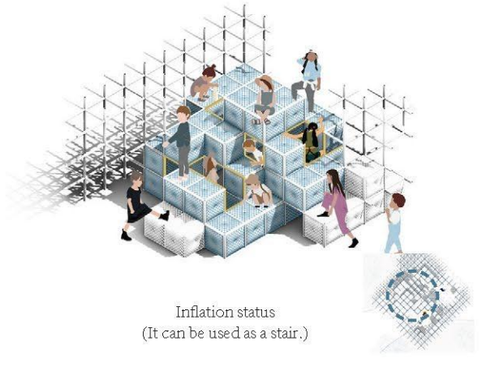
Deflation status



Deflation status

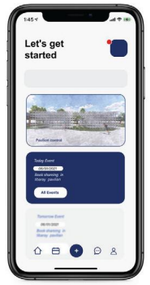
Inflation status
(It can be used as a sunshade.)



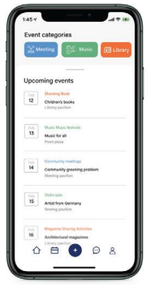


Inflation status
(It can be used as a stair.)

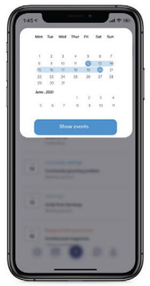
APPLICATION SYSTEM



Step 1



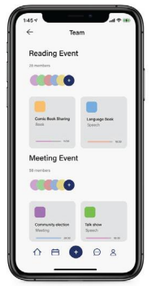
Step 2



Step 3

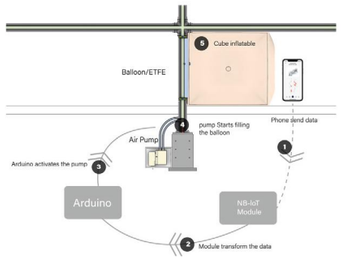
Step 4

Step 5

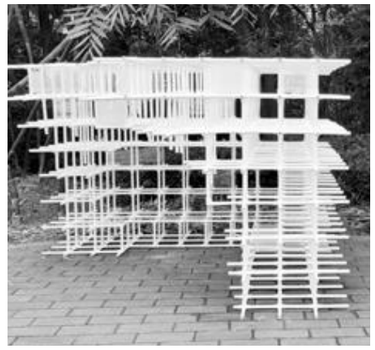


Step 6

DEVICE CONTROL LOGIC



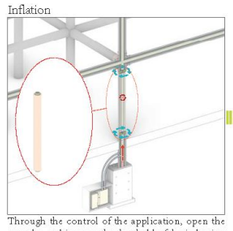
DEMOLITION OF PAVILION



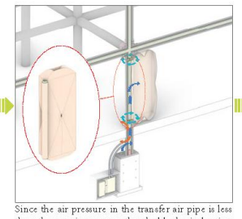
WORKING PRINCIPLE OF AIR BAG

Induction air pump | Transit trachea | ETFE membrane | Air valve | Induction barometer | Airbag

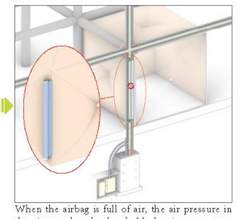
Inflation



Through the control of the application, open the gas valve and increase the threshold of the induction barometer,turn the shaft and eject the airbag.

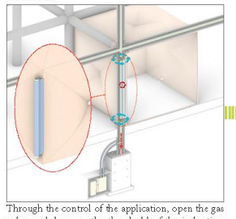


Since the air pressure in the transfer air pipe is less than the set air pressure threshold, the induction air pump starts to inflate the air bag and the shaft continues to turn.

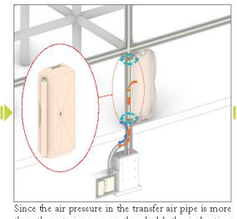


When the airbag is full of air, the air pressure in the pipe reaches the threshold, the air pump stops inflating, and the air valve is automatically closed.

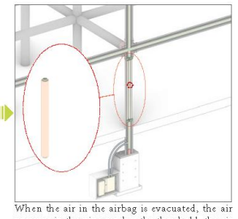
Deflation



Through the control of the application, open the gas valve and decrease the threshold of the induction barometer,turn the shaft and retract the airbag.

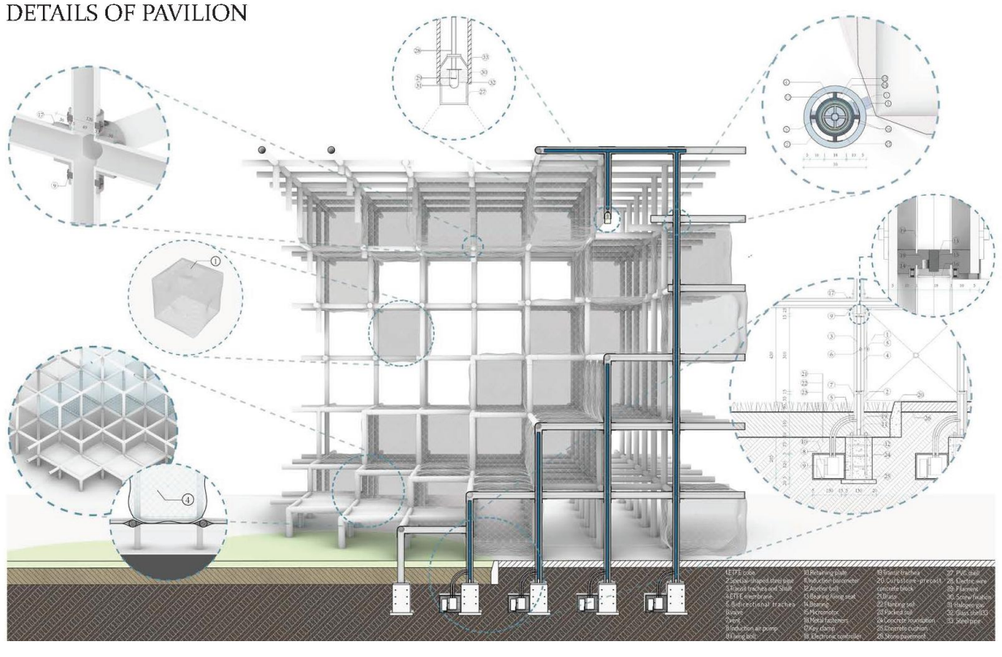


Since the air pressure in the transfer air pipe is more than the set air pressure threshold, the induction air pump starts to deflate the air bag and the shaft continues to turn.

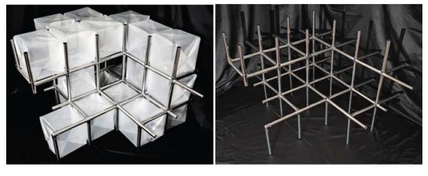


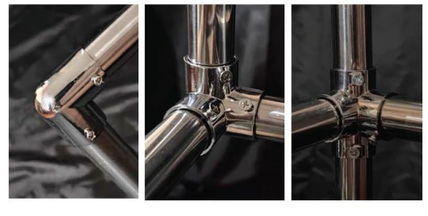
When the air in the airbag is evacuated, the air pressure in the pipe reaches the threshold, the air pump stops delating, and the air valve is automatically closed.

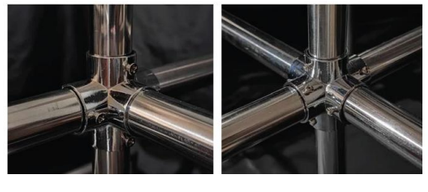
DETAILS OF PAVILION



DETAIL MOCK UP (1:2)



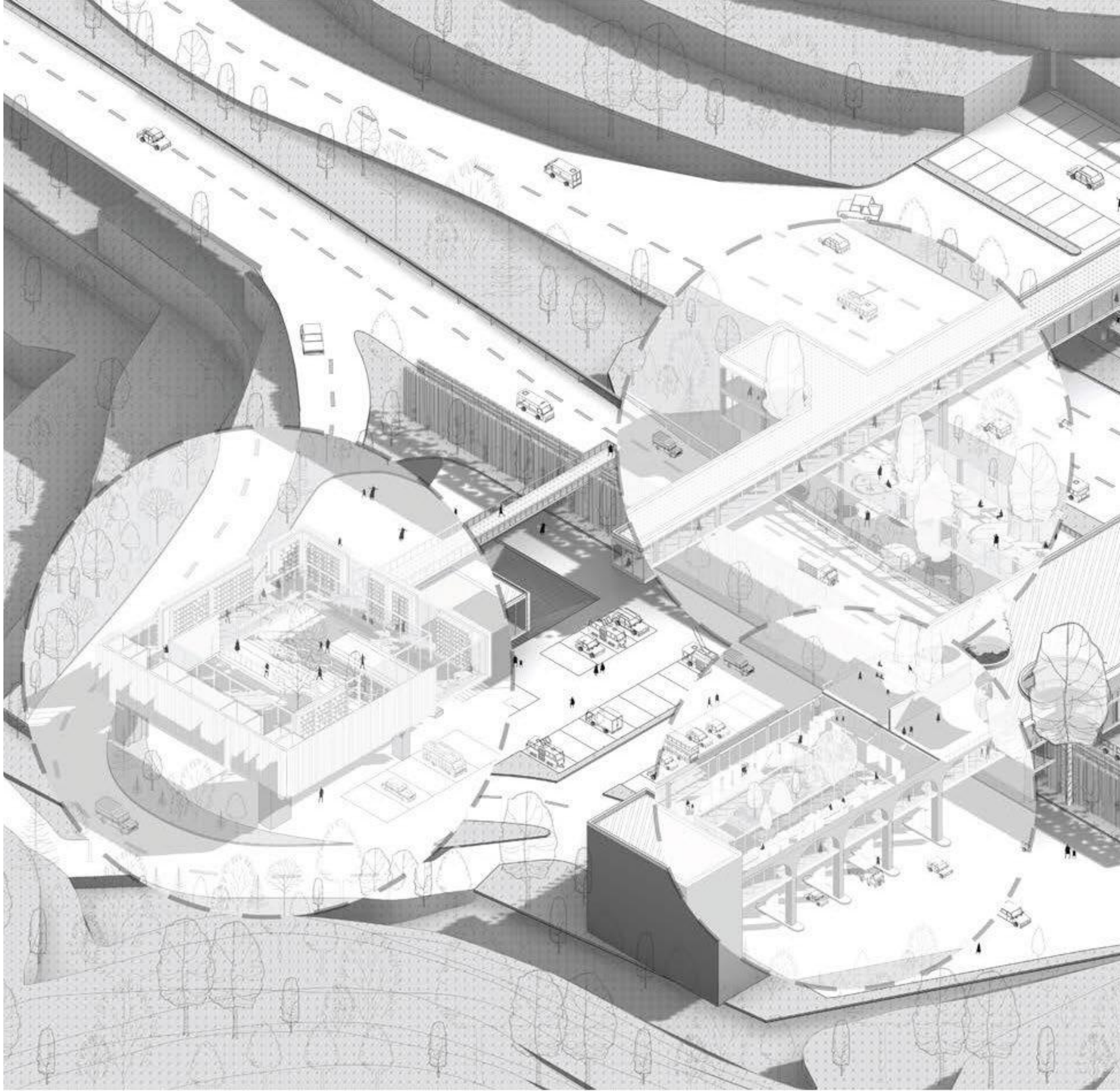


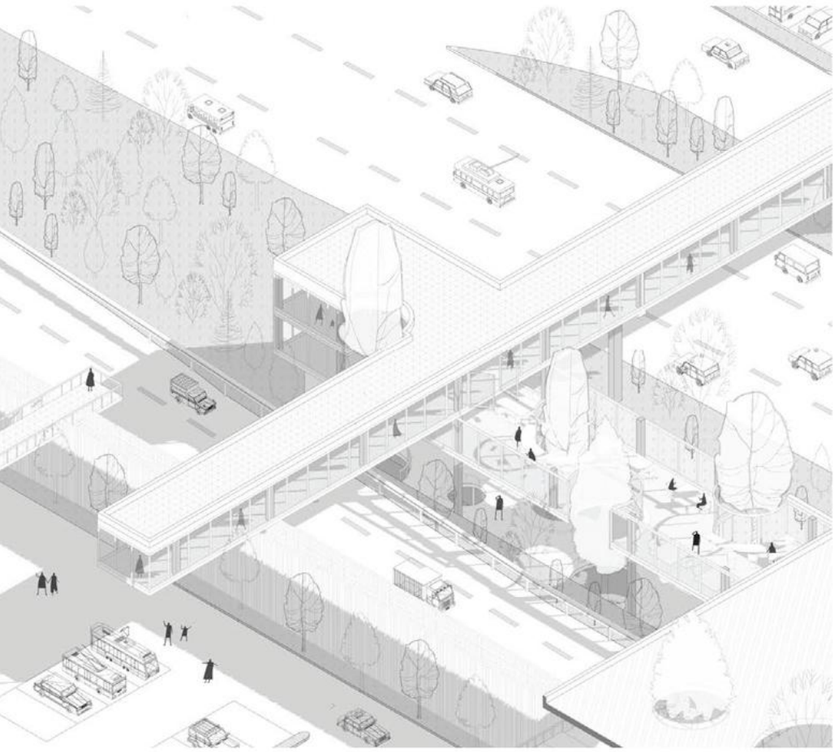


CAMPORA
Service Center

Highway service area design.

Zigi Cui, Shuqing Chen, Suofeiya Nanxi
2020 Fall
Faculty: Prof.Alessandro Rocca, Prof.Gian Luca Brunetti
Monica Manfredi, Luca Negrini, Francesca Zanotto

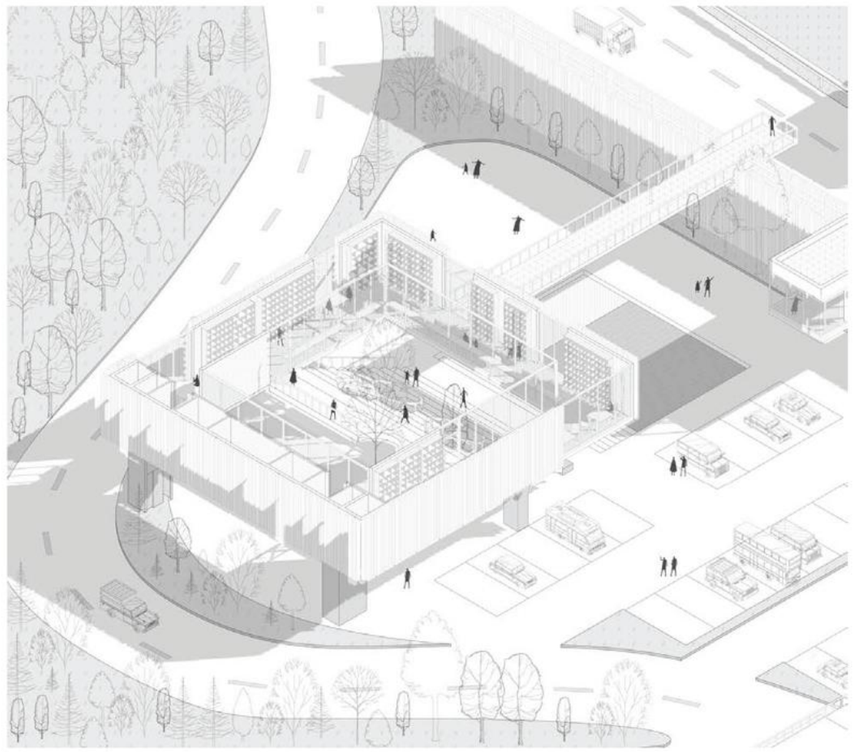




I

SERVICE CENTER

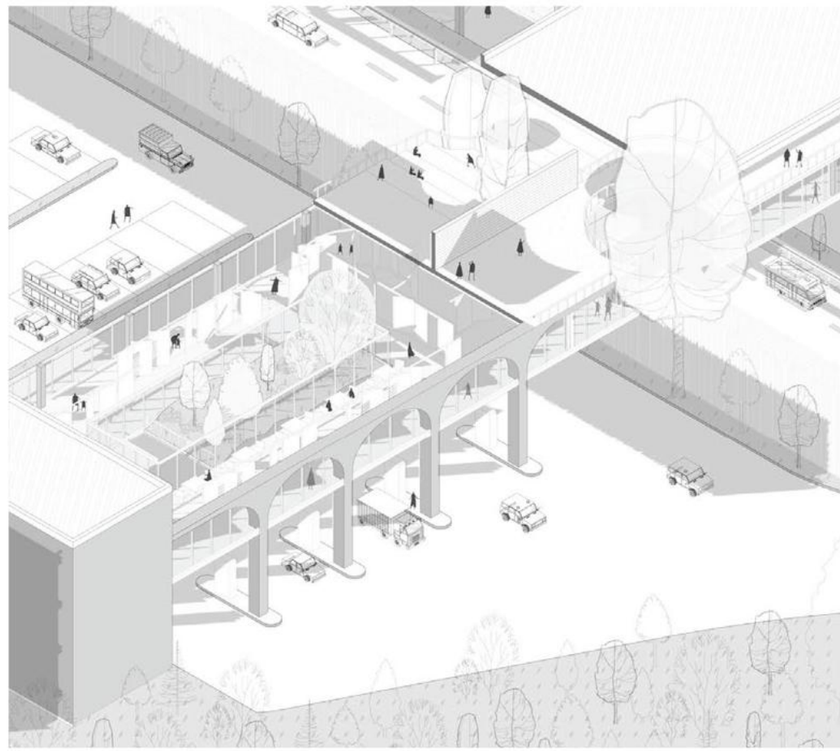
The service center is located in the middle of the site. It includes restaurant, outdoor garden, roof terrace, entertainment center and some service space.
In order to create a more comfortable environment, it uses a lot of glass to get a better view on both sides of the highway. And the introduction of green plants into the interior, making the building better combined with nature.



II

ARCHIVE CENTER

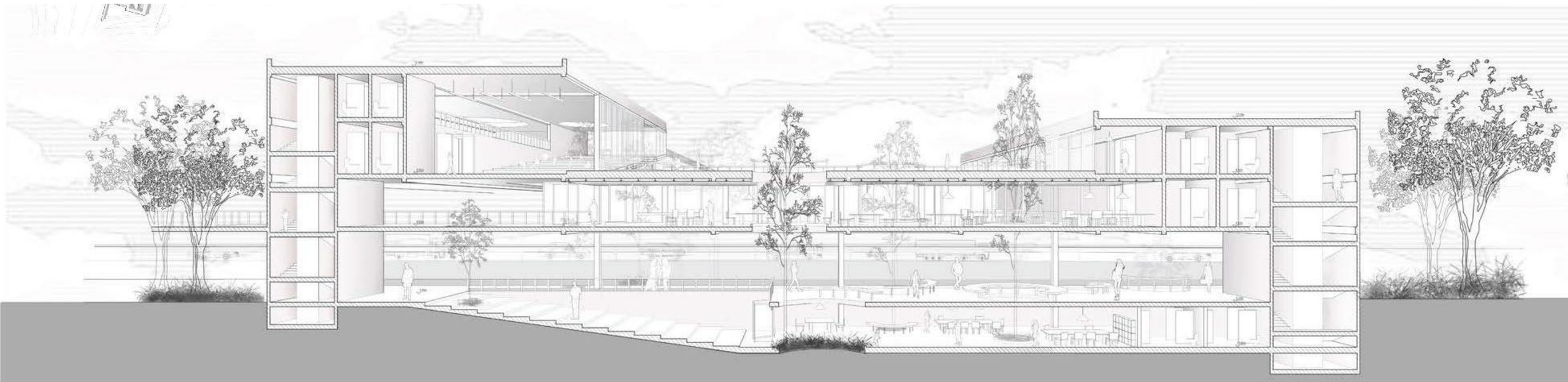
The archive center is located in the southeast of the site.
It is a small library which contains lots of reading materials about the history of highway for people to browse and read. Attracting visitors to stay, slow down people's travel speed, so that they can better understand the history of the Italian highway.



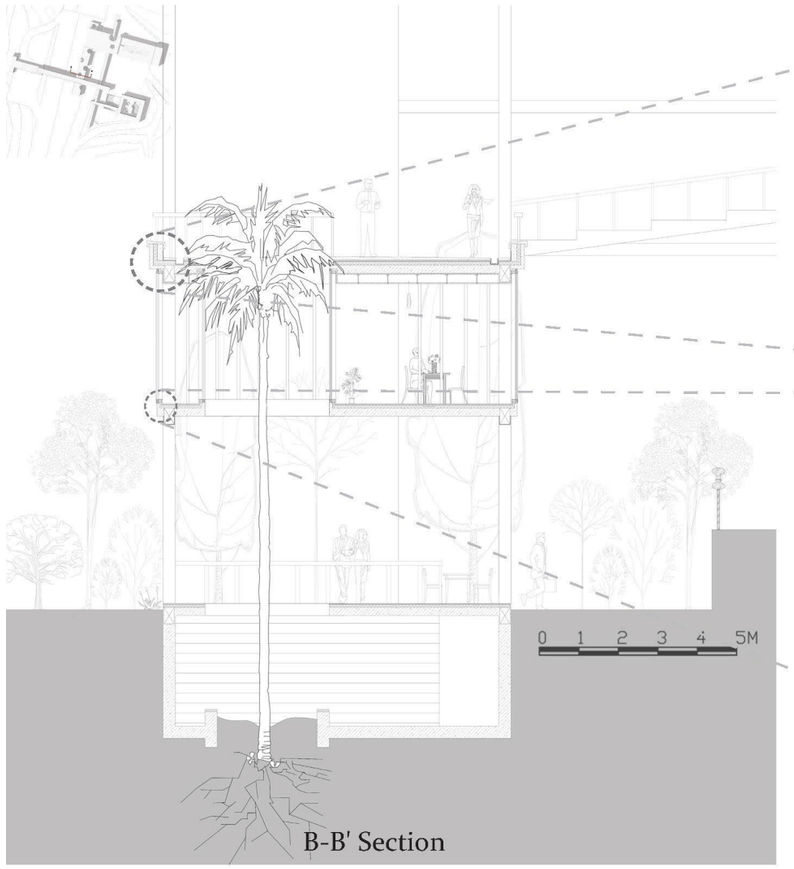
III

ACCIDENT MUSEUM

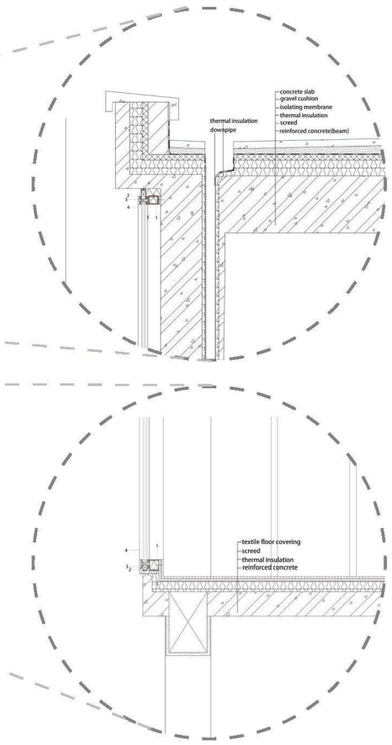
The accident museum is located in the northeast of the site.
This is an exhibition space to show tourists all kinds of accidents caused by highway damage in Italian history. It helps people understand the development history of Expressway and stimulate tourists' interest in high-speed travel.



D-D' Section

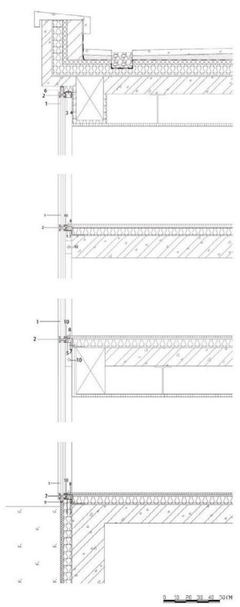


B-B' Section



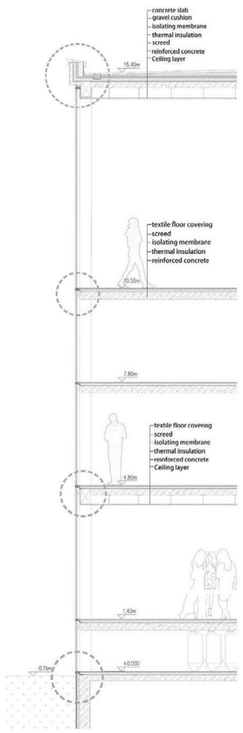
Detail Section 0 10 20 30 40 50CM

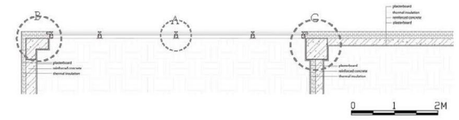
1, Insulated aluminium section, with thermal break
2, Plastic spacer block
3, Aluminium facade rail
4, Double glazing
5, Movement joint, permanently elastic seal



Detail Section

1, Double glazing
2, Aluminium facade rail
3, Insulated aluminium section, with thermal break
4, Movement joint, permanently elastic seal
5, Steel angle, hot-dip galvanized
6, Plastic spacer block
7, Sheet aluminium cover and vapour barrier
8, Thermal insulation
9, Insulated facade panel
10, Bolt in elongated hole



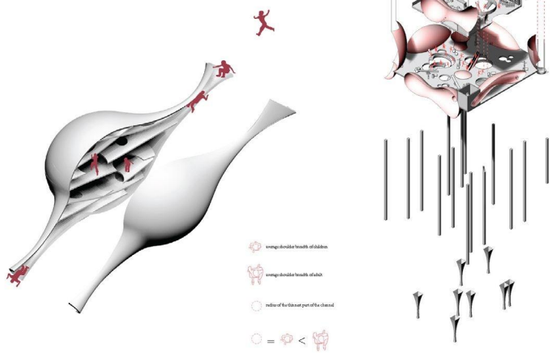


Rabbit hole for kids

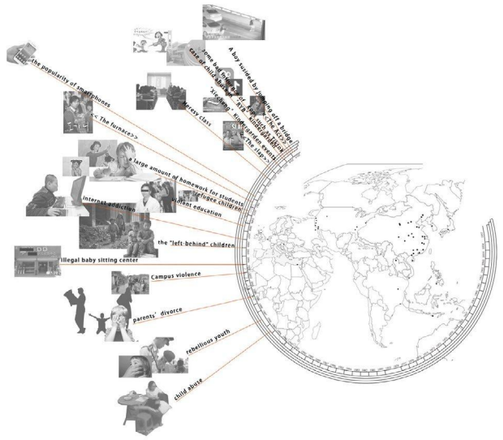
A physical and mental refuge from a shooting incident

Zigi Cui
2018 Fall
Architecture Design Studio
Faculty: Wei Wu

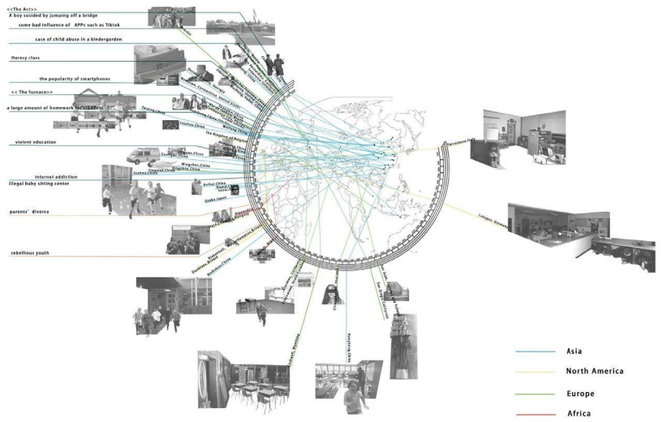
Sandy Hook shooting event lets people pay more attention to the Children protection. Comparing to teenagers who have more self-protection awareness, it is difficult for kids in kindergarten to protect themselves when they are in danger. Besides, children also face some increasing serious mentality problems. So, this project proposes a possible plan which can provide kids a useful protection in the school: when the danger is coming, channels around the building can let children exscape in a short time.These slide-like channels can take them into all kinds of irregular space which become a shelter for the moment. In normal times, as a relatively closed entertainment space, the irregular space can help children keep away from annoying reality and create a no teachers' pressure, completely free world.



A specific irregular space: It shows that how to use the irregular space to escape.

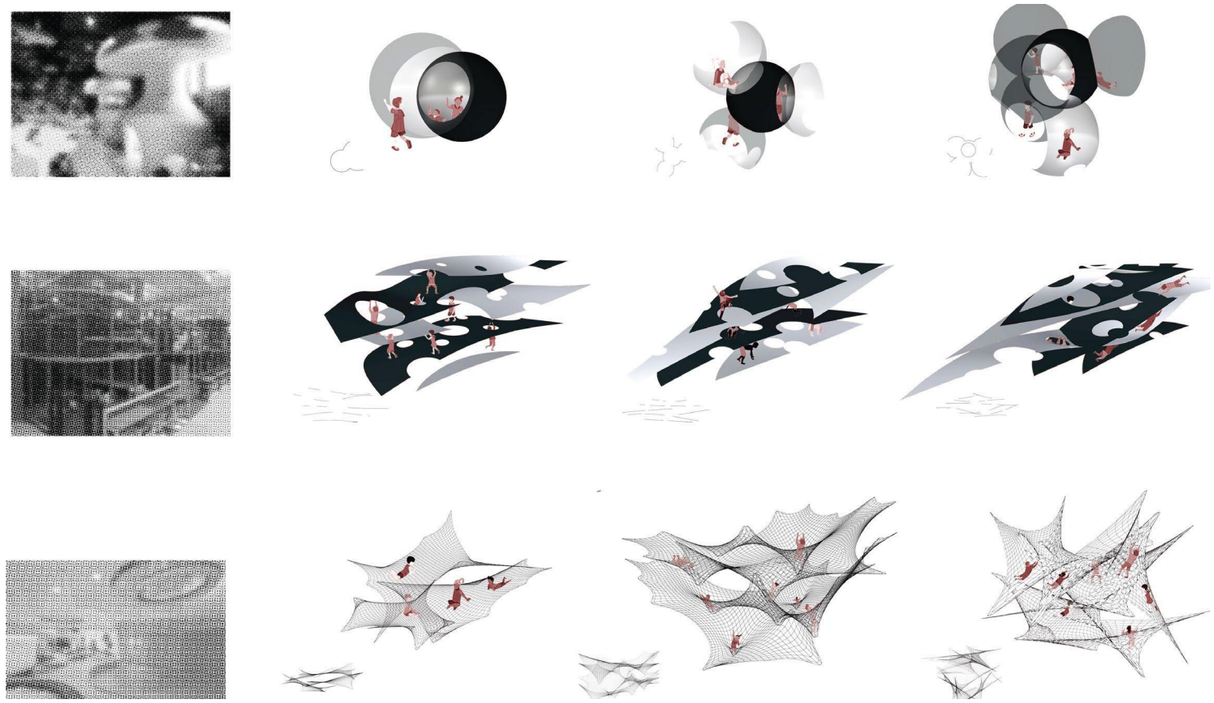


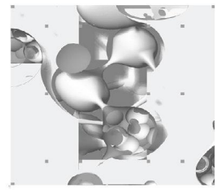
Other damage events for children (1995--2019.04 Incomplete statistics)

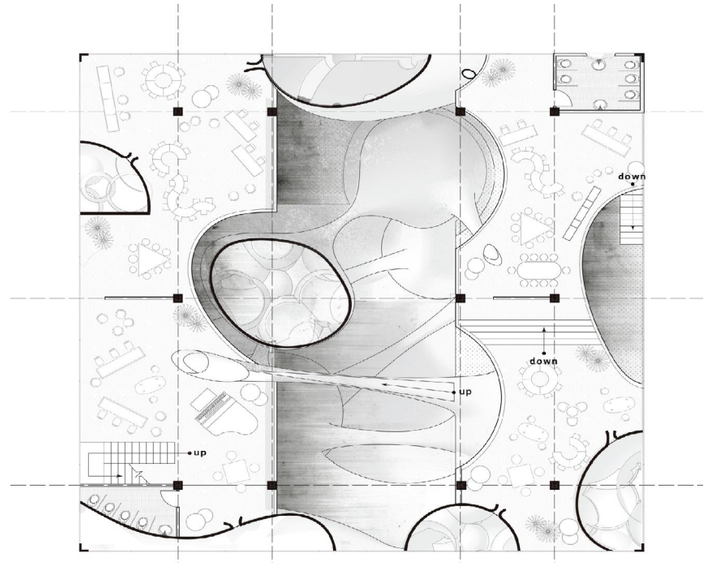


Primary school shooting events (1957--2019.04 Incomplete statistics)

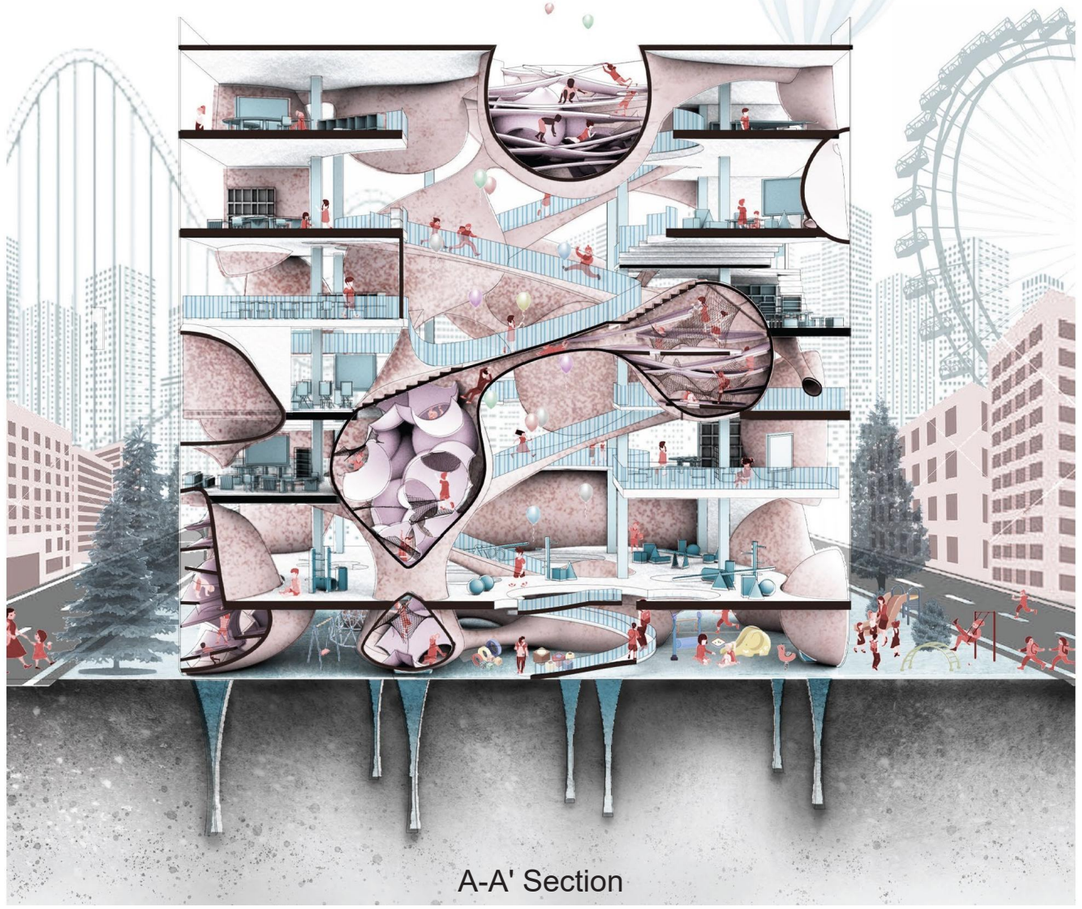
Concept Development



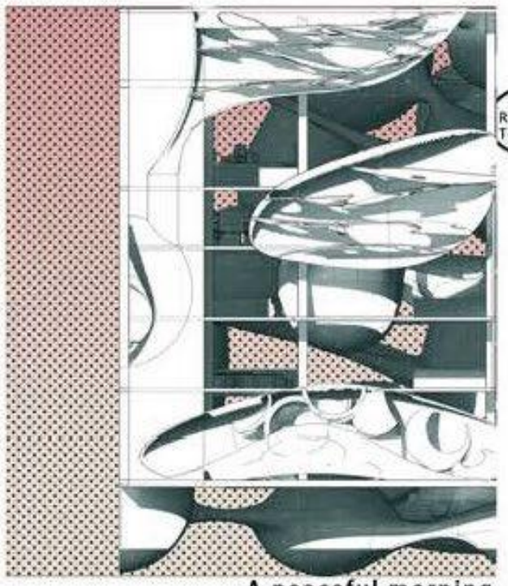




Fifth floor plan

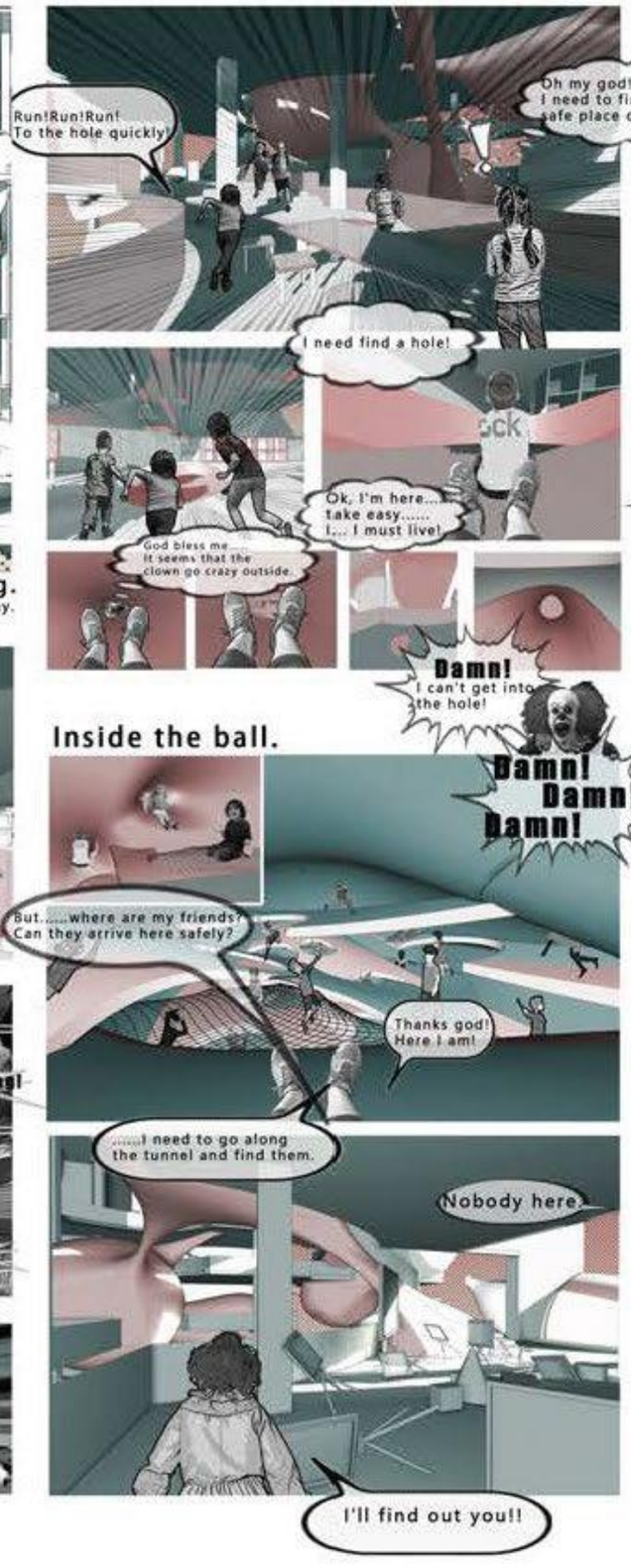


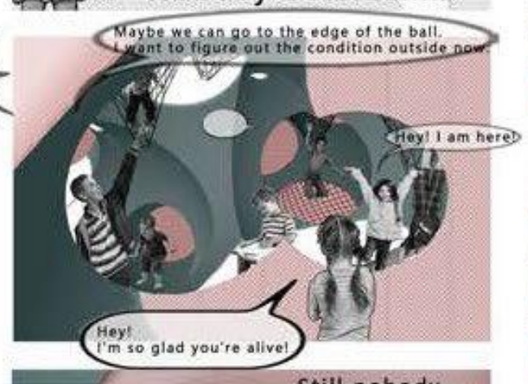
A-A' Section

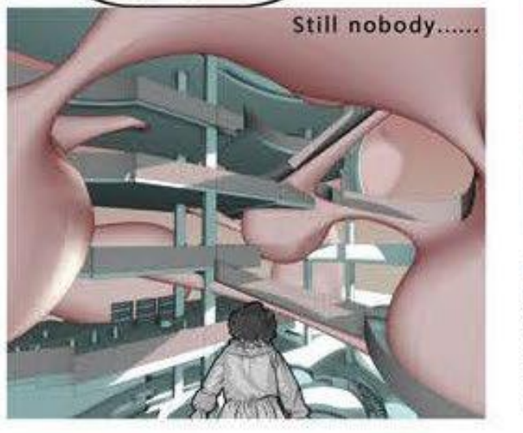


A peaceful morning.

Children go to their kindergarden and prepare for a new day.





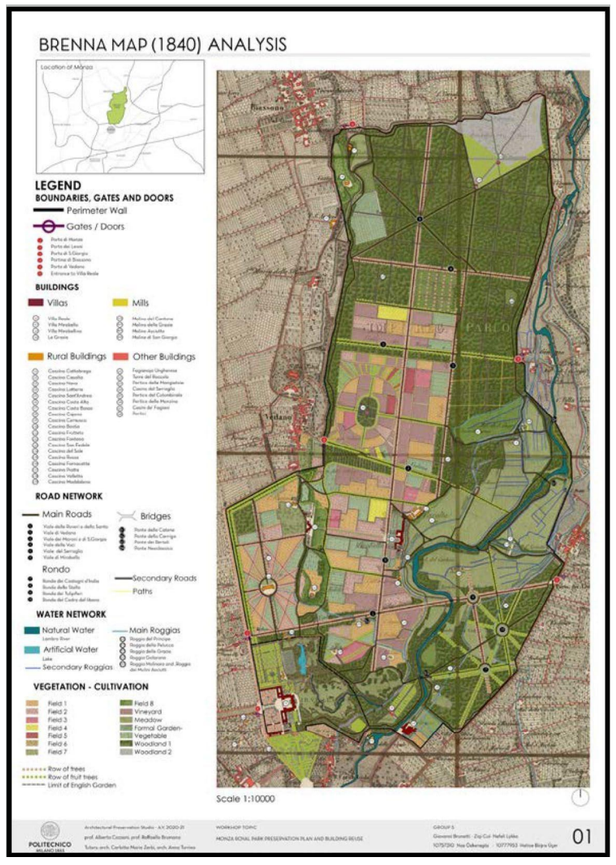


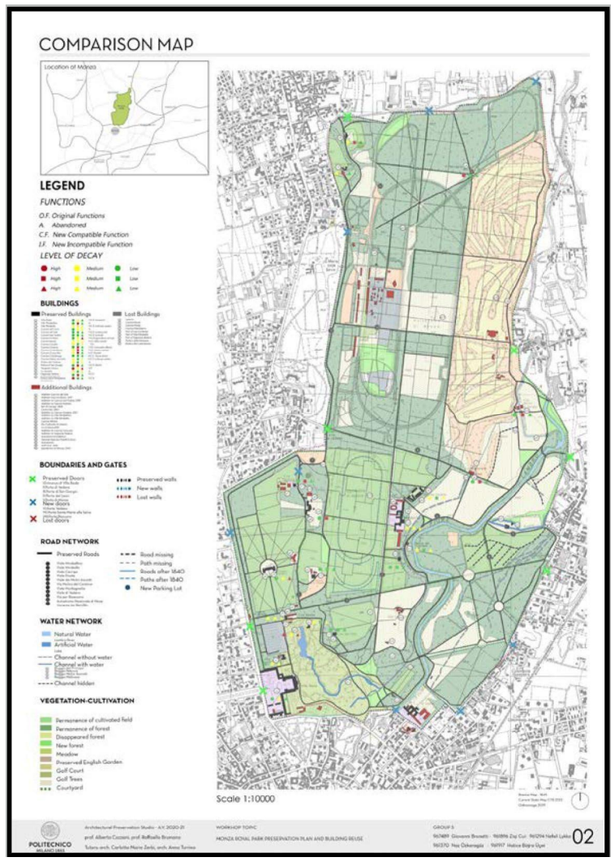
-The end-

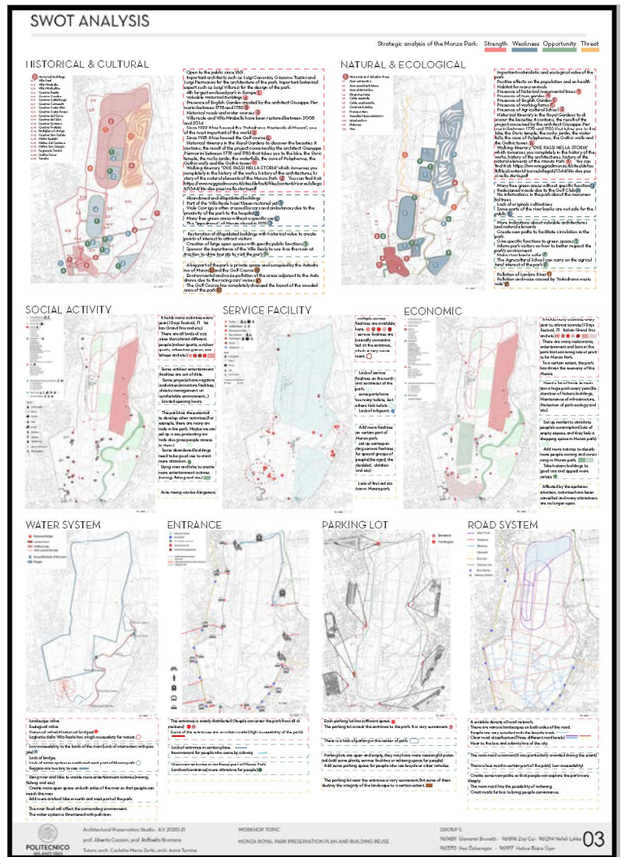
The MONZA Park

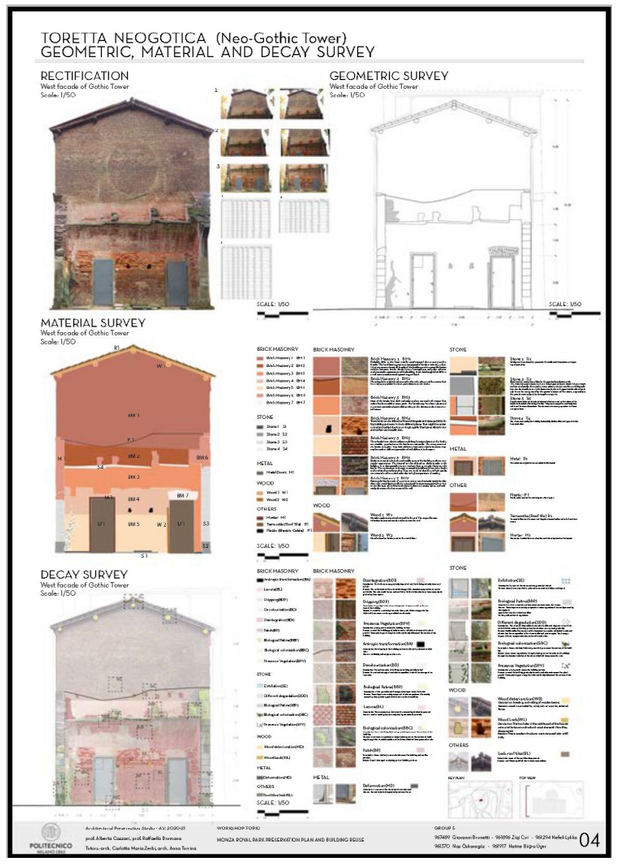
A preservation project of the royal park Monza

Giovanni Brunetti, Zigi Cui, Nefeli Lykka Naz Ozkaragoz, Hatice Busra Ucer
2020 Fall
Architecture Preservation Studio
Faculty: Prof.Alberta Cazzani, Prof.Raffaella Brumana

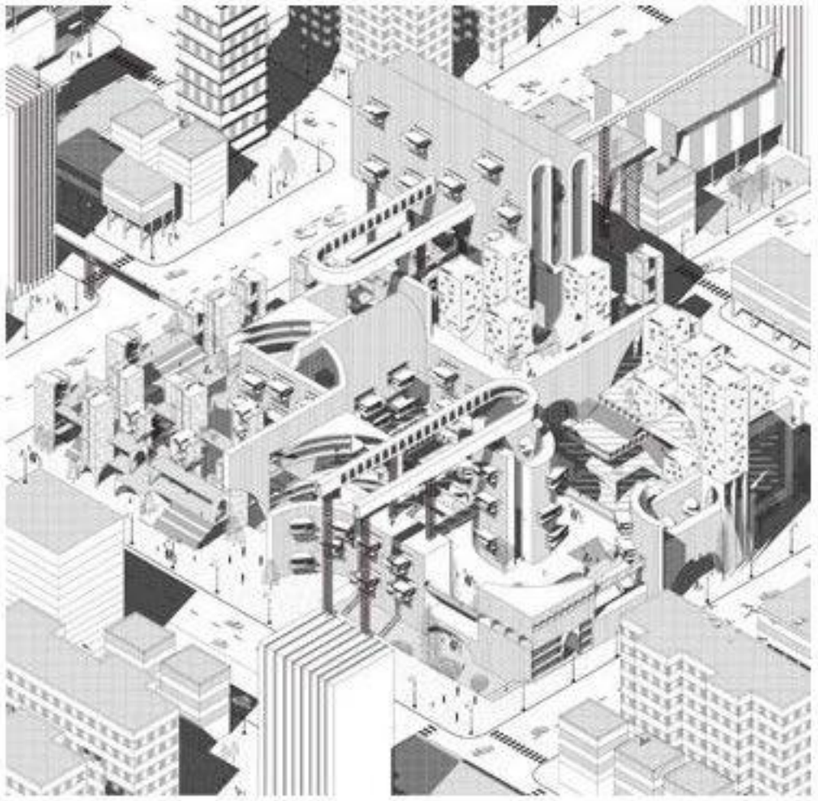


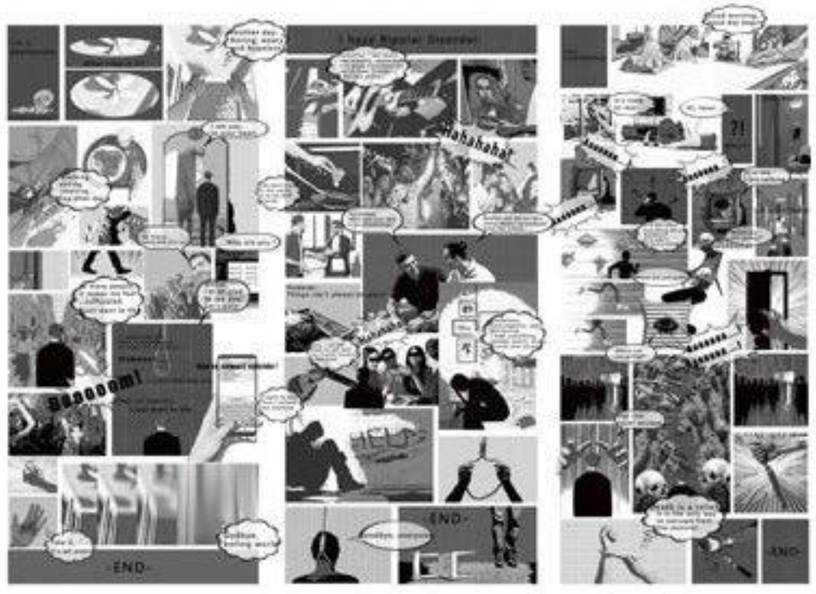






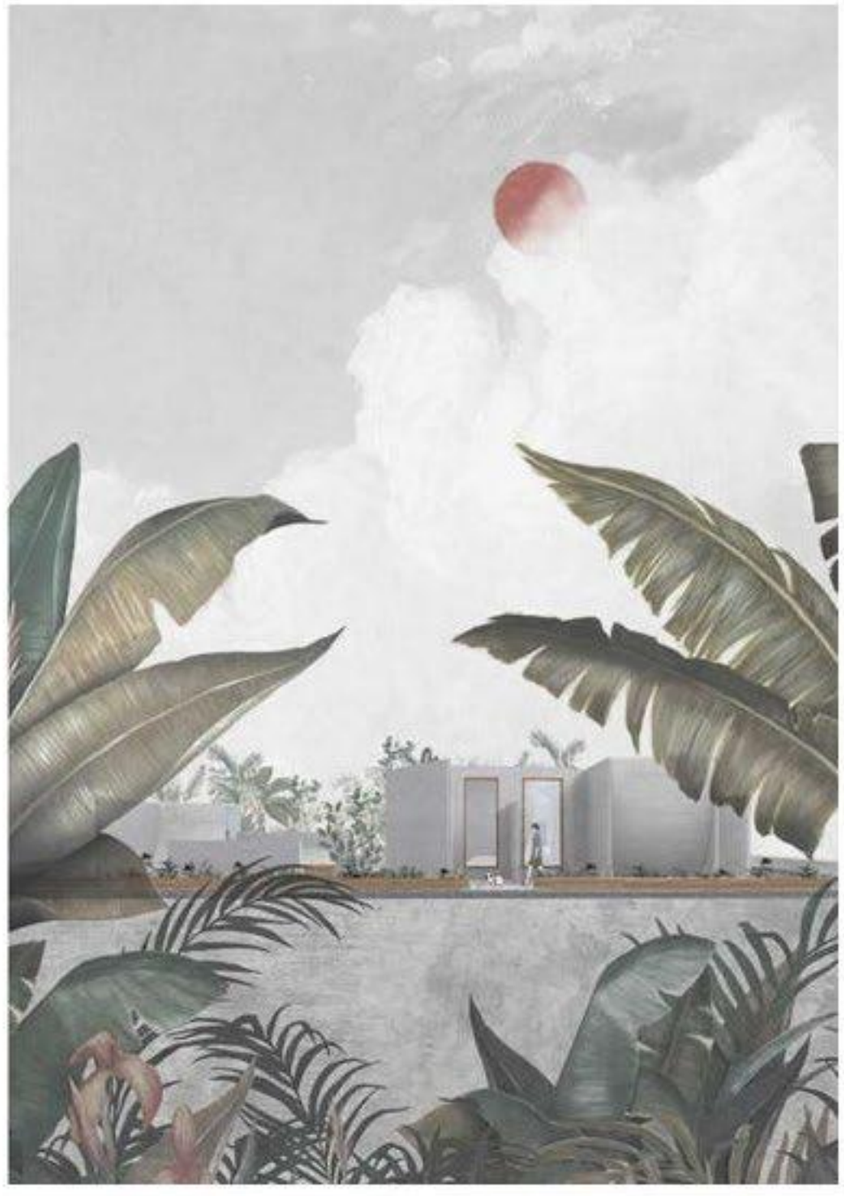
OTHER WORKS





Asylum

Individual Work/ 2019 Spring



Home for owners and pets

Individual Work/ 2019 Fall

PORTFOLIO

ZIQI CUI